Class 9

Biology

(Solutions)

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1. Diversity in Living organisms – Plants

Solutions

LEVEL-I

1. Correct option: (1) a-iii, b-iv, c-ii, d-i

Chlamydomonas is a unicellular alga. Cycas is a gymnosperm, Selaginella is pteridophyte. Sphagnum is bryophyte or a moss.

2. Correct option: (B) Whitaker

Five kingdom of classification was proposed by RH Whittaker in 1969. He divided all living organisms into five kingdoms based on following features: Complexity of cell structures. Complexity of body structures.

3. Correct option: (A) Metaphyta

The Kingdom Plantae is also called as kingdom Metaphyta. The Kingdom Plantae includes all types of eukaryotic, multicellular, photosynthetic plants found in this biosphere.

4. Correct option: (D) Bacteria

Cyanobacteria are sometimes considered algae, but they are actually bacteria (prokaryotic), where the term "algae" is now reserved for eukaryotic organisms. They also derive their energy through photosynthesis, but lack a nucleus or membrane bound organelles, like chloroplasts.

5. Correct option: (A) Linnaeus

Carolus Linnaeus gave the artificial system of classification which is based on few morphological characters.

6. Correct option: (C) Both (A) and (B)

Artificial systems of classification were based upon morphological characters {such as habit, habitat, colour, number and shape of leaves, etc.) for grouping of organisms. They were mainly based on vegetative characters or on the androecium structure.

7. Correct option: (C) Both (A) and (B)

Artificial systems were the earliest systems of classification that used only gross superficial morphological characters such as habit, colour, number and shape of leaves, etc. They were based mainly on vegetative characters or on the androecium structure (system given by Linnaeus).

8. Correct option: (D) Based on mainly vegetative characters (or) on the androecium structure

Natural classification systems were based on natural affinities among the organisms and consider, not only the external features, but also internal features, like ultrastructure, anatomy, embryology and phytochemistry. Such a classification for flowering plants was followed by George Bentham and Joseph Dalton Hooker.

9. Correct option: (A) Evolutionary relationships between the various organisms

Phylogenetic classification system is based on the evolutionary relationship. As it is based on the evolution of life and the evolutionary history of a group of genetically-related organisms is called a phylogeny. A phylogeny is usually represented by phylogenetic tree.

10. Correct option: (3) a-iii, b-iv, c-ii, d-i

Artificial systems of classification were based upon morphological characters for grouping of organisms. They were mainly based on vegetative characters or on the androecium structure. Natural classification system is based on natural affinities among the organisms. It considers external features along with internal features such as ultrastructure, anatomy, embryology and phytochemistry. Phylogenetic classification is based on evolutionary relationships between organisms. Cytotaxonomy is the system of classification of organisms which is based upon the cytological information of their cells like chromosome number.

11. Correct option: (B) Algae

Algae are chlorophyll-bearing, simple, thalloid, autotrophic and largely aquatic (both fresh water and marine) organisms.

12. Correct option: (2) i-c, ii-d, iii-a, iv-b

The science of classification is taxonomy, while the study of evolutionary connections between species is phylogeny. Numerical taxonomy is a classification system in biological systematics which deals with the grouping by numerical methods of taxonomic units based on their character states. Cytotaxonomy is the system of classification of organisms which is based upon the cytological information of their cells. These includes Chromosome number, morphology, and Behaviour.

13. Correct option: (4) a-iii, b-iv, c-i, d-ii

Spirogyra belongs to Chlorophyceae and has a filamentous structure. Colonial forms exist in case of volvox. Kelps have massive plant body reaching up to the surface of water body. Chlamydomonas are unicellular green algae.

14. Correct option: (D) All of the above

Vegetative reproduction usually takes place by fragmentation or by formation of different types of spores. Zoospore formation is rare. Sexual reproduction can be isogamous, anisogamous, or oogamous. Asexual reproduction can occur by flagellated zoospores produced in zoosporangia.

15. Correct option: (2) a-iv, b-ii, c-i, d-iii

Zoospores are motile asexual spores having two polar flagella. Fucus produce oogamous spores with single large female spore and small male spores. Chlamydomonas produces isogamous or similar types of motile gametes. Spirogyra produces anisogamous spores.

16. Correct option: (C) Both (A) and (B)

Certain marine brown and red algae produce large amounts of hydrocolloids (water holding substances), e.g., algin (brown algae) and carrageen (red algae) which are used commercially.

17. Correct option: (C) Gelidium and Gracilaria

Gracilaria has become the preferred seaweed for making food grade agar because it has been successfully cultivated in Chile and Indonesia. Gelidium continues to be the preferred seaweed for making bacteriological and pharmaceutical grade agars and agarose.

18. Correct option: (A) Chlorophyceae

The members of Chlorophyceae are commonly called green algae. The plant body may be unicellular, colonial, or filamentous. They are usually grass-green due to the dominance of pigments chlorophyll a and b.

19. Correct option: (D) Cellulose, pectose

Cell wall of green algae is thin, transparent and firm and consists of outer pectic and inner cellulosic layers. It is smooth but in most of species it gets thickened at the anterior end to form an apical papilla.

20. Correct option: (A) Both assertion (A) and Reasoning (R) are correct and (R) explains (A)

Pyrenoids are characteristically found in the chloroplast of algae. They are one or more storage bodies present mainly in Chlorophyceae (green algae). Pyrenoids contain protein and starch. RuBisCO enzyme is also present in pyrenoids. It helps in CO_2 fixation in plants.

21. Correct option: (A) Both assertion (A) and Reasoning (R) are correct and (R) explains (A)

Carotenoids, xanthophylls, chlorophyll a, and c are all present in Phaeophycean. Depending on the concentration of xanthophyll pigments, members of the phaeophycean family exhibit colour variations ranging from olive green to various hues of brown.

22. Correct option: (D) Both 1 and 2

Phaeophyceae are commonly known as brown algae. They possess chlorophyll a, c, xanthophylls, and carotenoids. Food is stored in the form of complex carbohydrates, mostly in the laminarin or mannitol form.

23. Correct option: (C) b and d

The members of Phaeophyceae or brown algae are found primarily in marine habitats. They show great variation in size and form. In the class phaeophyceae of algae, the plant body is usually attached to the substratum by a holdfast, and has a stalk, which is termed the stipe, and a leaf-like photosynthetic organ called the frond.

24. Correct option: (B) Statement- I is correct and II is also correct

In Algae, there are 3 types of sexual reproduction, Isogamy, Anisogamy, and Oogamy. In Isogamy male and female gametes have similar morphology. They have similar shape and size. In Anisogamy male and female gamete have different morphology. In Oogamy female gamete is significantly larger than the male gamete. In this, the female gamete is non motile and male gamete is motile. Gamete union can occur in the oogonium or water (oogamous species).

25. Correct option: (D) Laminaria, Ectocarpus, Sargassum

Brown algae are a group of algae belonging to class Phaeophyceae. They are named due to their colour, which varies from brown to olive green. They contain seaweeds, some of the notable examples are Ectocarpus, Fucus, giant kelps, Sargassum, etc.

26. Correct option: (C) Laminaria and sargassum

Laminaria and Sargassum are the brown algae (Phaeophyceae) and are cultivated as a crop.

27. Correct option: (B) red algae

Due to the dominance of red pigment r-phycoerythrin in the algae, the body appears to be red. These red algae are known as Rhodophyceae.

28. Correct option: (A) Both Assertion (A) and (R) are correct and (R) explains(A)

The red algae flourish and occur in abundance at great depths of sea where other plants do not occur in abundance. The red thalli of most of the red algae are multicellular. Some of them have complex body organisation.

29. Correct option: (C) Holdfast

In the class phaeophyceae of algae, the plant body is usually attached to the substratum by a holdfast, and has a stalk, which is termed the stipe, and a leaf-like photosynthetic organ called the frond.

30. Correct option: (A) Zoospores

Asexual reproduction in most brown algae is by biflagellate zoospores that are pear-shaped and have two unequals laterally attached flagella.

LEVEL-II

1. Correct option: (B) Taxonomy

The science of identification, naming, and classification of organisms is known as taxonomy.

2. Correct option: (C) Carolus Linnaeus

Linnaeus' gift to science was taxonomy: a classification system for the natural world to standardize the naming of species and order them according to their characteristics and relationships with one another.

3. Correct option: (B) Aristotle

Aristotle is known as the father of zoology. Zoology is a branch of biology that deals with animals' life, evolution, anatomy, physiology, and behaviour. Aristotle is also known as the father of biology; he was an ancient Greek philosopher.

4. Correct option: (C) Whittaker

Five kingdom of classification was proposed by RH Whittaker in 1969. He divided all living organisms into five kingdoms based on following features: Complexity of cell structures. Complexity of body structures.

5. Correct option: (D) species

In this system of classification, kingdom is always ranked the highest followed by division, class, order, family, genus, and species.

6. Correct option: (A) Cat

The Felidae family, comprised of 42 species including the domestic cat, is widely divergent in geographic range and phenotype, with body sizes ranging from 1.5 to 300 kg.

7. Correct option: (A) Thallus

Plants whose body is not differentiated into distinct roots, stems and leaves are called thallophytes.

8. Correct option: (A) Phycology

Phycology, also called algology, the study of algae, a large heterogeneous group of chiefly aquatic plants ranging in size from microscopic forms to species as large as shrubs or trees.

9. Correct option: (D) Lichens

Lichens: Symbiotic relationship between algae and fungi is called lichens. The fungi provide water and minerals to the alga, while the algae perform photosynthesis and supplies food in the form of sugars to the fungi.

10. Correct option: (B) Bryophytes

Bryophytes are called the amphibians of the plant kingdom. Amphibians are those organisms which live on both land and in water. Bryophytes are called amphibians of the plant kingdom because these plants need both water and soil for their survival.

11. Correct option: (A) Sporophyte

The main plant body of the bryophyte is haploid. It produces gametes, hence is called a gametophyte. The male and female gametes fuse together to form zygote. The zygote produces a multicellular body called a sporophyte which is diploid.

12. Correct option: (C) E. Haeckel

The term “Pteridophyta” was given by Ernst Haeckel in the year 1866. Pteridophytes are considered as the first land plants with a well-developed vascular system. The Greek word “Pteron” means feather and “Phyton” means plants.

13. Correct option: (A) Heterosporous

Pteridophytes like Selaginella and Salvinia produce two kinds of spore (microspores and macrospores). They are known as heterosporous.

14. Correct option: (A) Gymnosperms

Gymnosperms are non-flowering plants and bear naked seeds. For example - Pine.

15. Correct option: (C) Leaves

The leaves in gymnosperms are well adapted to withstand extremes of temperature, humidity, and wind. In conifers, the needle-like leaves reduce the surface area. Their thick cuticle and sunken stomata also help to reduce water loss.

16. Correct option: (A) Gymnosperms

Gymnosperms are the seed-producing plants, but unlike angiosperms, they produce seeds without fruits. These plants develop on the surface of scales or leaves, or at the end of stalks forming a cone-like structure. Pinus and Cycas are examples of gymnosperms.

17. Correct option: (D) Angiosperms

Angiosperms are plants that produce flowers and bear their seeds in fruits. They are the largest and most diverse group within the kingdom Plantae, with about 300,000 species.

18. Correct option: (B) 2

There are two types of root systems: taproots and fibrous roots (also known as adventitious roots) in angiosperms present in monocots and dicots respectively.

19. Correct option: (B) 2

The dicotyledons, also known as dicots (or, more rarely, dicotyls), are one of the two groups into which all the flowering plants (angiosperms) were formerly divided. The name refers to one of the typical characteristics of the group: namely, that the seed has two embryonic leaves or cotyledons.

20. Correct option: (D) flower

Angiosperms are commonly known as flowering plants. Flowers are the reproductive parts of the plant as it produces the specialised reproductive cells called gametes.

21. Correct option: (B) Pistil

The female part is the pistil. The pistil usually is in the centre of the flower and is made up of three parts: the stigma, style, and ovary. The stigma is the sticky knob at the top of the pistil. It is attached to the long, tubelike structure called the style.

22. Correct option: (B) Pistil

Pistil: The part of the flower which contains the ovary is called Pistil. It comprises style, stigma, and ovary. The ovary in flowering plants is a part of the female reproductive structure of the gynoecium.

23. Correct option: (A) Pollination

Pollination is the act of transferring pollen grains from the male anther of a flower to the female stigma.

24. Correct option: (D) Zygote

One male gamete fuse with the egg cell to form the zygote that gives rise to the embryo.

25. Correct option: (B) pollen grains

Anther, in flowering plants, the part of a stamen that produces and contains pollen. Each anther is generally borne at the tip of a long slender stalk known as a filament and consists of two lobes that each house a pair of pollen sacs (microsporangia) that produce pollen for pollination.

2. Diversity in Living Organisms – Animals

Solutions

LEVEL-I

1. Correct option: (B) & (C) Mollusca and Annelida

The body cavity between the body wall and the gut wall, lined by mesoderm, is called coelom. Animals possessing coelom are called coelomates, e.g., annelids, molluscs, arthropods, echinoderms, hemichordates, and chordates.

2. Correct option: (D) Aschelminths

Bilateral Symmetry refers to the division of a body into two equal halves in such a way that both halves are mirror images of each other or the same on both sides of an axis. Triploblastic refers to the organisms whose body is derived from three germ layer. Ctenophores are biracially symmetrical and diploblastic, whereas some species are triploblastic. Sponges are asymmetrical and diploblastic. Coelenterates are radially symmetrical and diploblastic. Aschelminths are bilaterally symmetrical and triploblastic.

3. Correct option: (A) Sponges

Phylum Porifera members also known as sponged have a water transport or canal system. Water enters through the minute pores of sponges called ostia in the body wall into a central cavity called spongocoel, from where it goes out through the osculum.

4. Correct option: (B) Echinodermata

Echinoderms have a water-driven tube system that the members of this phylum use for their locomotion. This system is also known as the water vascular system. It also helps in food capturing.

5. Correct option: (D) Eight

The body of a ctenophore typically bears eight external rows of ciliated comb plates, which help in locomotion.

6. Correct option: (C) Excretion and Osmoregulation

Flame cells are present in the phylum Platyhelminthes (flatworms). Flame cells are the specialized excretory cells that help in excretion in animals. These cells work like kidneys and help in removing waste materials from the body of the animals.

7. Correct option: (C) Echinodermata

8. Correct option: (A) A- Radially symmetrical B- Bilaterally symmetrical

Echinoderms are a phylum of marine animals. The adults are recognizable by their (usually five-point) radial symmetry, whereas the larvae are bilaterally symmetrical.

9. Correct option: (A) Hemichordate

Excretory organ in animals belonging to the phylum Hemichordate is the proboscis gland. A proboscis is an elongated appendage from the head of an animal, either a vertebrate or an invertebrate.

10. Correct option: (D) Ventral cord, nerve cord

Chordates are the group of animals which possess a notochord. They have a tubular hollow and dorsal nerve cord. The notochord is the rod like structure. It is made up of cartilage and extends along the length of the body. In certain groups, the notochord forms the tail like structure which helps in swimming in case of fishes. They have a ventral heart. There is presence of pharyngeal slits. There is an endostyle and a post anal tail for at least some period of their life.

11. Correct option: (A) Torpedo

Torpedoes are also known as electric ray. They have electric organs present that act as a defines mechanism against predators.

12. Correct option: (D) Limulus

Limulus commonly known as king crab belongs to phylum Arthropoda. It is a living fossil and large sized marine arthropod.

13. Correct option: (D) Choanocytes

Choanocytes (also known as "collar cells or flagellated cells ") are cells that line the spongocoel. The water current of the body of sponge is maintained by the movement of flagella of choanocytes.

14. Correct option: (A) i & ii

Parapodia are present in some annelid worms. These are a pair of flattened flesh lobes bearing numerous bristles. Radula (rasping organ) in molluscs is a structure involved in feeding.

15. Correct option: (A) Roundworms are pseudocoelomates

Platyhelminthes or flatworms are acoelomates i.e., they lack a coelom. Molluscs and arthropods are coelomates as they have a true coelom derived from the germ layer mesoderm. Aschelminths or roundworms are pseudocoelomates as the coelom is not derived from the mesoderm.

16. Correct option: (A) Both (A) and (R) are true and '(R)' is correct explanation of 'A'

Tetrapod is a superclass of animals that includes all limbed vertebrates that primarily live on land. Fishes are adapted to live in aquatic habitats. As a result, instead of limbs, they have paired fins that assist them in swimming, balance, stability, and steering. So, fishes are not considered as tetrapod's due to the presence of fins.

17. Correct option: (A) Catfish

Catfish is true fish having prominent barbels without scales. Jellyfish is a free-swimming marine animal that has an umbrella-shaped body. Cuttlefish is a mollusc. Silverfish is a wingless insect.

18. Correct option: (A) Both (A) and (R) are true and '(R)' is correct explanation of 'A'

Amphibian eggs do not have a hard protective shell-like bird or reptile eggs. Frogs and toads lay eggs in jelly-like masses, and unless the eggs are covered with water, they will dry up and die. This is why amphibians breed in water. The reproduction occurs by the process of external fertilization. The females release their eggs into the water which are then fertilized by the male sperm.

19. Correct option: (C) Crocodile

Crocodile is the only exceptional animal that belongs from class Reptiles and has a four-chambered heart.

20. Correct option: (A) Ectotherms

Ectotherms are cold-blooded animals that depend on the temperature of surroundings to change their own body temperature to survive. Endotherms are warm-blooded animals which able to maintain constant body temperature, even in extreme environments. Psychrotherms are animals that lives in the cold while thermophiles are animals that live in hot springs.

21. Correct option: (A) Both (A) and (R) are true and '(R)' is correct explanation of 'A'

Mammals and Aves are homeothermic animals, meaning they can maintain a constant body temperature. They have evolved various mechanisms to do so, sweating and shivering are some of the most common ones. Hence, they are able to cope up in extreme climatic conditions.

22. Correct option: (C) Bats

Echolocation is a technique used by bats, dolphins and other animals to determine the location of objects using reflected sound.

23. Correct option: (D) Viviparity is not found in species of lizards and snakes

Majority of mammals are viviparous, which means they give birth to young ones. The egg is retained and implanted in the mother's uterus, the eggshell does not form, and a placenta forms that serves to supply nourishment and gas exchange between the mother and embryo. Most reptiles and snakes give birth to live young. They are viviparous animals. They produce living young instead of eggs from within the body.

24. Correct option: (D) All of the above

The common ancestor of birds and mammals must have occurred at the split of the Sauropsida (Reptiles) and Synapsida (ancestors of mammal-like reptiles). The amnion, yolk sac, chorion, and allantois are four extraembryonic membranes found in reptile, avian, and mammalian embryos.

In an amniotic sac filled with amniotic fluid, the embryo is protected by amnion. Outer skin covering of reptiles, mammals and birds are watertight.

25. Correct option: (C) Platypus

Platypus (Ornithorhynchus) and Echidna (Tachyglossidae) are oviparous mammals. In oviparous animals, internal fertilization occurs but development takes place outside the body.

26. Correct option: (A) Hollow body

A sponge's body is hollow and is held in shape by the mesohyl, a jelly-like substance made mainly of collagen and reinforced by a dense network of fibres also made of collagen. The inner surface is covered with choanocytes, cells with cylindrical or conical collars surrounding one flagellum per choanocyte.

27. Correct option: (B) Coelenterate

Coelenterate animals are diploblastic animals. The mesoglea layer lies between the ectoderm and the endoderm. Mesogloea is helpful in providing buoyancy. Mesoglea is a non-cellular layer.

28. Correct option: (C) Physalia

The Portuguese-man-of-war, or the Physalis physalis, also known as bluebottle, is a marine cnidarian of the family Physaliidae. Its venomous tentacles can deliver a painful sting.

29. Correct option: (D) Mouth, pharynx, intestines, and Anus

The digestive system of a nematode is made up of three main parts: the stomodeum, the intestine, and the proctodaeum. The stomodeum is the beginning of the digestive tract where we find the mouth opening, esophagus (also called the pharynx), and the buccal cavity (the area inside the mouth).

30. Correct option: (D) Insects

Hemocoel is the cavity between the organs of arthropods and molluscs, through which the blood circulates.

31. Correct option: (B) Tail region

Urochordates, like Herdmania have notochord and nerve cord present only in nerve stages. Body is covered by a test made up of tunicin. Notochord is present only in tail. Tadpole undergoes retrogressive metamorphosis and notochord, and nerve cord is absent in adults, which is lost during metamorphosis.

32. Correct option: (D) Tortoise

A tortoise is not a true amphibian animal; it is a reptile. A tortoise can live on land as well as in water but its body is covered by the shell (provides protection), dorsally by carapace and ventrally by plastron. The presence of protective shells is not a characteristic feature of true amphibians. Amphibians are vertebrates that live on both land and water. They respire through gills and moist body surface but on land, they respire through lungs. Frogs, toads and salamanders are examples of amphibians.

33. Correct option: (C) Ornithorhynchus

Platypus (Ornithorhynchus) and Echidna (Tachyglossidae) are oviparous mammals. In oviparous animals, internal fertilization occurs but development takes place outside the body.

34. Correct option: (A) 4, 2, 3, 1, 4

Water canal system is a characteristic feature of phylum Porifera. The body of the poriferans have pores called ostia. The body has only one opening called the osculum. There is a continuous water movement from the exterior to the interior of the body of the poriferans. Water flows through the ostia into the spongocoel and goes out through the osculum. This helps in food gathering, exchange of gases and removal of waste.

35. Correct option: (D) 3 and 4

Most flatworms are hermaphrodite that is they posses both male and female sex organs. Fertilization is internal in case of them and occurs through cross-fertilization. Most of the flatworms have a direct mode of development. And polyembryony is quite common amongst them.

36. Correct option: (C) A is true, R is false

Ascaris belong to the phylum aschelminths in which fertilization is internal and development may be direct or indirect. Ascaris usually lay eggs and thus are oviparous.

37. Correct option: (D) 1, 2, and 3

Nematodes are dioecious with separate male and female individuals. They possess an internal fertilization. Nematodes may produce eggs (oviparous) or larvae (viviparous). Some lay eggs containing larvae which immediately hatch out (ovoviviparous). The life cycle consists typically of four larval stages and the adult form. The most common life cycle pattern among parasitic nematodes is the direct one.

38. Correct option: (D) They have a circulatory system

Flatworms breathe through their moist body surface (skin) and diffusion of oxygen takes place directly. By this diffusion process, each cell can meet its oxygen requirements thus there is no need for the circulatory system to circulate oxygen and nutrients.

39. Correct option: (C) They are protostomes

The echinoderms and the chordates are deuterostomes, while all other invertebrates are protostomes. The main difference between protostomes and deuterostomes is that the blastopore in protostomes is developed into a mouth while the blastopore in deuterostomes is developed into an anal opening.

40. Correct option: (B) Hairy skin and mammary glands

Two characteristics are used to define the mammal class. They are mammary glands and body hair (or fur). Female mammals have mammary glands.

LEVEL-II

1. Correct option: (B) Two

Animal kingdom can be broadly classified into two major groups i.e., vertebrates and invertebrates.

2. Correct option: (C) Chordata

Animals with notochord are called chordates. In some chordates, the notochord is modified as the vertebral column. Such animals are called vertebrates.

3. Correct option: (A) 2

Diploblastic animals have two cell layers and they are outer ectoderm and inner endoderm. These animals don't have the mesoderm whereas they have middle mesoglea which is an undifferentiated layer.

4. Correct option: (D) Sponges

The phylum Porifera comprises the sponges. Sponges are simple invertebrate animals that live in aquatic habitats.

5. Correct option: (D) Ctenophores

Ctenophora is a phylum including the animals that live in marine waters worldwide. Their most distinctive feature is the combs groups of cilia, which they use for swimming. They are the largest animals that swim by the means of cilia. Comb jellies belong to phylum Ctenophora.

6. Correct option: (A) External

Ctenophores reproduce sexually. Fertilisation occurs outside the body. Development is indirect.

7. Correct option: (B) Excretion

Excretion and osmoregulation in Platyhelminthes occur with the help of a protonephridia system having flame cells.

8. Correct option: (A, B) Aschelminths / Nematode

Incomplete digestive tract i.e., anus is absent, found in Platyhelminthes and Ctenophora Aschelminths/ Nematode and Annelida have both mouth and anus, which have complete digestive tract.

9. Correct option: (B) Nephridia

Nephridia is an excretory organ that eliminates nitrogenous waste from the body cavity and removes them through the pores outside the body. Numerous species have a couple of nephridia in each portion.

10. Correct option: (C) Arthropoda

The organ system level of organization is when two or more organs work together for a specific function. Annelids, Arthropods, Molluscs, Echinoderms and Chordates have this level of organization.

11. Correct option: (C) Arthropoda

Insects belong to the phylum Arthropoda. Arthropods move with the help of jointed appendages. The word "Arthropod" means "jointed feet".

12. Correct option: (A) Mollusca

Mollusca is the second largest phylum in the animal kingdom. Molluscs have the following characteristics: They are bilaterally symmetrical. They are triploblastic; have presence of three germ layers. This phylum includes a wide variety of marine, freshwater, and terrestrial species. It includes about 100,000 living species.

13. Correct option: (C) Sensory

The anterior head region has sensory tentacles in molluscs.

14. Correct option: (A) Excretory

The excretory organs of Mollusca are metanephridia. Metanephridia filters waste from internal body fluid.

15. Correct option: (B) Mollusca

The simplest sort of respiratory system found in molluscs is cutaneous respiration. The dorsal surface of the body is covered in papillae in the majority of the Aeolididae species. The papillae differ in size and are connected to the heart by veins.

16. Correct option: (B) Gill slits

Hemichordates consists of a small group of worm-like marine animals with an organ-system level of organization. They are no chordates. The circulatory system is of open type. The blood flows to a cavity known as hemocoel which bathes the organs directly with oxygen and nutrients and there is no distinction between blood and interstitial fluid. They respire with help of gill slits.

17. Correct option: (B) Excretory
Proboscis gland is the excretory organ of hemichordate.
18. Correct option: (A) hemichordate
Balanoglossus belongs to the phylum Hemichordate, the animals in this group are also called acorn worms.
19. Correct option: (D) Both are true
Animals in the phylum Chordata share four key features that appear at some stage during their development: a notochord, a dorsal hollow nerve cord, pharyngeal slits, and a post-anal tail. In some groups, some of these are present only during embryonic development.
20. Correct option: (C) 7
Broadly Vertebrata is divided into Agnatha and Gnathostomata; Class Cyclostomatous is under Agnatha and other 6 are under Gnathostomata including Chondrichthyes, Osteichthyes, Amphibia, Reptilia, Aves, and Mammalia.
21. Correct option: (D) Kidney
Cyclostomes have a pair of mesonephric kidneys that make up the excretory system of the organism.
22. Correct option: (A) Amphibians
The first amphibians evolved from a lobe-finned fish ancestor about 365 million years ago. They were the first vertebrates to live on land, but they had to return to water to reproduce. This meant they had to live near bodies of water.
23. Correct option: (C) 3
Heart of Amphibians: It has three chambers – one ventricle and two atria.
24. Correct option: (D) 4
Reptiles generally have a three-chambered heart, while crocodile is an exception with a four-chambered heart. There are two atrium and two ventricles in the heart forming the four chambers.
25. Correct option: (C) Lungs
The avian respiratory system exchanges oxygen and carbon dioxide between the gas and the blood utilizing a relatively small, rigid, flow-through lung, and a system of air sacs that act as bellows to move the gas through the lung.
26. Correct option: (D) 4
Mammal hearts have four chambers (two atria and two ventricles).
27. Correct option: (C) Both A and B
Generally, all mammals are viviparous that is they give birth to their young ones, except some monotremes like Platypus and Echidna that can lay eggs and are oviparous.
28. Correct option: (A, C) Aves, Mammalia
Aves are warm-blooded animals and have a four-chambered heart. All mammals and birds can generate this internal heat and are classed as homoiotherms, or warm-blooded animals. Normal temperatures for mammals range from 97° F to 104° F. Most birds have a normal temperature between 106° F and 109° F.
29. Correct option: (B) Cold blooded
Reptiles are cold-blooded, or ectothermic, animals. This means that they cannot produce heat in their own bodies and have to rely on their surroundings to keep warm. This is why most reptiles are found in hot or warm climates.

3. The Fundamental Unit of Life

Solutions

LEVEL – I

1. Correct option: (D) All of these

The cytoplasm is kept apart from the outside environment by the cell membrane, sometimes referred to as the plasma membrane. It is composed of a phospholipid bilayer with proteins incorporated into it. The hydrophobic tails of the lipids are facing each other and the hydrophilic heads are facing outward in the phospholipid bilayer, which is a double layer of lipids. The cell membrane may be selectively permeable because to its structure.

Many essential functions are carried out by the cell membrane, including the following:

- controlling the flow of materials into and out of the cell.
- putting up a wall between the cell and its surroundings.
- defending the cell against hazardous elements.
- maintaining the interior environment of the cell.

2. Correct option: (C) Robert Brown

Scottish botanist Robert Brown is well recognized for identifying the nucleus in plant cells in 1831. He was looking at plant cells under a microscope when he saw a tiny, solid mass in the middle of the cell. The "smallest resolvable part" of the cell, this structure was given the term "nucleus" by the scientist.

Nucleus, which meaning "kernel" or "seed" in Latin, is where the term "nucleus" derives from. Given that the nucleus is the core of the cell and houses the genetic material for the cell, the name "nucleus" is appropriate.

Camillo Golgi was an Italian physician and biologist who discovered the Golgi apparatus in 1898.

Matthias Schleiden was a German botanist who, along with Theodor Schwann, is credited with the discovery of the cell theory.

Rudolf Virchow was a German physician and pathologist who coined the term "cell theory".

3. Correct option: (D) Mitochondria

The mitochondria are the engine of the cell. Eukaryotic cells—those having a well-defined nucleus and numerous other membrane-bound structures—have mitochondria, which are membrane-bound organelles. These organelles are sometimes referred to as the "powerhouses" of the cell because they are crucial to the process of cellular respiration, which produces adenosine triphosphate (ATP), the main source of energy for the cell.

Plastids are organelles that are found in plant cells. They are responsible for photosynthesis, which is the process by which plants convert light energy into chemical energy.

Lysosomes are organelles that are found in all cells. They are responsible for breaking down cellular waste products.

RER stands for rough endoplasmic reticulum. It is a type of endoplasmic reticulum that is covered in ribosomes. Ribosomes are responsible for making proteins.

4. Correct option: (B) Cell wall

A plant cell's cell wall is the additional layer that exists outside of the cell membrane. The cell wall, which is mostly made of cellulose, gives the cell structural support and defense, keeping it from bursting or collapsing under diverse environmental circumstances. Animal cells, which lack this stiff outer layer and are therefore more flexible in shape and movement, are distinguished from plant cells by the absence of the cell wall, which is a distinguishing trait of plant cells.

Plasma membrane is the outermost layer of all cells, including plant cells. It is a thin, flexible layer that protects the cell and controls what enters and leaves the cell.

Epidermis is the outer layer of plant tissue. It is made up of cells that are tightly packed together. The epidermis protects the plant from the environment and helps to regulate water loss.

Biological membrane is a general term for the outer layer of all cells. It is made up of lipids and proteins.

5. Correct option: (B) Prokaryotes

Blue-green algae, often known as cyanobacteria, are prokaryotic organisms that fall within the realm of bacteria. Prokaryotes are unicellular creatures that are organelle-free and devoid of a genuine nucleus. Instead, a single circular DNA molecule found in the nucleoid region houses all of their genetic information. Similar to plants, cyanobacteria are photosynthetic microorganisms that can perform photosynthesis to create energy and oxygen. They contribute to the ecosystem's ability to produce oxygen and fix nitrogen, which makes them crucial for many aquatic habitats.

Eukaryotes, on the other hand, do contain a nucleus and other organelles that are attached to membranes.

Biological entities classified as mesokaryotes share traits with both prokaryotes and eukaryotes. Although their nucleus is there, a double membrane does not encircle it.

Two nuclei can be found in each cell of a species of organisms called dikaryotes.

6. Correct option: (C) Nucleoplasm

Eukaryotic cells have a unique fluid called nucleoplasm within their nucleus. It encloses and encircles the nucleus and chromosomes, among other genetic components. This liquid is crucial for preserving the nucleus's integrity and structural integrity. Proteins, enzymes, and other substances essential for nuclear and gene control are abundant in the nucleoplasm. In addition to being essential for the efficient operation and regulation of genetic information within the nucleus, it plays a significant role in coordinating the actions of the cell.

Although they are not exactly correct, the phrases "nuclear fluid" and "karyoplasm" are occasionally used interchangeably with nucleoplasm. While karyoplasm refers to the entire nucleus's contents, including the chromosomes, nucleoli, and nucleoplasm, nuclear fluid is a broader phrase that may be used to describe any fluid found in the nucleus.

So, the answer to the question is C) Nucleoplasm.

7. Correct option: (D) Both a & b

Lysosomes are known as both "A) Digestive vesicles" and "B) Suicidal bags."

The membrane-bound organelles known as lysosomes have enzymes that may degrade cellular trash, foreign objects, and even fragments of the host cell. They are sometimes referred to as "suicide bags" because the enzymes they contain have the ability to kill cells if they burst.

Lysosomes are also referred to as "digestive vesicles" since they are crucial for the breakdown of cellular waste. Lysosomes contain hydrolytic enzymes that help break down and digest various cellular materials, such as waste products, damaged organelles, and foreign particles.

8. Correct option: (A) 150 times

In comparison to the nucleus, mitochondria are around 150 times smaller. The biggest organelle in a cell is the nucleus, which houses the DNA for that cell. Because they are significantly smaller than the nucleus, mitochondria provide the majority of the energy in a cell. They play a crucial role in cellular respiration, generating energy in the form of ATP through a series of biochemical reactions.

The size of a mitochondrion can vary depending on the type of cell, but they are typically about 1-2 micrometers in length and 0.5-1 micrometers in diameter. The nucleus, on the other hand, is typically about 5-10 micrometers in diameter.

9. Correct option: (B) Endoplasmic reticulum

A network of membranes called the endoplasmic reticulum (ER) is present in the cytoplasm of eukaryotic cells. It is in charge of a number of processes, such as detoxification, lipid synthesis, and protein synthesis.

Rough endoplasmic reticulum (RER), which has ribosomes attached to its surface, and smooth endoplasmic reticulum (SER), which is ribosome-free, are two different forms of ER. The endoplasmic reticulum's network of membranes offers a significant surface area for several cellular activities and makes communication between diverse parts of the cell easier.

10. Correct option: (D) Both a & c

Plant and algal cells include double-membrane organelles called plastids. They may participate in photosynthesis and are in charge of the synthesis and storage of food. Chloroplasts, chromoplasts, and leucoplasts are the three primary categories of plastids. The most typical kind of plastids are chloroplasts. They contain chlorophyll, the substance that gives plants their green color. Photosynthesis, the process by which plants turn light energy into chemical energy, is carried out by chloroplasts.

Chromoplasts are plastids that include pigments besides chlorophyll. Fruits and flowers have their distinctive colors thanks to these pigments. Pigments are created and stored by chromoplasts.

11. Correct option: (C) Both lipids and proteins

The cell membrane, also known as the plasma membrane, is composed of both lipids and proteins. It is a thin, flexible barrier that surrounds the cell and separates its internal environment from the external environment.

The main components of the cell membrane are phospholipids, which form a lipid bilayer, with hydrophilic heads facing outward and hydrophobic tails facing inward. Proteins are embedded within this lipid bilayer, either spanning the entire membrane (integral proteins) or loosely associated with the surface (peripheral proteins).

Together, the lipids and proteins in the cell membrane play essential roles in maintaining the cell's structure, regulating the movement of molecules in and out of the cell, and facilitating cell communication and recognition.

Cellulose, mentioned in option D, is a carbohydrate found in the cell walls of plants but not a part of the cell membrane.

12. Correct option: (A) Nucleus

The component of the cell that contains the genetic code is called the nucleus. The DNA of the cell, which holds the genetic instructions for the cell's construction, function, and reproduction, is what gives it the nickname "control center" of the cell. The nucleolus, which is located inside the nucleus and is important in ribosome production, is there.

The nucleolus is a small, dense region within the nucleus that is involved in the production of ribosomes. Ribosomes are the organelles that synthesize proteins.

Ribosomes are not found in the nucleus, but they are produced there. They are then exported to the cytoplasm, where they carry out protein synthesis.

The Golgi complex is a stack of flattened sacs that is involved in the processing and packaging of proteins. It is not involved in the storage of genetic information.

13. Correct option: (D) Both Bacterium and Cyanobacteria

Cyanobacteria and bacteria are both prokaryotic organisms. Cells without a nucleus or other membrane-bound organelles are known as prokaryotic cells. Their genetic material occurs in the cytoplasm as a single circular DNA molecule rather than being contained

within a membrane-bound nucleus. Prokaryotic creatures include bacteria and cyanobacteria

A eukaryotic cell is Paramecium sp. A nucleus and other membrane-bound organelles are features of eukaryotic cells. They are present in all three domains of life, with the exception of Archaea, and are more complicated than prokaryotic cells.

Hence the answer is D) Both Bacterium and Cyanobacteria

14. Correct option: (A) Endoplasmic reticulum

Intracellular transport does definitely entail the endoplasmic reticulum (ER). The enormous network of membranes in the cell is responsible for transporting proteins and other materials. The ribosome-filled rough ER makes proteins and aids in their folding and transport. In addition to helping with the transit of these chemicals, the smooth ER is engaged in lipid production. As a result, the endoplasmic reticulum is crucial for the movement of proteins and lipids inside the cell.

A network of membranes called the endoplasmic reticulum (ER) is present in the cytopl asm of eukaryotic cells.

It is in charge of a number of processes, such as detoxification, lipid synthesis, and protein synthesis. In the process of intracellular transport, the ER is crucial.

The Golgi complex receives and processes the proteins that are produced on the riboso mes of the rough ER.

The proteins are subsequently put into vesicles by the Golgi complex and sent to vario us cell locations.

So, the answer to the question is A) Endoplasmic reticulum. The endoplasmic reticulum participates in intracellular transport.

15. Correct option: (B) Lipids

Lipid synthesis is carried out by the smooth endoplasmic reticulum (SER). Under a microscope, it appears smooth because its surface is devoid of ribosomes. Lipid metabolism, which includes the production of numerous lipids including phospholipids, cholesterol, and steroid hormones, is one of its primary tasks.

The smooth ER also stores calcium ions, which are necessary for cellular signaling and muscle contraction, and is in charge of detoxifying certain medications and poisons from the cell. The smooth endoplasmic reticulum (ER) is an expert in lipid-related activities, whereas the rough endoplasmic reticulum (RER) is predominantly involved in protein synthesis.

16. Correct option: (A) Smooth endoplasmic reticulum

The smooth endoplasmic reticulum (SER), found in vertebrate liver cells, is essential for the detoxification of several toxins and medicines. It has cytochrome P450 enzymes, which detoxify medications, poisons, and other hazardous substances. These enzymes change these chemicals, making them more water-soluble and simpler for the body to remove via urine or bile. The smooth ER found in liver cells is an important part of the body's detoxification system, which depends on the liver for metabolic and detoxifying functions.

An endoplasmic reticulum that contains ribosomes connected to it is known as the rough endoplasmic reticulum (RER). It contributes to the creation of proteins.

Organelles with a membrane around them, lysosomes have enzymes that digest cellular trash.

Organelles with a membrane around them, vacuoles store waste, nutrients, and water.

17. Correct option: (B) 150 – 3000

Depending on the kind of cell and its level of activity, different types of cells have different numbers of mitochondria. For instance, because muscle cells require a lot of energy to

contract, they have several mitochondria. Red blood cells, on the other hand, lack mitochondria since they do not require energy to function.

There are around 150–3000 mitochondria per cell on average. Some cells, nevertheless, contain more or fewer mitochondria than this. For instance, although a human sperm cell only has one mitochondrion, a human liver cell has roughly 2000 mitochondria.

The cellular energy currency, ATP, is created by mitochondria. Therefore, a cell's energy needs are correlated with the quantity of mitochondria present. Muscle cells and red blood cells contain more mitochondria than cells with lower energy requirements like white blood cells.

18. Correct option: (B) Photosynthesis

Chloroplasts perform photosynthesis as their main activity. The function of chloroplasts, which are organelles present in plant cells and certain algae, is to absorb light energy and transform it into chemical energy in the form of glucose. In this procedure, known as photosynthesis, light energy is absorbed by chlorophyll, a green pigment found in chloroplasts. Utilizing the energy of the collected light, glucose and oxygen are created by combining carbon dioxide and water.

Respiration is the process by which cells use oxygen to break down carbohydrates and other molecules to produce energy. Nutrition is the process by which cells take in nutrients from the environment and use them to build and maintain the cell. Transport is the process by which cells move materials around inside the cell and between cells.

So, the answer is B) Photosynthesis. The primary function of chloroplasts is to carry out photosynthesis.

19. Correct option: (C) Both Schleiden & Schwann

Both Theodor Schwann and Matthias Schleiden put out the cell idea. They were eminent scientists who contributed significantly to our understanding of cells and living things in the 19th century.

German botanist Matthias Schleiden made the claim that plants are made up of cells in 1838. Schleiden's theories were expanded upon by another German scientist, Theodor Schwann, who in 1839 postulated that animal tissues also contain cells

Together, their efforts laid the groundwork for the cell hypothesis, which contends that since all living things are made up of cells, they are life's fundamental building blocks.

The third scientist, Rudolf Virchow, also made substantial contributions to cell theory, but he is best remembered for the phrase "Omnis cellula e cellula" (every cell derives from another cell), which emphasizes the idea of cell division and the continuity of life.

20. Correct option: (A) Nucleoplasm

Nucleoplasm refers to the liquid that fills a eukaryotic cell's nucleus. The genetic material of the cell, including the nucleolus and chromosomes, is encased in and contained by nucleoplasm, a gel-like substance. The nuclear membrane, which encloses it and divides it from the cytoplasm, surrounds it.

The fluid that surrounds the cell's nucleus but is still within, known as the cytoplasm (option B), includes a variety of organelles and is the scene of several cellular activity.

Historically, the live components of a cell, which include the cytoplasm and nucleus, were referred to as protoplasm (option C).

The term "germplasm" (option D) refers to the genetic material in the germ cells, which are in charge of transmitting genetic information to progeny.

21. Correct Matching is: a-T;b-P;c-S;d-Q;e-R

a) Robert Hook – T) Living cell

The word "cell" was first used by British scientist Robert Hooke in 1665. In cork tissue, he noticed cells, which he characterized as "little boxes".

b) Robert Brown – P) Nucleus

The nucleus was found in 1831 by Scottish botanist Robert Brown. He examined plant cell nuclei and referred to them as "highly refractive bodies."

c) Schwann – Q) Zoologist

Theodor Schwann was a German zoologist who proposed that all animal cells are made of cells. He published his findings in 1839.

d) Schleiden – Q) Botanist

Matthias Schleiden was a German botanist who proposed that all plant cells are made of cells. He published his findings in 1838.

e) Camillo Golgi – R) Golgi apparatus

Camillo Golgi was an Italian physician and histologist who discovered the Golgi apparatus in 1898. He described the Golgi apparatus as a "network of canals".

22. Correct Matching is: a-d;b-a;c-b;d-e;e-c;

a) Selectively permeable membrane - d) Plasma membrane

Selectively permeable membrane is a characteristic of the plasma membrane, which is the outer layer of all cells. The plasma membrane controls the movement of materials into and out of the cell.

b) Traps solar energy - a) Chloroplast

Traps solar energy is a function of the chloroplasts, which are organelles found in plant cells. Chloroplasts use sunlight to convert carbon dioxide and water into glucose, which is the plant's food source.

c) Cyanobacteria - b) Prokaryotic

Cyanobacteria are a type of prokaryote, which means they do not have a nucleus. Cyanobacteria are photosynthetic, meaning they can trap solar energy and use it to produce food.

d) Detoxification - e) SER

Detoxification is the process of removing harmful substances from the body. SER (Smooth Endoplasmic Reticulum) is involved in detoxification by breaking down harmful substances.

e) Endoplasmic reticulum – c) Transport of minerals

By promoting the transfer of materials from its own membrane to other cellular sites, the smooth endoplasmic reticulum contributes to intracellular transport. To guarantee appropriate sorting and transport of lipids and proteins to their ultimate locations inside the cell, it can also interact with other organelles, such as the Golgi apparatus.

23. Correct option: (C) Leucoplasts

Plant cells have plastids as organelles. They are in charge of the manufacture and storage of numerous compounds. Chloroplasts, chromoplasts, and leucoplasts are the three primary categories of plastids.

A kind of plastids known as leucoplasts lacks pigment and is consequently colorless. They have a major role in the storage of proteins, lipids, and starches. Leucoplasts are widespread in the non-photosynthetic tissues of plants, including the roots, tubers, and seeds, where they are essential for storing and supplying the plant with nutrients.

The green plastids called chloroplasts are in charge of photosynthesis. Chlorophyll, a green pigment found in them, absorbs sunlight and uses it to change water and carbon dioxide into glucose. Chromoplasts are the colored plastids that give flowers, fruits, and other plant components their distinctive hues.

Therefore, the only colorless plastids are leucoplasts. So, the answer is C.

24. Correct option: (D) Chloroplast to chromoplasts

Chloroplasts, the organelles in unripe green fruits that are essential for photosynthesis, are present. Chloroplasts are made of the green pigment chlorophyll, which utilizes sunlight to absorb carbon dioxide and water and turn them into glucose.

The green pigment chlorophyll is destroyed when a fruit ripens, and the chloroplasts start to disintegrate. This makes the carotenoids and anthocyanins, two additional fruit pigments, more noticeable. These pigments are what give mature fruits their distinctive colors, including red, yellow, orange, and purple.

Thus, the conversion of the fruit's chloroplasts to chromoplasts accounts for how an unripe, green fruit changes color as it ripens. A particular class of plastid that has carotenoid pigments called chromoplasts. The deterioration of the chloroplasts and the buildup of carotenoid pigments are responsible for the fruit's color shift.

25. Correct option: (B) Plasmolysis

In a hypertonic solution, plasmolysis is the contraction of the cytoplasm away from the cell wall. This happens because osmosis—the flow of water from an area of high-water concentration to an area of low water concentration—causes the water inside the cell to escape the cell.

A solution that contains more solutes than the cell has a hypertonic solution. This indicates that there is less water present in the cell than there is in the solution. Osmosis causes water to leave the cell as a result, which causes the cytoplasm to contract.

The plasma membrane is detached from the cell wall during frontolysis. The destruction of the cell wall by acids is known as acidolysis. The disintegration of a cell by enzymes is known as allolysis.

26. Correct option: (B) They help with the process of cell division

Organelles that are membrane-bound and present in plant cells are called vacuoles. The vacuole serves several other essential functions in plant cells. These include storing toxic metabolic wastes (option A), helping to maintain turgidity (option C), and providing structural support (option D). The vacuole is a large, fluid-filled organelle in plant cells that plays a crucial role in maintaining the cell's shape, turgor pressure, and storing various substances.

Vacuoles don't participate in the division of cells, nevertheless. A cell can split into two new cells through a process called cell division. Many different genes regulate this intricate process. This procedure excludes the use of vacuoles.

27. Correct option: (C) Endoplasmic reticulum

All cells include the endoplasmic reticulum (ER), which is a network of membranes.

Rough ER, which has ribosomes and synthesizes proteins, and smooth ER, which lacks ribosomes and is involved in the synthesis of lipids, are the two different forms of ER.

The Golgi apparatus is where the ER-produced proteins and lipids are processed, packed, and delivered after being carried there.

Organelles called lysosomes are in charge of breaking down cellular trash. Chromosomes are biological entities that house a cell's genetic material. Organelles called mitochondria are in charge of a cell's ability to produce energy.

28. Correct option: (C) Osmosis

Water travels across a semi-permeable membrane known as an osmotic membrane by the process of osmosis from an area of greater water concentration (lower solute concentration) to a region of lower water concentration (higher solute concentration). Until the water concentration on both sides of the membrane is equal or balanced, this water flow continues. It is essential for many biological activities and is crucial for preserving the water balance in cells.

Options A and B are incorrect because diffusion is the movement of particles, including solute molecules, from a region of high concentration to an area of low concentration, whereas evaporation is the process of converting liquid into vapor.

29. Correct option: (A) Robert Hooke

The word "cell" was first used by English scientist Robert Hooke in 1665. He observed what he characterized as "little boxes" when examining a thin slice of cork under a microscope. He referred to these containers as "cells" since they reminded him of the honeycomb's cells. The finding made by Hooke marked a significant advance in our knowledge of life. It demonstrated how cells are the fundamental building block of life and how they are present in all living things.

The first person to use a microscope to view cells was Anton van Leeuwenhoek. He did not, however, invent the word "cell".

Himmler was a top Nazi official who was in charge of the Holocaust. From 1985 until 1991, Gorbachyey served as the Soviet Union's president. He had nothing to do with the discovery of cells, and they weren't scientists.

30. Correct option: (D) ATP

Adenosine triphosphate, or ATP, is frequently described as the "energy currency" of the cell. Within cells, it is a molecule that stores and transmits energy. The breakdown of glucose and other energy-dense compounds occurs during cellular respiration, notably in the mitochondria. This process results in the production of ATP. One of ATP's high-energy phosphate bonds can be broken by cells to swiftly release energy when they need it for different tasks.

This energy fuels cellular activity. Endoplasmic reticulum (Option A) and oxygen (Option B) are incorrect since they are not directly linked to being the energy unit of the cell. Option C (Mitochondria) is partially correct in the sense that mitochondria are the sites where ATP is produced, but ATP itself is the actual energy carrier in the cell.

LEVEL - II

1. Correct option: (A) Anton Van Leeuwenhoek

When Anton Van Leeuwenhoek first observed a cell, he said it was alive. Dutch scientist Anton Van Leeuwenhoek is frequently referred to be the founding father of microbiology. He made important contributions to microscopy in the 17th century and created powerful lenses that made it possible for him to examine microbes for the first time. He discovered little, single-celled creatures in 1674 that he referred to as "animalcules" or "little animals." He recognized that these bacteria were living things despite not completely understanding their nature and meticulously recorded his findings.

Robert Hooke first used the word "cell" to refer to the tiny spaces he saw in cork tissue. Schleiden suggested that all plants are composed of cells, whereas Robert Brown argued that plant cells have nuclei. They both missed seeing living cells, though. So, Anton Van Leeuwenhoek first saw a cell and described it as a live cell.

2. Correct option: (C) Cytology

Studying the structure, behavior, and functions of cells is known as cytology. Kytos, which means "hollow vessel," and logos, which means "study of," are the roots of the term "cytology." It is a field of biology that focuses on comprehending the cell, the building block of life. Cytologists inspect and thoroughly examine cells using a variety of methods and equipment, including microscopes.

Cardiology is the study of the heart. The heart is an organ, not a cell. Cosmology is the study of the universe. The universe is made up of many cells, but it is not itself a cell.

Histology is the study of tissues. Tissues are made up of cells, but they are not themselves cells.

Cytologists examine cells, which helps us comprehend the intricacies of living things and how they work at the microscopic level. So, the study of cell is known as Cytology.

3. Correct option: (D) All the above

Three scientists, Matthias Schleiden, Theodore Schwann, and Rudolf Virchow, worked together to develop the hypothesis of the cell.

German botanist Matthias Schleiden investigated plant tissues and came to the conclusion that all plants are made up of cells. German naturalist Theodore Schwann expanded on this notion and claimed that all animal tissues likewise consist of cells. The first two elements of the cell theory were developed jointly by Schleiden and Schwann.

German physician Rudolf Virchow believed that all cells develop from pre-existing cells.

Thus, the three scientists who worked together to develop the cell hypothesis were important in its development.

4. Correct option: (D) Cell

The smallest unit of life is a cell. All living things, from small unicellular creatures like bacteria to sophisticated multicellular organisms like plants and animals, including humans, are made up of these fundamental building components. Cells perform vital tasks required for the survival and operation of living creatures.

Each cell is a standalone organism with a unique combination of structures and organelles that collaborate to support life activities. Organs are made up of different types of tissues functioning together, tissues are grouped together to create tissues, and organ systems are made up of multiple organs working together to carry out particular body duties.

5. Correct option: (B) Ostrich

The biggest known single cell in the animal kingdom is the egg of an ostrich. Ostrich eggs may weigh up to 1.4 kilograms (3 pounds), which is quite a sizeable egg. These huge eggs are designed to support the weight of the incubating parent while providing the developing ostrich embryo with the nourishment and security it needs until it hatches. Native to Africa, ostriches are flightless birds that lay and nurture their eggs on the ground.

The other answer choices are incorrect. The eggs of elephants, camels, and cows are all much smaller than the ostrich egg.

6. Correct option: (A) RBC

In the human body, red blood cells (RBCs) are the most prevalent type of cell. They are in charge of delivering oxygen to the tissues and eliminating carbon dioxide. RBCs have a diameter of around 7.0 μm, which is comparable to a bacterium. They can effectively carry out their job in the bloodstream because to their small size and distinctive biconcave shape, which maximizes the surface area available for gas exchange.

The blood also contains white blood cells (WBCs); however, they are considerably smaller than RBCs. WBCs are in charge of preventing infection. A WBC's diameter typically ranges between 10 and 15 μm.

RBC is the only option that is accurate as a result.

7. Correct option: (D) None

The correct answer is not provided in the options. The cell membrane, also known as the plasma membrane, is the outermost membrane of the cell. It serves as a semi-permeable barrier around the cell and regulates how things enter and exit the cell.

A phospholipid bilayer, or two layers of lipids, composes the plasma membrane. The hydrophilic (water-loving) heads of the lipids face the inside and outside of the cell, whereas the hydrophobic (water-fearing) tails of the lipids face each other.

The jelly-like material that fills a cell is called cytoplasm. It includes all of the organelles, which are the minute structures that carry out the operations of the cell. The control center of the cell is the nucleus. It has the DNA of the cell, which is the genetic material responsible for a cell's structure and function. Proteins and other chemicals are transported throughout the cell via the endoplasmic reticulum, a network of membranes.

So, the outer most membrane of the cell is the plasma membrane, and none of the other answer choices are correct.

8. Correct option: (D) All the above

The plasma membrane is a very important part of the cell. It has a number of functions, including:

Providing structure and shape to the cell. The plasma membrane is a thin, flexible barrier that surrounds the cell and gives it its shape.

- Acting as a mechanical barrier for protection. The plasma membrane helps to protect the cell from its environment. It prevents harmful substances from entering the cell and keeps the cell's contents from leaking out.
- Regulating the movement of substances across the membrane. The plasma membrane controls what substances can enter and leave the cell. This is important for maintaining the cell's internal environment.

So, option D is the correct answer as it includes all the functions of the plasma membrane.

9. Correct option: (B) Hypertonic

The concentration of solutes outside a cell is higher than inside the cell when the cell is submerged in a hypertonic solution. As a result, water osmotically diffuses from the cell into the surrounding solution, shrinking the cell. In an effort to balance the concentrations on both sides of the cell membrane, water moves from a region of lower solute concentration (inside the cell) to an area of greater solute concentration (outside the cell).

A cell will enlarge when submerged in a hypotonic solution because water will enter the cell through osmosis. This is true because water always moves from a region with low solute concentration to one with high solute concentration.

A solution that contains the same amount of solute as the cell is said to be isotonic. When a cell is submerged in an isotonic solution, there is no net water flow into or out of the cell.

So, the answer to the question is Hypertonic.

10. Correct option: (A) Exocytosis

Exocytosis is the process through which substances, such as waste products or proteins, are moved from inside the cell to the outside. Exocytosis is the discharge of components from vesicles into the extracellular space after they have fused with the cell membrane. Numerous cellular processes, such as the release of hormones, neurotransmitters, and other cellular byproducts, depend on this mechanism.

The process by which materials travel from the exterior of the cell to the interior is known as endocytosis. It is thus a form of active transportation. The process of osmosis involves the transfer of water molecules from a region with low solute concentration to one with high solute concentration.

So, the answer to the question is Exocytosis.

11. Correct option: (B) Endocytosis

The process of endocytosis is how cells absorb substances from their surroundings. Because it is a form of active transport, the components are transported inside the cell using energy. Phagocytosis and pinocytosis are the two forms of endocytosis.

Phagocytosis is the procedure through which cells take in big objects like bacteria. The particle is enclosed by the cell membrane, which creates a vacuole around it. The particle is broken down by the enzymes, and the cell then takes the nutrients in.

Exocytosis is the process by which cells discharge items to the surrounding environment from inside. Vesicles, which contain the materials and fuse with the cell membrane to release their contents, are used to package the materials.

So, the process by which materials move into the cell by engulfing of the food is called endocytosis. The answer is B.

12. Correct option: (A) Plant

The plasma membrane of plant cells is protected on the outside by the cell wall.

Plants have a protective, hard structure called a cell wall that is mostly made of polysaccharides like cellulose and hemicellulose. It gives the plant cell structural support and defense, assisting in maintaining its form and preventing it from bursting or collapsing under various circumstances.

Animal cells, however, do not have a cell wall; instead, they are covered and protected by a flexible plasma membrane.

13. Correct option: (A) Robert brown

Robert Brown, a Scottish botanist, made the discovery that cells had nuclei in 1831. He was using a microscope to examine orchid cells when he noticed a tiny, dark-staining body in the center of the cell. He referred to it as the nucleus.

Brown's finding was significant since it demonstrated that the nucleus was a separate cell component and not merely a collection of proteins. It is now understood that the nucleus, which houses the cell's DNA, serves as its administrative hub.

An English scientist named Robert Hooke is credited with discovering cells in 1665. He failed to find the nucleus, though.

Robert Brown was the scientist who gave the nucleus the name "Nucleus" in 1831.

14. Correct option: (C) Two

Rough endoplasmic reticulum (RER) and smooth endoplasmic reticulum (SER) are two separate compartments within the endoplasmic reticulum (ER).

Ribosomes, the organelles in charge of protein synthesis, are embedded in the rough endoplasmic reticulum (RER). Proteins that will be exported from the cell are created with the help of the RER.

The ribosome-free smooth endoplasmic reticulum (SER) is present. The SER is engaged in a number of processes, including as calcium ion storage, lipid synthesis, and the detoxification of poisonous chemicals.

Hence, Endoplasmic reticulum is divided into C) Two distinct compartments.

15. Correct option: (D) Ribosomes

The endoplasmic reticulum's (ER) rough surface is caused by the presence of ribosomes on its cytoplasmic surface. Ribosomal RNA (rRNA) and proteins make up the small, intricate structures known as ribosomes. They are in charge of the cell's protein production.

The endoplasmic reticulum, also known as the "rough endoplasmic reticulum" or RER, has a rough look when ribosomes are connected to the cytoplasmic side of the ER.

The RER is involved in the synthesis of proteins that are destined for secretion from the cell, incorporation into the cell membrane, or transport to other organelles.

16. Correct option: (A) Proteins

In the cell, proteins are made and transported through the Rough Endoplasmic Reticulum (RER). It is a membrane-bound organelle that has a cytoplasmic surface covered in ribosomes.

The ribosomes on the RER synthesize the proteins straight into the lumen (interior) of the RER when a cell wants to manufacture proteins. In order to achieve their functional structure, the freshly synthesized proteins are next processed, folded, and changed in the RER. The RER assists in moving proteins to their ultimate locations inside the cell or for secretion outside the cell once they have been appropriately digested.

The Rough Endoplasmic Reticulum does not directly synthesize or transport carbohydrates. Depending on the individual kind of carbohydrate, they are largely synthesized in the cytoplasm or other organelles such as the smooth endoplasmic reticulum and chloroplasts (in plant cells).

17. Correct option: (D) Mitochondria

The crucial function they play in generating energy has led to the nickname "powerhouse of the cell" for mitochondria. Eukaryotic cells include these organelles, which are in charge of producing adenosine triphosphate (ATP) via cellular respiration. The energy-storing and energy-transfer molecule ATP gives the fuel for many cellular operations by storing and transferring energy inside the cell.

Golgi complexes are crucial in the packaging and processing of proteins, but they don't make ATP. Water, nutrients, and waste materials are all stored in vacuoles, which are storage organelles. Plastids are exclusively present in plant cells and are essential for photosynthesis. Thus, mitochondria are the organelle referred to as the powerhouse of the cell.

18. Correct option: (D) None

In the cytoplasm of the cell, amino acids and fatty acids are created. The cytoplasm, which makes up the majority of a cell's volume but excludes the nucleus, is a jelly-like material that houses all of the organelles.

Although they are important in the packaging and processing of proteins, Golgi bodies do not produce fatty or amino acids. Water, nutrients, and waste materials are stored in vacuoles, which are storage organelles. Only present in plant cells, plastids play a role in photosynthesis.

As a result, none of the organelles mentioned above are involved in the production of amino acids or fatty acids.

19. Correct option: (B) Proteins

Protein synthesis occurs at ribosomes. All cells, both plant and animal, have them in the cytoplasm. Two subunits, a big subunit and a small subunit, make up ribosomes. A functioning ribosome is created by the union of the two subunits.

The two steps involved in protein synthesis. First, messenger RNA (mRNA) is synthesized from DNA. The genetic code for a protein is contained in a molecule called mRNA. The mRNA is translated into a protein at the ribosome in the second step.

In order to put together the amino acids into a protein, the ribosome reads the mRNA codons. Peptide bonds are used to connect the amino acids. After that, the ribosome releases the finished protein into the cytoplasm.

So, the organelles that are the site of protein synthesis are ribosomes. The answer is B.

20. Correct option: (C) Lysosomes

Lysosomes are referred to as "suicidal bags of the cell" because they contain hydrolytic enzymes that can disassemble biological components. Lysosomes contain various hydrolytic enzymes that are capable of breaking down and digesting cellular waste, foreign materials, and even damaged organelles. These enzymes have the power to harm

or kill a cell if a lysosome ruptures. This is the reason why lysosomes are frequently referred to as "suicide bags."

Because they lack hydrolytic enzymes, ribosomes are not referred to as "suicidal bags." Protein synthesis is carried out by ribosomes, which are organelles.

Rough endoplasmic reticulum has the initials RER. Ribosomes cover a specific kind of endoplasmic reticulum in this cell.

So, the organelles that are called as "suicidal bags of the cell" are lysosomes. The answer is C.

21. Correct option: (B) Plant

Plant cells, some varieties of algae, and protists have plastids as organelles. Animal cells don't contain them. Plastids are necessary for many cellular functions, such as photosynthesis, the storage of starch and pigments, and the production of lipids and amino acids.

Chlorophyll, a green pigment that absorbs light energy during photosynthesis, is a component of chloroplasts, a particular form of plastid. Leucoplasts, which are involved in the manufacture and storage of numerous compounds, such as starch and oils, and chromoplasts, which store the pigments that give fruits and flowers their vivid colors, are other types of plastids.

As opposed to plants, which undertake photosynthesis and have diverse organelles to suit their metabolic demands, animal cells do not have plastids or chloroplasts.

So, plastids are present only in plant cells. The answer is B.

22. Correct option: (A) Cytosomes

In 1965, Christian de Duve used the word "cytosome" to refer to these organelles. Because they are located in the cytoplasm, he picked this nomenclature.

Cytosomes are another name for microbodies. The cytoplasm of eukaryotic cells contains these tiny, membrane-bound organelles. They take part in several metabolic processes, such as the oxidation of lipids, the creation of hydrogen peroxide, and the detoxification of poisonous chemicals. The two most well-known varieties of microbodies are peroxisomes and glyoxysomes.

23. Correct option: (D) Centrioles

Animal cells have structures called centrosomes that house two centrioles, which are cylindrical organelles. These granule-like centrioles are essential for mitosis and meiosis, the processes of cell division. The centrioles travel to the cell's opposing ends during cell division and assist in the creation of the mitotic spindle, which is necessary for the appropriate distribution of chromosomes to the daughter cells.

A matrix of proteins and other molecules that play a role in organizing the cell's microtubules may also be found in the centrosome in addition to centrioles. Not all cell types include centrioles, and some eukaryotic cells do not. Despite the absence of centrioles in plant cells, the microtubule-organizing center (MTOC) in these cells serves a similar purpose.

24. Correct option: (A) Rudolf Von Virchow

A German physician and pathologist named Rudolf Virchow is credited with creating cellular pathology. He studied animal cell division in 1855 and put out the idea that all cells are descended from earlier cells. The term for this idea is the "omnis cellula e cellula principle," which is derived from the Latin phrase "every cell comes from a cell."

An English scientist named Robert Hooke is credited with discovering cells in 1665. He did not, however, see cell division.

Robert Brown, a Scottish botanist, is credited with discovering cells' nuclei in 1831. Additionally, he missed seeing cell division.

Rudolf Von Virchow, then, was the scientist who first noticed cell division in 1855. The response is A.

25. Correct option: (A) Prokaryotic

Prokaryotic cells are those that do not contain a nucleus. Bacteria and archaea are home to the most basic sort of cell, the prokaryotic cell. The nucleus, mitochondria, and chloroplasts are examples of organelles that are membrane-bound yet absent in them. The cytoplasm, the jelly-like material that makes up the inside of prokaryotic cells, is where the DNA is located.

A nucleus and other membrane-bound organelles are present in eukaryotic cells, on the other hand. Each and every other type of life, such as plants, mammals, fungi, and protists, has eukaryotic cells.

So, the cell without nucleus is known as prokaryotic cell. The answer is A.

4. Tissues

Solutions

Animal Tissues

LEVEL-I

1. Correct option: (B) 120 Days

Red blood cells (RBCs), sometimes referred to as erythrocytes, have a lifetime in the circulation of around 120 days. These specialized cells are responsible for transferring oxygen from the lungs to the body's tissues and organs and returning carbon dioxide to the lungs for removal throughout their lives. In order to maintain a healthy balance and guarantee appropriate oxygen supply throughout the body, old RBCs are ultimately eliminated from circulation by the spleen and liver as they get older and less effective. This makes room for new RBCs to be created by the bone marrow.

2. Correct option: (A) Cartilage

The connective tissue that links bones at joints is known as cartilage. It is a tough, flexible tissue that helps to cushion the bones and allow them to move smoothly against each other.

Bones can move easily against one another and absorb stress when moving because it offers a soft and supple surface between them. Additionally, cartilage aids in minimizing friction and guards against harm to the bones in the joint.

Tendons (B) are a different kind of connective tissue that joins muscles to bones rather than connecting bones to bones. Basophils (D) and neutrophils (C) are two different categories of white blood cells, not connective tissues.

3. Correct option: (B) Monocytes

White blood cells known as monocytes are known as scavengers because they are essential components of the immune system. Monocytes are drawn to the location of infection or inflammation in the body, where they undergo a macrophage conversion. These macrophages take on the role of scavengers, ingesting foreign substances, decomposing cells, and other detritus. A distinct kind of white blood cell called a lymphocyte (A) is essential to the adaptive immune response. White blood cells called neutrophils (C) are also principally responsible for battling bacterial infections. Basophils (D) are not often thought of as scavengers, although they do play a role in allergic responses and the body's response to parasites.

4. Correct option: (D) 7200 liters

At rest, the typical adult heart pumps 5 liters of blood each minute. Accordingly, the heart pumps roughly 3000 liters of blood every hour and 7200 liters per day.

Depending on the level of activity, the heart may pump different amounts of blood. For instance,

increased heart rate and stroke volume during exercise might result in a considerably greater

cardiac output.

Therefore, D) 7200 liters is the right response.

5. Correct option: (C) Squamous

Squamous epithelial cells have a scale-like appearance and are flat and thin. They can be discovered throughout the body, including in the bladder, blood vessels, alveoli, and

the lining of the lungs. Skin contains squamous epithelial cells as well, which aid in shielding the body from the outside elements.

The cube-shaped, uniformly sized cuboidal epithelial cells are comparable in height and breadth. Urine and digestive enzymes are among the compounds secreted by cuboidal epithelial cells.

Columnar epithelial cells resemble columns and are taller than they are broad. Mucus and digestive enzymes are secreted by columnar epithelial cells.

6. Correct option: (C) 'AB' group

People with AB blood type is called universal recipients because they can receive red blood cells from any blood type. This is because their blood lacks any antibodies to the A or B antigens.

Only red blood cells from blood types A or O may be given to people with blood type A. Only B or O blood types can donate red blood cells to those with the B blood type.

O blood types are referred to as universal donors since anybody can get their blood. This is because persons with other blood types will not have an immunological reaction to their blood since it lacks the A or B antigens.

7. Correct option: (A) Fats

Specialized connective tissue called adipose tissue serves largely as a depot for storing fats, sometimes referred to as triglycerides. Adipocytes, which are cells that store and release fat molecules, make up its structure. Adipose tissue is found all over the body and acts as an energy reserve, providing a source of stored energy for times when the body needs additional fuel in between meals or during times of elevated energy consumption. Adipose tissue also supports, cushions, and insulates the body's organs. When the body needs energy, the stored fats in adipose tissue can be broken down and released into the circulation for utilization by numerous tissues and organs.

8. Correct option: (C) Heart

The heart has striated involuntary muscles. We refer to them as cardiac muscles. Cardiac muscles are involuntary since they are not under conscious control, despite the fact that they appear to be striped and are striated.

The two choices, the iris of the eye and the colon, both have smooth muscles. Involuntary muscles without stripes are known as smooth muscles.

Therefore, the correct answer is C) Heart.

9. Correct option: (D) Both A and B

Both calcium phosphate (A) and calcium carbonate (B) are components of bone. These are the two primary minerals responsible for the strength and hardness of bones. Together with collagen fibers, these minerals provide bones their strength and flexibility, helping to support the body's structure and safeguard interior organs. Bone remodeling, a process that requires ongoing deposition and resorption of these minerals to maintain strong bones that can withstand a variety of mechanical pressures, is the name given to the processes of bone production and maintenance.

10. Correct option: (A) Heparin

Blood clotting in blood arteries is prevented by a chemical called heparin. Since it is an anticoagulant, it prevents blood clots from forming. Heparin prevents the production of fibrin, a crucial component of blood clots, by interfering with the function of certain clotting factors in the blood. It is frequently used in medical settings, including as

hospitals, to prevent and treat illnesses like deep vein thrombosis (DVT), pulmonary embolism (PE), and during operations to lower the danger of blood clots.

11. Correct option: (A) Epithelial tissue

The interior and exterior surfaces of an animal's body are both covered and shielded by epithelial tissue. It creates the skin's outer layer and lines the body's numerous organs and cavities, acting as a defense against physical harm, infections, and dehydration. Absorption, secretion, and sensing are other functions of epithelial tissue. By acting as a barrier between the internal environment and the outside world, it is essential for preserving the integrity and homeostasis of the animal body.

The functions of connective, muscular, and nerve tissues differ from those of epithelial tissue. The body's structure and support are provided by connective tissue. Movement is governed by muscular tissue. Signals are sent throughout the body through nervous tissue.

12. Correct option: (B) Connective tissue

The body's many organs and structures are joined and supported by connective tissue. It consists of a variety of tissues, including blood, tendons, ligaments, cartilage, bone, and adipose tissue. Connective tissue helps the body stay structurally sound, fills in the voids between organs, and contributes to the preservation of the body's general form and integrity. It also makes it easier for nutrients, waste products, and other materials to move about the body.

The other tissues mentioned, such as epithelial, muscular, and neurological tissue, do not link organs. The lining of organs and the outside of the body are both covered in epithelial tissue. Movement is governed by muscular tissue. Signals are sent throughout the body through nervous tissue.

13. Correct option: (C) Adipose tissue

Areolar tissue, a loose connective tissue that is present all throughout the body, contains a particular type of cell called fibroblasts. Organs are supported and cushioned by areolar tissue, which also aids in tying diverse tissues together. The extracellular matrix of the areolar tissue, which is composed of fibers and ground material and gives the tissue its structural integrity, is created and maintained by fibroblasts.

These cells are crucial for tissue repair, wound healing, and preserving the connective tissue's general form and function. Although fibroblasts are present in many connective tissues, such as cartilage and bone, they are particularly prevalent in areolar tissue.

14. Correct option: (A) Cartilage

Cartilage is the tissue that lines the ends of ribs, noses, external ears, and the trachea. It is also present in the joints of bones. It is a sort of connective tissue that helps keep certain bodily parts' shapes and offers structural support. It is hard and flexible. Cartilage, which has a smooth, rubbery feel, is important for minimizing friction in joints, supporting some bodily structures, and serving as a cushion in places where flexibility and shock absorption are needed.

The skeleton contains bone, a form of connective tissue. It is a stiff, firm tissue that gives the body stability and structure. All across the body, there is a form of loose connective tissue called areolar tissue. It is a flexible, soft tissue that aids in tying together various organs and tissues. Bones are joined together by a form of thick connective tissue called a ligament. It is a robust, fibrous tissue that aids in maintaining joint stability.

15. Correct option: (B) White

Blood from cockroaches is white in color. Hemolymph, a colorless or white fluid found in cockroaches, is comparable to blood in vertebrates in terms of function. The red pigment hemoglobin, which is present in vertebrate blood, is absent from hemolymph, making it look white or colorless. In cockroaches, the hemolymph performs a number of jobs, such as moving nutrients and waste away from the body and assisting the immune system.

16. Correct option: (D) Snail

Blue blood is seen in various invertebrates, including spiders and snails. Hemocyanin, a pigment with a copper base that carries oxygen, is present in their blood. When hemocyanin is linked to oxygen, it is blue; when it is not, it is colorless. The blue color is caused by the formation of ox hemocyanin, which is created when oxygen binds to hemocyanin in the snail's blood. Snails' adaptability makes it possible for their circulatory system to transfer oxygen effectively.

Frogs, whales, and earthworms all have red blood. Hemoglobin, an iron-based oxygen-carrying pigment, is present in their blood. When hemoglobin is bonded to oxygen, it is red; when it is not, it is dark red.

17. Correct option: (A) 5 Liters

The amount of blood in an adult's body is around 5 liters. This sum can differ according on a person's size, gender, and state of health.

Plasma, red blood cells, white blood cells, and platelets are the components of human blood. The liquid component of blood, called plasma, accounts for around 55% of its total volume. The volume is mostly made up of red blood cells, which also transport oxygen to the tissues. White blood cells, which make up around 1% of the total volume, assist in preventing infection. As blood clots and bleeding stops, platelets play a role.

Blood transports oxygen and nutrition to the tissues, removes waste, and fights illness, thus the amount of blood in the body is crucial.

18. Correct option: (A) Hemoglobin

Red blood cells (erythrocytes) are red in color due to the presence of hemoglobin. Red blood cells contain a protein called hemoglobin, which binds to oxygen in the lungs and transports it to other bodily areas. Oxyhemoglobin, a substance created when oxygen attaches to hemoglobin, is what gives red blood cells their distinctive red color. The ability of red blood cells to transfer oxygen efficiently to tissues where it is required for a variety of physiological activities depends on hemoglobin's presence.

Blood contains fibrinogen, prothrombin, and plasma, yet none of these substances is responsible for the red color of red blood cells. A protein called fibrinogen aids in blood clotting. Fibrinogen is a precursor of thrombin. Blood's liquid component, called plasma, includes water, proteins, nutrients, and waste materials.

19. Correct option: (A) Kari Landsteiner

Blood types were first identified by Kari Landsteiner, a biologist. In 1901, he discovered the presence of several blood types and categorized them into the ABO system. He was an Austrian immunologist and pathologist.

Landsteiner investigated how various sera affected red blood cells in 1900. Red blood cells occasionally clumped together when he combined the sera of several individuals, he discovered. He understood that this was the result of antibodies in the sera reacting

with the antigens on the red blood cells. Blood transfusions have been made safe and successful because to Landsteiner's discovery of blood groups, which represented a major advance in our knowledge of the process. In 1930, he was awarded the Nobel Prize in Physiology or Medicine for his work.

20. Correct option: (C) Both Camel and Ilama

Both camels and llamas (llama is a frequent misspelling of llama) have red blood cells with a nucleus. These two species have nucleated red blood cells, unlike the majority of mammals.

These mammals have nuclei in their red blood cells that serve to store water since they live in arid settings. They have developed this adaption to help them endure these challenging circumstances.

Red blood cells in other animals, including humans, lack nuclei. This is due to the fact that the nuclei would make the red blood cells too big to fit through the capillaries since they would occupy an excessive amount of space.

21. Correct Matching is: a-U; b-S; c-P; d-Q; w-T

a) Cartilage - U) Collagen

The joints, the tip of the nose, and the ears are just a few areas in the body where cartilage, a form of connective tissue, may be found. The major substance in cartilage is collagen. It is a protein that contributes to the flexibility and strength of cartilage.

b) White colored blood - S) Cockroach

Hemoglobin, the pigment that gives blood its red color, is absent from the blood of cockroaches. Instead, their blood is either white or colorless. Hemolymph is the name for cockroach hemolymph.

c) Blue colored blood - P) Snail

Blue blood is seen in various invertebrates, including spiders and snails. Hemocyanin, a pigment with a copper base that carries oxygen, is present in their blood. When hemocyanin is linked to oxygen, it is blue; when it is not, it is colorless.

d) Pus - Q) Dead WBC and pathogens

Pus is a viscous fluid that the body produces in reaction to illness. It can be thick and white or yellow. Pathogens, dead white blood cells, and other detritus make up this substance.

e) Cardiac muscles - T) Heart

Only the heart contains a particular sort of muscular tissue known as cardiac muscles. They control how the heart beats and relaxes, causing the heart to pump blood all over the body.

22. Correct Matching is: a-S; b-P; c-Q; d-T; e-R;

a) ‘O’ group - S) Universal donors

Blood can be donated to individuals of any blood group by blood group O donors. Because blood type O individuals lack A or B antigens on their red blood cells, this is the case. Therefore, those with other blood types won't have an immunological reaction from their blood.

b) ‘AB’ group - P) Universal recipients

Blood donors of any blood group are able to provide it to persons with blood group AB. Those who belong to blood type AB contain both A and B antigens on their red blood

cells, which explains why. Therefore, those with other blood types won't have an immunological reaction from their blood.

c) Myelin sheath - Q) Axon

Axons, which are the long, thin fibers that carry nerve impulses, are surrounded and protected by a layer of fatty tissue called the myelin sheath. Nerve impulses are transmitted more quickly thanks to the myelin coating.

d) Land Steiner - T) Blood groups

In 1901, the Austrian immunologist and physician Karl Landsteiner made the ABO blood type system public. The Rh blood group system was also discovered by him in 1940. Blood transfusions are now safe thanks to his research on blood types.

e) Granulocytes - R) Eosinophils

White blood cells known as granulocytes play a role in the immunological response. They contain granules that have enzymes and other anti-infection agents in them. Granulocytes of the eosinophil kind are highly active during allergic responses.

Plant Tissues

LEVEL – I

Meristematic tissues

1. Correct option: (C) It is composed of cells that can perform cell division

Cells that divide and differentiate into new cells frequently make up meristematic tissues. The cytoplasm is sparse, the nuclei are big, and these cells are tiny. In addition, they are frequently undifferentiated, which means that they have not yet developed into a specific type of cell.

The tips of roots and shoots contain meristematic tissues, which are in charge of the plant's growth. Additionally, they are located in other regions of the plant, such as the cambium, where they are in charge of promoting the development of new tissues.

So, C is the right response. Option A is wrong since meristematic tissues are made up of dividing cells rather than non-dividing cells. Option B is wrong since meristematic tissues are made up of several cell types, not just one.

2. Correct option: (A) Meristematic cells

Meristematic cells are the tissues that help plants develop and heal overall. Meristematic tissues, which are areas of active cell division in plants, are in charge of sustaining the growth and development of the various plant structures. These cells constantly divide to create new cells, which can subsequently undergo differentiation to become other specialized cell types, such as vascular, dermal, and ground tissues. Meristems, which are often found at the terminals of roots and stems and in the cambium layer of woody plants, are places where meristematic cells may be discovered. Depending on their location, meristematic tissues are classified as apical, lateral, or intercalary.

3. Correct option: (C) Both A and B

Meristematic tissues are present in the growing tips of both roots and shoots. Meristems are regions of actively dividing cells that are responsible for the growth and development of plant structures.

The apical meristem, which is found at the tip of the root, is where meristematic tissues are found in roots. The lengthening of the root is caused by the apical meristem.

The apical meristem, lateral meristem, and intercalary meristem of shoots include meristematic tissues. The lengthening of the shoot is caused by the apical meristem. The lateral meristem is in charge of the shoot's girth expansion. The development of the leaves is controlled by the intercalary meristem.

4. Correct option: (B) Dermal tissue

Dermal tissues in plants are those that create their outer coats. Dermal tissue covers and shields the surfaces of a plant and is the outermost tissue layer. It consists of the epidermis, the cell layer that covers the leaves, stems, and roots.

A waxy layer called the cuticle, which is frequently present in the epidermis, serves to stop water loss and shields the plant from different environmental dangers like diseases, insects, and physical harm.

Specialized tissues like stomata, which are tiny pores in leaves that permit gas exchange during photosynthesis and respiration, among other processes, are also found in dermal tissue.

5. Correct option: (C) Ground tissue

Ground tissues are the tissues that make up the majority of the plant's body. The bulk of a plant's body is made up of ground tissue, a type of tissue that performs a variety of tasks, including photosynthesis, storage, and support. The three forms of ground tissue are parenchyma, collenchyma, and sclerenchyma.

Photosynthesis, the storage of nutrients, and other metabolic processes are all carried out by parenchyma. Collenchyma gives the plant adaptable support, particularly in growing areas. The mature plant tissues are supported mechanically and protected by sclerenchyma.

Ground tissue is essential for the overall structure and function of the plant, and it constitutes the majority of the plant's non-vascular tissue.

6. Correct option: (B) Lateral meristem

The lateral meristem is a type of meristematic tissue that is found in the stems and roots of plants. It is in charge of the plant's girth expansion. Other names for the lateral meristem include the vascular cambium and cork cambium.

The growth in diameter or thickness of plant organs, such as stems and roots, is caused by lateral meristems, often referred to as secondary meristems. The plant undergoes secondary growth as a result, giving rise to secondary tissues such secondary xylem (wood) and secondary phloem. With time, this lateral development enables the plant to enlarge the diameter of its stem, offering support and enabling it to develop stronger and higher stems.

The tissues known as xylem and phloem, which move water and nutrients throughout the plant, are grown by the vascular cambium. The outer layer of the stem and root, known as the cork, grows because of the cork cambium.

7. Correct option: (A) Apical meristem

Meristematic tissues present at the growing tip of plants that bring about growth in length are called apical meristems. The growth in diameter or thickness of plant organs, such as stems and roots, is caused by lateral meristems, often referred to as secondary meristems. The plant undergoes secondary growth as a result, giving rise to secondary tissues such secondary xylem (wood) and secondary phloem. With time, this lateral

development enables the plant to enlarge the diameter of its stem, offering support and enabling it to develop stronger and higher stems.

Lateral meristem is responsible for the growth of the plant in girth. Intercalary meristem is responsible for the growth of leaves and internodes.

8. Correct option: (D) Intercalary meristem

If the tip of a sugar cane plant is removed from the field, it can continue to grow in length due to the presence of intercalary meristem. In grasses like sugar cane, intercalary meristems are found at the base of nodes, which are the places where leaves are joined to the stem. The intercalary meristem at the base of the node continues to function and generate new cells after the tip is cut off, lengthening the stem. This is how sugar cane can regenerate and keep lengthening even if the top is cut off or otherwise harmed. In grasses and several other plants, intercalary meristems are responsible for growth and elongation.

Simple permanent tissues

9. Correct option: (D) Schelerenchyma

Sclerenchyma tissue makes up the husk of a coconut. The mature plant tissues that make up sclerenchyma cells are supported and shielded mechanically by their thick, stiff cell walls. Sclerenchyma cells support the stiff and fibrous characteristics of coconut husk, which make it enduring and force-resistant.

Sclerenchyma is a type of plant tissue that is made up of cells that have thick, lignified cell walls. Sclerenchyma cells are very strong and provide support to the plant. The husk of a coconut is composed of sclerenchyma cells, which is why it is so tough and durable.

10. Correct option: (D) Both B and C

Thin-walled and loosely packed cells are present in both Chlorenchyma and Aerenchyma.

Chlorenchyma is a parenchymal tissue type that has chloroplasts. Plants perform photosynthesis in their chloroplasts, where light energy is transformed into chemical energy (glucose). Chlorenchyma often develops in the mesophyll layer of stems and leaves, where it is crucial to photosynthesis.

Aerenchyma: A form of parenchyma tissue, aerenchyma cells have enormous air gaps or cavities that are filled with air. Aquatic plants are buoyant due to their aerenchyma, which enables them to float in water. The aerenchyma's air gaps aid in both the release of carbon dioxide and other gases from the plant as well as the diffusion of oxygen to its submerged regions.

Collenchyma cells are elongated and have thickened cell walls.

Despite having similarly specialized parenchyma tissues and loosely distributed cells as chlorenchyma and aerenchyma, each has unique roles and placements in the plant.

11. Correct option: (B) Sclerenchyma

The tissue with dead cells among the choices is sclerenchyma. Sclerenchyma cells have been specifically designed to support and shield mature plant tissues mechanically. These cells undergo programmed cell death as they develop, turning into non-living tissues with thick, stiff cell walls. Due to the high lignin content of these thick walls, the cells are exceptionally robust and long-lasting. Plant components including stems, vascular bundles, and seeds are supported structurally by sclerenchyma cells.

Parenchyma, collenchyma, and epithelial tissue are all made up of living cells

12. Correct option: (C) Apical and intercalary meristems are permanent tissues

Primary meristems, such as the apical and intercalary meristems, are only active during a plant's active growth phase. They are not permanent tissues. The length of plant components like roots and shoots is increased by these meristems, which are engaged in basic growth. They adopt a certain position and lose the capacity to differentiate. They develop a permanent tissue as a result. They differentiate into permanent tissues like parenchyma, collenchyma, and sclerenchyma, depending on their location and function, after aiding in the elongation and basic growth of the plant. Permanent tissues lack the ability for active cell division found in meristematic tissues since they are fully differentiated and specialized.

13. Correct option: (A) Chlorenchyma

Chlorenchyma is a kind of parenchyma that has chloroplasts and can thus carry out photosynthesis. Chlorenchyma is a particular kind of parenchyma tissue that may be found in the green stems and leaves' mesophyll layer. Chloroplasts are necessary for chlorenchyma cells to perform photosynthesis, which transforms light energy into chemical energy (glucose), which is necessary for the plant's development and energy needs.

14. Correct option: (B) Collenchyma

In collenchyma, there are longer, thick-walled cells. A kind of plant tissue called collenchyma gives developing plant elements like stems and petioles (leaf stalks) flexible support. Collenchyma cells have thicker cell walls, especially at the corners, and their elongated form enables them to offer structural support while yet allowing for some flexibility in the developing plant. When mechanical support is needed for vigorous growth, such as in young stems and petioles, collenchyma tissue is often present immediately below the epidermis.

Parenchyma, aerenchyma, and chlorenchyma are all types of plant tissue that are made up of thin-walled cells.

15. Correct option: (D) a-S; b-R; c-P; d-Q; e-T;

a) Dermal tissue - S) Epidermis

The outermost layer of tissue in plants is called dermal tissue. It is composed of densely arranged cells that have a thick cuticle. The plant's waxy cuticle acts as a barrier against infection and water loss.

b) Aerenchyma - R) Helps plant to float

A form of plant tissue called aerenchyma is composed of cells with significant air gaps. Aquatic plants have aerenchyma in their stems and roots, which aids in the plant's buoyancy and aeration.

c) Mechanical support - P) Collenchyma

Collenchyma cells serve as mechanical supports. Plant stems and leaves include collenchyma cells, which help the plant by being present there.

d) Tracheid’s - Q) Xylem

Cells known as tracheid’s are found in the xylem tissue of plants. Long, thin cells with thicker cell walls are called tracheids. Tracheid’s move water and nutrients from the plant's roots to its leaves.

e) Transpiration - T) Stomata

Through the act of transpiration, plants release water vapor through their leaves. The leaves' stomata, which are microscopic holes, allow water vapor to escape.

Complex permanent tissues

16. Correct option: (D) Vascular tissue

Vascular tissue refers to plant tissues that aid in the movement of materials. Transporting water, minerals, and nutrients throughout the plant is the function of the specialized form of plant tissue known as vascular tissue. There are two primary cell types present in it:

Xylem: Xylem is in charge of moving water and minerals upward from the roots to the leaves. It has unique cells called tracheids and vessel components that, when they reach maturity, die and form continuous tubes for conducting water.

Phloem: Phloem is in charge of moving organic nutrients from the leaves to the remainder of the plant. These nutrients include sugars and amino acids. It has specialized cells called companion cells and sieve tube components that cooperate to build sieve tubes for the transportation of nutrients.

Vascular tissue is essential for the overall functioning and survival of plants, as it ensures the distribution of vital resources to various parts of the plant.

17. Correct option: (D) All

Food is transported from the leaves to the rest of the plant through a specific form of vascular tissue in plants called phloem. Sieve tubes, companion cells, and phloem fibers are the three primary cell types that compose it.

The long, slender cells that convey food are called sieve tubes. Sieve plates, which are perforated regions in the cell walls, link them end to end.

Small, rounded cells known as companion cells are connected to sieve tubes. They aid in moving nutrients and supporting the sieve tubes.

Long, dense cells called phloem fibers sustain the phloem. They assist in the transportation of water as well.

So, the correct option is D) All. Phloem is a complex tissue composed of various cell types that work together to transport organic nutrients throughout the plant.

18. Correct option: (C) 330 ft

A tree species known as rosewood has a maximum height of 330 feet. Rosewood's xylem is in charge of moving water and minerals from the roots to the leaves. In rosewood, the xylem is composed of long, thin cells organized in a vertical column. This enables the xylem to move water and minerals all the way to the tree's summit. Any plant's ability to convey water to a certain height is constrained by physical laws like gravity and the cohesion-tension theory.

19. Correct option: (A) 200

In eucalyptus plants, the xylem is made up of long, narrow cells that are stacked in a vertical column. As a result, the xylem may move water and minerals all the way to the tree's summit. Eucalyptus trees have xylem that can carry water up to a height of 200 feet, but the tree itself can become much taller than that. The cohesion-tension hypothesis states that most plants can only transport water up to a height of around 100 meters, while eucalyptus trees, which are extremely tall, can transport water far higher. But in eucalyptus, the normal height for water conveyance is around 200 feet.
Match The Following .

20. Correct option: (D) a-R; b-S; c-Q; d-P; e-T;

a) Chlorenchyma - R) Chloroplasts

Chloroplasts are found in a particular kind of plant tissue called chlorenchyma. The organelles in plant cells that are in charge of photosynthesis are called chloroplasts. Plants employ the process of photosynthesis to convert carbon dioxide and water into glucose and oxygen.

b) Aerenchyma - S) Large inter cellular gaps

A form of plant tissue called aerenchyma is composed of cells with significant air gaps. Aquatic plants have aerenchyma in their stems and roots, which aids in the plant's buoyancy and aeration.

c) Collenchyma - Q) Living mechanical tissue

Plant tissue known as collenchyma is composed of elongated cells with thicker cell walls. Support for the plant is provided by the thicker cell walls. Plants' stems and leaves contain collenchyma.

d) Xylem - P) Conduction of water

The vascular tissue known as xylem in plants is responsible for moving water and nutrients from the roots to the leaves.

e) Phloem - T) Transport of sucrose

Food is transported from the leaves to the rest of the plant through a specific form of vascular tissue in plants called phloem.

21. Correct option: (A) Tracheid's

Cell walls in tracheid's are not perforated. The xylem, a vascular tissue responsible for the upward movement of water and minerals in plants, contains a kind of water-conducting cell known as a tracheid. Tracheid's feature tapered ends and pits in their cell walls, albeit these pits are not perforated like those seen in vessels, as is the case with vessels.

Vessels have perforated end walls called vessel elements that allow for more efficient water flow through the xylem. Sieve tubes have sieve plates, which are perforated end walls that facilitate the flow of nutrients in the phloem. Companion cells are associated with sieve tubes and have a perforated cell wall connecting them to the sieve tubes.

22. Correct option: (C) Conduction of water

In plants, the epidermis is the outermost layer of tissue. It is constructed from a single layer of densely packed cells. The epidermis serves a variety of purposes, such as:

It possesses a cuticle, which is a waxy coating that aids in preventing water loss under harsh situations.

The plant and its surroundings can exchange gases thanks to the epidermis. Gases can interchange through the epidermis's tiny stomata, or pores.

Plants exhale water vapor through their leaves through a process known as transpiration. By managing the stomata's opening and shutting, the epidermis aids in controlling transpiration.

The epidermis does not conduct water. Water is conducted through the xylem, which is a type of vascular tissue.

23. Correct option: (B) Xylem parenchyma

The tissue known as xylem parenchyma aids in the sideways conduction of water in a tree's branches. Living cells called xylem parenchyma cells are located in the xylem tissue and play a role in the storage and flow of water laterally throughout the tree. Although xylem parenchyma contributes to the radial and lateral transport of water across the xylem tissue, xylem vessels are the principal water-conducting cells in the xylem. Xylem parenchyma cells' live nature and capacity to sustain metabolic processes make them ideal for lateral water transfer.

Although collenchyma and parenchyma are kinds of tissue found in the xylem, they do not aid in the sideways movement of water. The plant's support tissue is called collenchyma. Food and water are stored in tissues called parenchyma. The sideways conduction of water does not include the xylem vessels.

24. Correct option: (A) Sieve tubes

Instead of xylem tissue, sieve tubes are present in phloem tissue. Transporting water and minerals from the roots to the leaves is done by xylem tissue.

The phloem tissue, which is in charge of moving organic nutrients like sugars throughout the plant, includes sieve tubes.

The xylem tissue contains the tracheid's, vessels, and the other three possibilities. Food and water are stored in a kind of tissue called xylem parenchyma. Long, thin cells called tracheids and vessels carry water up the plant's stem.

25. Correct option: (A) Tracheid's

The primary water-conducting tissue of gymnosperms is the tracheid. They have thicker cell walls and are long, thin cells. The thicker cell walls aid in the movement of water up the plant's stem.

In gymnosperms, which include conifers like pine, spruce, fir, and cedar trees, they are the main water-conducting cells. Tracheids are the primary water-conducting cells of gymnosperms, although they are also present in angiosperms (flowering plants).

Vessels, which are more prevalent in the xylem of angiosperms, are absent in gymnosperms. Xylem fibers are specialized cells that offer mechanical support but are not principally engaged in water conduction, and sieve tubes are a component of the phloem tissue, not the xylem.

26. Correct option: (D) All the above

The vascular tissue known as xylem in plants is responsible for moving water and nutrients from the roots to the leaves. It is composed of tracheid's, vessels, xylem parenchyma and xylem fibers which are the three primary cell types.

A) Tracheids and vessels: These xylem tissue components carry water. Long, tapered cells called tracheid's have pits on their borders that allow water to travel between them more easily. In order to move water effectively, vessels are made up of shorter, broader cells that are arranged end to end to form continuous tubes.

B) Xylem fibers: Xylem fibers are unique cells that give the plant mechanical support. They contribute to the overall stiffness and strength of the plant's structure because they contain dense cell walls.

C) Xylem parenchyma: A form of live cell that can be found in xylem tissue. Within the xylem, these cells are engaged in lateral water storage and transport.

So, the correct answer is D) All the above.

Ground tissues

27. Correct option: (A) Dermal layer

The dermal layer of plants secretes gum. The outermost layer of cells in plant organs including stems, leaves, and roots is referred to as the dermal layer or epidermis. It acts as a defense against infections, harmful environmental stressors, and excessive water loss. As a part of its defense systems, the dermal layer may release compounds like gum, resin, and wax. The dermal layer may secrete gum in reaction to wounds or as a defense mechanism against herbivores or infections.

Meristematic cells are undifferentiated cells with the ability to proliferate through cell division. Water and minerals are moved from the roots to the leaves through a kind of vascular tissue called xylem. Food is moved between the plant's leaves and other parts through a type of vascular tissue called phloem. They do not produce gums.

So, the answer is A) Dermal layer

28. Correct option: (B) Dermal tissue

Dermal tissues in plants are those that create their outer coats. Dermal tissue covers and shields the surfaces of a plant and is the outermost tissue layer. It consists of the epidermis, the cell layer that covers the leaves, stems, and roots.

A waxy layer called the cuticle, which is frequently present in the epidermis, serves to stop water loss and shields the plant from different environmental dangers like diseases, insects, and physical harm.

Specialized tissues like stomata, which are tiny pores in leaves that permit gas exchange during photosynthesis and respiration, among other processes, are also found in dermal tissue.

29. Correct option: (A) Epidermis

Stomata, which are tiny pores in the epidermis of leaves, allow for the exchange of gases between the plant and its surroundings. The epidermis of leaves, stems, and other aerial portions of plants are where stomata are most frequently found. They participate in transpiration as well, the process by which plants expel water vapor via their leaves.

The layers of mesodermis and endodermis do not have stomata. The middle layer of cells in the root is known as the mesoderm. The root's deepest layer of cells is called the endodermis. The movement of water and nutrients throughout the plant is facilitated by both of these layers.

30. Correct option: (B) Guard cells

Specialized cells called guard cells around stomata. In leaves and other sections of aerial plants, these cells make up the epidermis. The stomatal hole is surrounded by guard cells, which also regulate its opening and closure.

The guard cells become turgid and open the stomatal hole when they absorb water, facilitating the exchange of gases and water vapor between the plant and the environment. In contrast, the stomatal hole shuts to stop water loss by transpiration when the guard cells start to lose water and become flaccid.

Secretory cells, gland cells, and parenchymatous cells are not involved in the opening and closing of stomata.

31. Correct option: (A) Epidermis

The top layer of tissue of a plant is called the dermis. It is made up of closely packed cells

that are arranged in a single layer. The epidermis shields the plant from its surroundings and aids in maintaining the proper balance of water within the plant.

The other choices are wrong. In a plant, the mesodermis is the intermediate layer of tissue.

The deepest tissue layer of a plant is called the endodermis. These two layers are neither composed of cutaneous tissue.

Therefore, the answer is A, dermal tissue forms epidermis.

32. Correct option: (B) Root hairs

Root hairs are protrusions that grow from a plant's root that are very long and thin. They greatly expand the root's surface area, which improves the root's ability to absorb water and nutrients from the soil. Although they can be many, root hairs are normally just a few micrometers long. They are also necessary for effective root activity.

Root kinds include lateral roots, tap roots, and rootlets, however none of them resemble hair. Small roots that sprout from the main root are called rootlets. Although they are likewise tiny roots, lateral ones originate from the principal root or from other roots. The primary root of a plant is its tap root.

33. Correct option: (B) Nehemiah Grew

The term "parenchyma" was first used in 1671 by Nehemiah Grew, an English botanist and physician. Parenchyma, according to him, is "the pithy part of plants, composed of soft, succulent, and somewhat angular cells, which are mostly roundish, and not much compressed." Nehemiah Grew is frequently credited with founding plant anatomy and microscopy and made substantial advances to the study of botany.

Although they all made major contributions to the study of botany, Bichat, Robert Brown, and Aristotle did not invent the word parenchyma.

34. Correct option: (A) Cuticle

Due to the presence of the cuticle, a waxy covering on the surface of leaves and stems, water loss is slowed down in desert plants. Water loss from the surface of the plant is minimized by the cuticle's function as a waterproof barrier. Desert plants can conserve water because to this adaptation, which also helps them survive in dry climates.

The other options are incorrect. Stomata are small openings in the epidermis of leaves that

allow for the exchange of gases and water vapor. Lignin is a complex organic compound that is found in the cell walls of some plant cells. Suberin is a waxy substance that is found

in the cell walls of cork cells.

So, the correct answer is A) Cuticle.

35. Correct option: (A) Directly participates in the process of photosynthesis

On the surface of plant leaves and stems, stomata are microscopic holes that control gas

exchange. They principally control the uptake of carbon dioxide (CO_2) for photosynthesis and the release of oxygen (O_2) and water vapor (H_2O). Therefore, by enabling the entry of CO_2, a necessary raw material for photosynthetic processes, stomata indirectly aid the

process of photosynthesis. Stomata themselves do not, however, actively take part in photosynthesis. The chloroplasts of the leaf cells are where the real photosynthesis processes take place. So, option A is not a function of stomata. The correct option is A) Directly participates in the process of photosynthesis.

The other options are correct. Stomata help with the exchange of gases by allowing carbon dioxide to enter the leaf and oxygen to escape. They also help to regulate the plant's water balance by controlling the amount of water vapor that escapes from the leaf.

Therefore, the answer is A, stomata do not directly participate in the process of photosynthesis.

36. Correct option: (C) Suberin

A waxy compound called suberin makes the cork cells, which make up the outer bark of woody plants, resistant to water and gases. The hydrophobic and airtight substance suberin makes the outer bark waterproof and shields the plant's interior tissues from infections and desiccation by preventing the flow of water and gases through the cork cells. The protection and survival of the plant in a variety of environmental circumstances depend on this impermeable covering.

The other choices are false. All plant cells include the structural carbohydrate cellulose. All plant cells contain lipids, a sort of fatty material. Although some plant cells contain lignin, which is a complex organic substance, cork cells do not.

37. Correct option: (B) Conducting tissue

The survival of plants in the terrestrial environment has been made possible by the presence of conducting tissue, particularly the xylem and phloem. The xylem is in charge of moving water and minerals from the roots to the leaves, ensuring a steady supply of vital nutrients for the plant. On the other side, the phloem moves organic substances, such sugars, from the photosynthetic sites (leaves) to other sections of the plant for development, growth, and storage. In order for a plant to survive and develop on land, these conducting tissues are essential for distributing water, nutrients, and food throughout the plant.

The other choices are false. The intercalary meristem is in charge of the plant's increase in length. The apical meristem is in charge of the plant's growth in both length and height. All portions of the plant include a form of live tissue known as parenchymatous tissue. It is not in charge of moving nutrients and water.

LEVEL - II

1. Correct option: (C) Both A & B

Collenchyma is often referred to as "Living mechanical tissue" because it fulfills both conditions mentioned in options A and B.

A) Its cells give mechanical support and it's living tissue: Collenchyma cells have thicker main cell walls, particularly at the corners, which give the plant's expanding components mechanical support and strength. Young stems and petioles are supported by these living cells, which can extend as the plant matures.

B) Deposition of this tissue will not arrest the growth of a plant: Living plant tissue known as collenchyma allows the plant to grow and expand continuously without being

constrained. The plant can develop and adapt to its surroundings without difficulty because to the flexibility and elasticity of collenchyma cells.

Therefore, both aspects of collenchyma being a living tissue providing mechanical support and not arresting the plant's growth are the reasons why it is referred to as "Living mechanical tissue."

2. Correct option: (B) Sclerenchyma

Sclerenchyma is the tissue that is principally involved in the growth of wood. Highly specialized, non-living cells called sclerenchyma cells sustain and give mechanical strength to plant structures because of their thick, lignified secondary cell walls. The xylem, which transports water and nutrients from the roots to the rest of the plant, is one of the woody tissues made mostly of these cells.

Although parenchyma, collenchyma, and meristematic tissues play a number of additional roles in the plant, they are not directly involved in the production of wood. Collagenchyma cells support expanding plant parts mechanically, parenchyma cells participate in a variety of metabolic processes, and meristematic tissues drive ongoing growth and development.

3. Correct option: (A) Resin

Xylem parenchyma is a subset of parenchyma cells that are alive and present in the xylem of vascular plants. Its primary role is the distribution and storage of different molecules including starch, lipids, and other organic components. However, xylem parenchyma does not generally hold resin.

Resin, a viscous, sticky material, is frequently created by specialized resin-producing cells in different plant tissues, including resin canals and resin ducts. In addition to helping the plant heal from wounds, resin acts as a defense against infections and herbivores.

Starch and lipids are among the nutrients that are stored in xylem parenchyma; resin, however, is not one of them. Typical storage locations for resin in plants include specialized cells or ducts.

4. Correct option: (D) None of the given

Meristematic tissue is temporary due to the fact that its cells gradually lose the capacity to proliferate and become specialized for other purposes. The meristematic tissue will ultimately be replaced by other forms of tissue, proving that it is not a permanent structure.

Meristematic tissue can be found throughout the plant, yet its precise location is unknown. Meristematic tissue is sometimes known as non-permanent tissue; however, this is not the cause for this.

Meristematic tissue also increases the plant's depth and height. Meristematic tissue is sometimes known as non-permanent tissue; however, this is not the cause for this.

The correct response is D since none of the suggested explanations explains why meristematic tissue is also referred to as non-permanent.

5. Correct option: (A) Tannin

A particular class of live parenchyma cells called phloem parenchyma may be found in the phloem of vascular plants. It serves to distribute sugars and other nutrients, among other organic materials, throughout the plant.

Tannins are a type of chemical compound that are found in many plants. They have astringent properties and are used in tanning leather. Tannins are not stored by phloem parenchyma. Instead, they are stored in specialized cells called tannin cells. Tannin cells are found in the phloem, but they are not the same as phloem parenchyma cells.

The other choices are wrong. Option B, which states that mucilage is held by phloem parenchyma, is accurate since mucilage is a common sort of gel-like material in plants. The right answer is Option C, which states that latex cells, a particular kind of specialized cell found in the phloem, store latex. phloem parenchyma stores resin, which is Option D.

6. Correct option: (C) It possesses companion cells which has nucleus

Transporting food and other organic materials in the phloem are long, tube-like cells called sieve tubes. Despite not having a nucleus, they are live cells. The nucleus is instead found in the companion cells, which are small, elongated, and intimately related to the sieve tubes.

The sieve tubes are fed and given the energy they require to function by the partner cells. Additionally, they aid in controlling the flow of organic materials like food through the sieve tubes.

The other choices are false. Option A, according to which sieve tubes solely contain cytoplasm, is untrue since partner cells also have a nucleus in sieve tubes. Option B, that sieve tubes solely contain nuclei, is similarly false since sieve tubes also include cytoplasm in addition to their nuclei. Option D, which states that sieve tubes have both cytoplasm and a nucleus, is false since the partner cells, not the sieve tubes themselves, contain the sieve tubes' nucleus.

Therefore, the answer is C, sieve tubes are considered as living components because they possess companion cells which has nucleus.

7. Correct option: (B) Sclerenchymatous pericycle

B) The medullary rays are not formed by the sclerenchymatous pericycle. The parenchymatous cells that make up the medullary rays spread radially from the pith to the outside of the stem or root. A layer of cells called the pericycle encircles the vascular tissues in the roots and stems of plants. Sclerenchyma cells may be present in the pericycle of some plants, but they are not necessary for the development of medullary rays. The parenchymatous cells of the pericycle play a major role in the development of medullary rays.

8. Correct option: (D) It stimulates movement of leaf

Specialized epidermal cells called bulliform cells, often referred to as motor cells, are present in the leaves of several plants. In reaction to variations in water supply, they are in charge of leaf swaying or rolling. These bulliform cells lose turgor pressure during periods of water stress or drought, which causes the leaf to curl or roll inward. Rolling the leaf lowers its exposed surface area, reducing water loss via transpiration and preserving water within the plant. The bulliform cells restore turgor pressure when water supply increases, forcing the leaf to unfurl and return to its regular posture. This movement is a way for the plant to adapt to changing water conditions and maximize water usage effectiveness.

9. Correct option: (C) Bark

The area called as bark contains dead plant cells. Phellem (cork), phelloderm, and phloem are some of the layers that make up the protective outer coat of woody stems

and roots, which is known as the bark. The bark's top layer, called phellem (cork), is made up of dead cells with substantial walls. It shields the plant from diseases, water loss, and mechanical harm. The layer of live cells known as phelloderm is situated immediately underneath phellem. Consequently, the bark exhibits the presence of dead cells.

10. Correct option: (D) Radial

Roots include radial vascular bundles, not leaves. For vascular bundles of leaves, the other choices—unicollateral, closed, and conjoint—are all accurate.

In leaves, vascular bundles are normally organized in a certain way called "conjoint" or "closed" vascular bundles. Both xylem and phloem are present in conjoint vascular bundles, and they are often encircled by a bundle sheath of specialized cells. This configuration enables the efficient movement of carbohydrates, nutrients, and water within the leaf.

Therefore, the answer is D, radial vascular bundles are not found in leaves.

11. Correct option: (A) V.B are scattered

The ground tissue system of the maize stem is not differentiated into cortex, endodermis, pericycle, etc. because the vascular bundles are scattered throughout the stem. The ground tissue in between the vascular bundles in plants with a dispersed pattern is not segmented into the cortex, endodermis, and pericycle. Instead, the ground tissue lacks the distinct layers seen in plants with a ring of vascular bundles and is more uniform overall.

For monocot stems like maize, the dispersed distribution of vascular bundles is a distinguishing property. of contrast, the ground tissue of dicot stems with a ring of vascular bundles is divided into separate areas like cortex, endodermis, pericycle, etc. because a continuous cylinder of vascular bundles is present in these stems.

12. Correct option: (C) Both A & B

Collenchyma cells provide flexibility to plants due to both of the given reasons:

A) They have polygonal cells: Collenchyma cells frequently feature polygonal shapes and irregularly thickened cell walls. Collenchyma is found in the stems and leaves of plants. The polygonal shape of collenchyma cells allows them to bend and flex without breaking.

B) They have pectin deposition: Pectin, a component that gives cell walls increased elasticity, is present in collenchyma cells. The collenchyma cells' flexibility, which allows them to bend and stretch without breaking, is a result of the pectin deposition.

Collagen deposition and the polygonal form of collenchyma, which lends plants flexibility, make C the correct response.

13. Correct option: (A) The root dies first

In a ring girdled plant, the removal of the bark and phloem disrupts the bidirectional transport of nutrients and sugars from the leaves (photosynthetic sites) to other plant parts, including the roots. As a result, the roots are deprived of the essential nutrients and energy sources, leading to their death overtime due to starvation. The shoot, on the other hand, can still receive nutrients from the leaves through downward transport, but the roots cannot survive without the supply of food from the leaves. Therefore, the root dies first in a ring girdled plant. However, if the shoot is not able to get water and nutrients from the roots, it will eventually die as well.

14. Correct option: (D) All of the given

Dicot leaves are referred to be bilateral because of differences in the distribution of stomata on each epidermis, the kind of mesophyll tissue distributed between the two epidermises, and the intensity of the green color on each epidermis.

Mesophyll in bilateral leaves is divided into palisade and spongy parenchyma. On the top side of the leaf lies the palisade parenchyma, while the bottom side is home to the spongy parenchyma. On the epidermis, the stomata are likewise dispersed unevenly. The bottom side of the leaf has more stomata, which aids in absorbing more carbon dioxide from the atmosphere.

On each epidermis, the green color is not the same strength. Greener may be seen on the leaf's top side than its underside. This is due to the palisade parenchyma's higher concentration of chloroplasts, the organelles that aid in photosynthetic activity.

Therefore, the answer is D, all of the given reasons are why dicot leaves are known as bilateral.

15. Correct option: (D) All of the given

Monocot leaves are known as isobilateral because the distribution of stomata on both the epidermis is equal, the distribution of chlorenchyma between two epidermis is of same, and the intensity of green color on both epidermis is same.

Mesophyll does not develop into palisade and spongy parenchyma in isobilateral leave.

The mesophyll contains an equal distribution of chlorenchyma.
This explains why the intensity of the green colour on each epidermis is the same.
Additionally, the stomata are uniformly spaced across both epidermises.
This explainswhy there are equal numbers of stomata on both epidermises.

Therefore, the answer is D, all of the given reasons are why monocot leaves are known as isobilateral.

16. Correct option: (C) Xylem is endarch

The xylem in the maize-root stele is exarch rather than endarch. This indicates that the metaxylem is positioned towards the pericycle while the protoxylem is positioned towards the pith. The proto- and meta-xylem should be positioned in a V-shape on the stele of a maize root, with two proto and one meta-xylem placed in each unit of the vascular bundle.

The other choices are the correct. Because the xylem in the stele of maize-root is arranged in a V-shaped pattern, Option A, "Proto and meta-xylem are arranged in V-shaped," is true. In option B, each and every unit of V contains two proto and one meta-xylem. B is accurate because each and every vascular bundle unit in the stele of maize-root has two protoxylem poles and one metaxylem pole. In the stele of maize-root, the protoxylem is placed towards the pericycle, making Option D—protoxylem is arranged towards pericycle—correct.

17. Correct option: (D) Interfascicular cambium – Interfascicular cambium

The interfascicular cambium and interfascicular cambium are the two tissues that make up the cambium ring. The cellular layer between the vascular bundles in the stem is known as the interfascicular cambium. A layer of cells that is located inside the vascular bundles is known as the interfascicular cambium.

Interfascicular and interfascicular cambiums combine to produce the cambium ring. For secondary growth, we may thank the cambium ring. The stem can thicken because it generates additional xylem and phloem cells.

The other options are wrong. The cambium ring is not formed by phellogen and phelloderm, hence Option A, phellogen-phelloderm, is wrong. Because the interfascicular cambium and the interfascicular cambium, not the phellogen, create the cambium ring, Option B—cambium ring—phellogen—is wrong. The phellogen only contributes to the development of the periderm, not the cambium ring, hence Option C, phellogen-phellogen, is wrong.

18. Correct option: (B) Pith transforms into cambium

Some roots show secondary growth, even though they have closed vascular bundles. This is because the pith of these roots can transform into cambium.

A layer of cells called the cambium is in charge of secondary growth.

During the early stages of growth, the cambium ring in certain roots may be active, creating secondary xylem (wood) and secondary phloem (inner bark), which causes the root to grow in girth or thickness. The vascular bundle may shut as the root grows, and cambium activity may cease.

The other options are false. The cambium ring is created from the procambium, not the pith, hence Option A, that they previously possessed one, is untrue. Because medullary rays don't generate fresh xylem and phloem cells, Option C—that they undertake secondary growth—is false.

19. Correct option: (C) It is conducting in nature

Heartwood, commonly referred to as duramen, is the core, non-living portion of the tree trunk. Sapwood is transformed to create it, and it has lost its conducting properties. When sapwood turns into heartwood, the cells lose their live protoplasm and the xylem vessels, which act as conducting components and carry water and nutrients, get clogged with resin and other materials. Duramen is usually darker in color due to the accumulation of various substances, but it is not conducting in nature as compared to the living sapwood.

20. Correct option: (A) Collenchyma – Division and differentiation

An elongated cell type with thicker cell walls is called collenchyma, which is a form of plant tissue. Plant stems and leaves are both covered in collenchyma. It gives the plant assistance and adaptability. No division or differentiation occurs in collenchyma cells. These cells have taken on their final shape and are permanent.

Cell division occurs in meristematic tissues, which are present in areas of dynamic growth like the development of axillary buds. The flexible plant tissue known as parenchyma serves a number of purposes, including nutrition storage, photosynthesis, and substance secretion. Because it has strong, lignified cell walls, sclerenchyma is a kind of plant tissue that supports the plant mechanically.

Therefore, the answer is A is mismatched, collenchyma cells do not divide or differentiate, so they cannot perform the function of division and differentiation.

21. Correct option: (C) Helpful moisture from air

Root hairs are small, thread-like projections that extend from the root tips of plants. The primary purpose of root hair is to increase the surface area of the root, which aids in

water and mineral absorption from the soil. They participate in the process of absorbing moisture from the soil but do not directly absorb moisture from the air.

They are not multicellular (option A). Root hairs are found on underground roots, not on aerial organs (option B). Root hairs are not covered with cuticle (option D). Instead, they have thin cell walls that facilitate the uptake of water and nutrients from the soil.

Therefore, the answer is C, root hairs are helpful in absorbing moisture from the air.

22. Correct option: (D) 4

Trichomes and stomata are both important structures in plants. They have 4 sub-layers of ground tissue:

• Epidermis: The epidermis is the outermost layer of cells in a plant. It is made up of a single layer of cells that are tightly packed together. The epidermis protects the plant from the environment and helps to regulate the plant's water balance.

• Cuticle: The cuticle is a waxy layer that covers the epidermis. It helps to protect the plant from the environment and helps to prevent water loss.

• Mesophyll: The mesophyll is the layer of tissue that lies between the epidermis and the vascular tissue. It is made up of parenchyma cells that are involved in photosynthesis.

• Vascular tissue: The vascular tissue is the layer of tissue that transports water and nutrients throughout the plant. It is made up of xylem and phloem.

Therefore, the answer is D, there are 4 sub-layers of ground tissue of trichome (stem hair) and stomata.

23. Correct option: (B) Cortex- endodermis- pericycle- conjunctive tissue- pith

The ground tissue system is the tissue system that makes up the bulk of the plant body. It is made up of three types of tissue: cortex, endodermis, pericycle, conjunctive tissue and pith.

The ground tissue's outermost layer is called the cortex. Parenchyma cells that are loosely packed together make up its structure. Storage cells, air pockets, and vascular bundles are found in the cortex.

A layer of cells called the endodermis encircles the stele. The Casparian strip, a unique layer of waxy substance in the endodermis, inhibits water and solutes from readily passing through the cell walls.

A layer of cells called the pericycle, immediately within the endodermis, is where lateral roots first appear.

In the ground tissue system of plants, the vascular bundles are surrounded by a layer of tissue called conjunctive tissue. Sclerenchyma cells make up this structure, which supports the vascular bundles.

The pith is the central region of ground tissue. It is made up of parenchyma cells that are loosely packed together. The pith contains storage cells and air spaces.

This sequence of ground tissues plays essential roles in various plant functions, including nutrient transport, storage, and structural support.

24. Correct option: (B) It has lenticels

Epidermis is multi layered is also a characteristic of hygroscopic roots of orchids. The epidermis of hygroscopic roots is multi-layered, which helps to trap moisture.

The roots of hygroscopic plants take in moisture from the atmosphere. Epiphytes, or plants that grow on other plants, are where you may find them. One kind of epiphyte is

the orchid. Orchid roots have been altered so they can take up moisture from the atmosphere. A spongy tissue that helps to retain moisture covers them. The roots' broad surface area also makes it easier for them to absorb moisture.

There are no lenticels on orchid roots. Lenticels are tiny holes that allow for the exchange of gases in a tree's bark. Because orchids lack bark, they are not in need of lenticels.

The other possibilities—a multilayered epidermis, a lack of a cuticle, and the ability to absorb moisture from the environment—have all been seen in the hygroscopic roots of orchids. As a result, option B is correct since orchids' hygroscopic roots lack lenticels.

25. Correct option: (A) Open

When referring to a vascular bundle that lacks phloem parenchyma, the phrase "open" is not appropriate. The phrase "open" in reference to vascular bundles denotes the existence of cambium between the xylem and phloem, which permits the bundle to undergo secondary growth. The bundle would not be regarded as "open" for secondary growth if there is no phloem parenchyma since it is likely that there is no cambium as well.

Contrarily, whether phloem parenchyma is present or absent, alternatives B) Endarch, C) Unicollateral, and D) Conjoint are still relevant properties of a vascular bundle.

26. Correct option: (C) Location – Maize stem, Function – Mechanical support

The maize stem contains a sclerenchymatous bundle sheath, which serves as the plant's mechanical support system. Plant tissue known as sclerenchyma is stiff and powerful because of its thick, lignified cell walls. It aids in giving the plant structural support, enabling it to remain erect and fend off bending or breaking, by wrapping vascular bundles in the stem. For the stability and integrity of the stem, particularly in tall, quickly-growing plants like maize, this mechanical support is crucial.

The other options are not correct. Maize roots do not have sclerenchymatous bundle sheath, and they do not store water-minerals. The function of sclerenchymatous bundle sheath in maize stems is to provide mechanical support, not to store water-minerals.

27. Correct option: (A) Collenchyma- Deposition of pectin- flexibility and elasticity

Plant tissue known as collenchyma is composed of elongated cells with thicker cell wall s. Plants' stems and leaves contain collenchyma.

It provides the plant stability and flexibility. Collenchyma cells are found in the hypodermis of aerial parts of a plant. The stem and leaf's outermost layer of tissue is known as the hypodermis.

The alternate choices are incorrect. The tips of roots and shoots contain meristematic tissue. It is in charge of the plant's growth and development. All portions of the plant have parenchyma cells. They perform a number of tasks, such as photosynthesis, storage, and transportation. Plant stems, leaves, and fruits contain sclerenchyma cells. But it is not present in hypodermis. They provide the plant stability and strength.

So, the answer is A, collenchyma is the tissue that is present in the hypodermis of aerial parts of a plant. It provides flexibility and elasticity to the plant.

28. Correct option: (D) All the given

A) Sieve plate is present in sieve tube, but not in sieve cell

Food may be transported from one sieve cell to another via the sieve plate, which is a perforated area in the end walls of sieve cells. In sieve tubes, there are no sieve plates.

B) Companion cell is present in sieve tube, but not in sieve cell

Specialized parenchyma cells known as companion cells are connected to sieve tubes. They may aid in moving food from the sieve tubes to other areas of the plant. Sieve cells don't contain companion cells.

C) Conduction efficiency of sieve tube is more than sieve cell

The sieve tube has a higher conduction efficiency than the sieve cell. Long, continuous cells called sieve tubes are joined by sieve plates. This makes it possible to carry food across great distances in an effective manner. Sieve plates do not link sieve cells, which are shorter. As a result, food transport efficiency in sieve cells is constrained.

29. Correct option: (C) a – iv; b – ii; c – i; d – iii

a) Collenchyma - iv) Only in aerial organ - Pectin deposited cells

Collenchyma tissue gives developing plant parts mechanical support. It is only present in aerial organs since it is mostly located in the corners of stems and leaves. In some aerial organs, collenchyma cells may have pectin deposits, which adds to the organs' support.

b) Parenchyma - ii) Air-exchange

Photosynthesis, storage, and transport are just a few of the tasks that parenchyma cells do. Additionally, they exchange air. Stomata, which are little holes in the epidermal of leaves, allow gases to be exchanged between the plant and its surroundings. Around the stomata are parenchyma cells, which aid in controlling the stomata's opening and shutting.

c) Sclerenchyma - i) Lignified cells - in all organ

Plant tissue known as sclerenchyma is made up of cells with thick, lignified secondary cell walls. All plant organs have them, and they give the plant solid structural support.

d) Phloem - iii) Albuminous cells are present

In the phloem of gymnosperm wood, albuminous cells, which are specialized parenchyma cells, may be found next to sieve cells. As a source of nitrogen for the sieve cells, they include the protein albumin. Companion cells seen in angiosperms are similar to albuminous cells.

30. Correct option: (A) Subsidiary cell- stomatal apparatus, triangular, one pair

Companion cell- phloem, one, Irregular

On either side of a stoma (a hole in the leaf), there are two tiny, kidney-shaped cells called subsidiary cells. They are believed to aid in controlling the stoma's opening and shutting. The group of cells known as the stomatal apparatus, which surrounds a stoma, contains subsidiary cells.

Small, living cells called companion cells are found in the phloem and are connected to sieve tubes. They may aid in moving food from the sieve tubes to other areas of the plant. In the phloem, companion cells are present.

Subsidiary and companion cells differ from one another in terms of their location, number, composition, and purpose. Stomata include subsidiary cells, whereas phloem contains companion cells. Companion cells are solitary, whereas subsidiary cells are paired. While companion cells are asymmetrical, subsidiary cells are kidney-shaped. Companion cells assist in moving food from sieve tubes to other sections of the plant, whereas subsidiary cells assist in controlling the opening and shutting of stomata.

Therefore, the answer is A, subsidiary cells are found in the stomata and are paired, while companion cells are found in the phloem and are single.

5. Why do we fall ill?

Solutions

LEVEL-I

1. Correct option: (D) Elephantiasis

Lymphatic filariasis, commonly known as elephantiasis, is a painful and profoundly disfiguring disease. Elephantiasis can last for a very long time and has long-term effects on the body. Therefore, it is a chronic disease.

2. Correct option: (C) Jaundice

Jaundice happens when too much bilirubin is made in the blood or when the liver is damaged and cannot get rid of bilirubin from the body. Other than yellow skin, people with jaundice may have stomach pain, nausea, fever, weight loss, or itching.

3. Correct option: (A) Microbes

nfectious diseases are disorders caused by organisms — such as bacteria, viruses, fungi or parasites. Many organisms live in and on our bodies. They're normally harmless or even helpful. But under certain conditions, some organisms may cause disease.

4. Correct option: (C) Cancer

Mutations or genetic changes on DNA results in changing genetic makeup of a cell resulting in altering the behaviour and metabolism of cells which in return cause cells to divide indefinitely.

5. Correct option: (C) Robin Warren

Barry J. Marshall and Robin Warren, two Australian researchers who discovered the bacterium Helicobacter pylori and deciphered its role in gastritis and peptic ulcer disease, have been awarded this year's Nobel Prize in Physiology or Medicine.

6. Correct option: (B) 50

Upon analysis of negative-stained SARS-CoV-2 articles by electron microscopy, different researchers have had varying results, but the diameter of the virus has been found to range between 50 nm to 140 nm.

7. Correct option: (A) Small intestine

Roundworms are small organisms that can live in your intestine, which is part of your digestive system. Roundworms can live in the human small intestine for a long time and can cause infections. Ascaris is an intestinal parasite of humans. It is the most common human worm infection. The larvae and adult worms live in the small intestine and can cause intestinal disease.

8. Correct option: (A) bacteria

There are many organisms that can cause skin infections including bacteria, viruses and fungi. Bacteria like Staphylococcus aureus may cause skin infections, Fungi including Trichophytons cause ringworms, and viruses like Herpes cause cold sores. Most prominent infections are caused by bacteria.

9. Correct option: (B) tuberculosis

TB is spread through the air from one person to another. TB bacteria are passed through the air when someone who is sick with TB disease of the lungs or throat coughs, speaks, laughs, sings, or sneezes. Anyone near the sick person with TB disease can breathe TB bacteria into their lungs.

10. Correct option: (A) Mosquitoes

Most of the time mosquitoes feed on sugary substances like plant nectars. However, when it's time for female mosquitoes to lay eggs, they need the extra protein that comes from blood supplies. Only female mosquitoes bite living hosts such as humans and other warm-blooded animals.

11. Correct option: (B) AIDS

HIV (human immunodeficiency virus) is a virus that attacks the body's immune system. If HIV is not treated, it can lead to AIDS (acquired immunodeficiency syndrome). There is currently no effective cure. Once people get HIV, they have it for life. But with proper medical care, HIV can be controlled.

12. Correct option: (A, B, D) Small pox, Chicken pox and AIDS

Small pox, Chicken pox and AIDS are viral diseases caused by Variola virus, varicella zoster virus and HIV respectively. Tuberculosis is a bacterial infection caused by Mycobacterium tuberculosis.

13. Correct option: (D) Diabetes

Diabetes is one of the four major types of noncommunicable diseases (cardiovascular disease, diabetes, cancer and chronic respiratory diseases). It is a chronic condition that occurs when the body either does not produce enough insulin or cannot effectively use the insulin it does produce.

14. Correct option: (D) Female Anopheles mosquito

In the case of malaria, the vector is the female anopheline mosquito and the disease-causing organism is the malaria parasite. Humans and anopheline mosquitoes are both considered to be the parasite's hosts.

15. Correct option: (C) TB

BCG, or bacille Calmette-Guerin, is a vaccine for tuberculosis (TB) disease. Many foreign-born persons have been BCG-vaccinated. BCG is used in many countries with a high prevalence of TB to prevent childhood tuberculous meningitis and miliary disease.

16. Correct option: (A) Cholera

The cholera bacteria is passed through feces (poop). It is spread by eating or drinking food or water contaminated by the feces (poop) of an infected person. This occurs more often in underdeveloped countries lacking proper water supplies and sewage disposal.

17. Correct option: (D) Bed bugs

Plague is an infectious disease caused by the bacteria Yersinia pestis, a zoonotic bacterium, usually found in small mammals and their fleas. It is transmitted between animals through fleas. Humans can be infected through: the bite of infected vector fleas.

18. Correct option: (D) TB

Tuberculosis is a chronic, progressive mycobacterial infection, often with an asymptomatic latent period following initial infection. TB bacteria can live in the body without making you sick. This is called latent TB infection.

19. Correct option: (B) Alexander Flemming

In 1928 Dr Alexander Fleming returned from a holiday to find mould growing on a Petri dish of Staphylococcus bacteria. He noticed the mould seemed to be preventing the bacteria around it from growing. He soon identified that the mould produced a self-defence chemical that could kill bacteria and later on named it penicillin.

20. Correct option: (D) Cholera

Cholera is an acute diarrheal illness caused by infection of the intestine with Vibrio cholerae bacteria. People can get sick when they swallow food or water contaminated with cholera bacteria.

LEVEL-II

1. Correct option: (B) Increases

It can be hard to follow the work schedule, but the good thing is that you are at the comfort of your home, so utilize it at its best. Take a good 10-minute walk every 2 or 3

hours or do stretching and take a good diet and sanitary hygiene. This will not only increase your productivity but would also prevent you from diseases.

2. Correct option: (A) Good economic

Social harmony and good economic conditions are necessary for good health.

3. Correct option: (D) All the above

Community health is defined as the group of individuals living in a society who aims to maintain, protect and improve the health of the people. The whole community aims to maintain the health of people by providing safe drinking water, proper disposal of sewage, by providing proper medical facilities, and by providing proper health education to the people in the community.

4. Correct option: (C) Disturbed

The term "disease" is a condition in which the normal functioning of the body is disturbed. Many times, such conditions are not very serious and get cured, but sometimes they become serious and even fatal.

5. Correct option: (A) Short

Acute diseases come on rapidly, and are accompanied by distinct symptoms that require urgent or short-term care, and get better once they are treated.

6. Correct option: (D) Both A and B

Acute diseases last for a shorter period of time. They do not have any long-term effect on the patient. E.g., common cold, cough, etc.

7. Correct option: (B) Chronic

Reducing a person's ability to work efficiently and experiencing difficulties in learning at school are symptoms typically associated with chronic conditions or long-term issues rather than acute ones. They develop over time and can have a gradual onset. Symptoms like reduced work efficiency and learning difficulties can be linked to chronic conditions that affect a person's overall health, cognitive abilities, or emotional well-being over an extended period.

8. Correct option: (B) Chronic

Chronic diseases are typically long-lasting conditions that develop gradually and persist over time. Tuberculosis (TB) is caused by the bacterium Mycobacterium tuberculosis and primarily affects the lungs. It often presents with symptoms such as cough, weight loss, fatigue, and fever. If left untreated, TB can persist for an extended period, making it a classic example of a chronic disease.

9. Correct option: (B) Pathogens

The disease-causing organisms like bacteria, viruses, fungi, protozoans, worms, etc. are referred to as pathogens.

10. Correct option: (D) virus

Chickenpox is caused by the varicella-zoster virus (VZV). This virus is responsible for the development of chickenpox (varicella) during the initial infection and can later reactivate to cause shingles (herpes zoster) in some individuals.

11. Correct option: (B, D) Typhoid and Cholera

Cholera is caused by the bacterium Vibrio cholerae and primarily affects the digestive system, leading to severe diarrhea and dehydration. Typhoid fever is indeed caused by the bacterium Salmonella typhi. It's a bacterial infection that can lead to high fever, abdominal pain, and other symptoms.

12. Correct option: (D) None of the above

Leprosy, Syphilis, and TB are all bacterial diseases caused by Mycobacterium leprae, Treponema pallidum and Mycobacterium tuberculosis respectively.

13. Correct option: (B) Unhealthy

A nutritious diet provides essential nutrients that the body needs to function properly, maintain good health, and prevent various health issues. When a person lacks proper nutrition due to an inadequate diet, their body may not receive the necessary vitamins, minerals, proteins, and other nutrients it requires.

14. Correct option: (A) Congenital

Congenital refers to a condition or trait that is present at or from birth. It can refer to both genetic conditions and conditions that might have been influenced by prenatal factors.

15. Correct option: (B) Acquired

Acquired diseases are those that are not present at birth and are typically developed or contracted due to various factors such as infections, lifestyle choices, environmental influences, and so on. These diseases are not inherited and do not pass from one generation to another through genetic means.

16. Correct option: (A) Communicable

A communicable disease (also known as an infectious or contagious disease) is a type of disease that can be transmitted from one person, animal, or object to another through various means. This can include direct contact, respiratory droplets, contaminated food or water, vectors like insects, and more. Some examples of communicable diseases include the common cold, influenza, tuberculosis, and COVID-19.

17. Correct option: (B) Infectious

Tuberculosis (TB) is an infectious disease that most often affects the lungs and is caused by a type of bacteria. It spreads through the air when infected people cough, sneeze or spit.

18. Correct option: (B) Non-infectious

Scurvy is a disease resulting from a lack of vitamin C (ascorbic acid). Early symptoms of deficiency include weakness, fatigue, and sore arms and legs. Thus, it is non-infectious.

19. Correct option: (C) Common cold

Colds are primarily transmitted from person to person via hands that are contaminated with nasal secretions. Less often, viruses that cause colds can be transmitted by touching a contaminated surface or via sneezing or coughing. Direct contact — People with colds typically carry the cold virus on their hands

20. Correct option: (C) Antibiotic

Penicillin is an antibiotic that is it is used to treat bacterial infections. It was discovered by Sir Alexander Fleming in the year 1920.

21. Correct option: (D) None

AIDS is a viral disease caused by HIV virus through blood transfusion via infected needles, sexual contact, and mother to infant through breast feeding. It is present in body fluids.

22. Correct option: (A) Water borne

Drinking water should be filtered and boiled, ozonised, and treated with chlorine before making it potable, thereby avoiding water borne infections. This ensures that no pathogenic bacteria like typhus and cholera are persisting there.

23. Correct option: (A) transmissible

Communicable diseases, also known as infectious diseases or transmissible diseases, are illnesses that result from the infection, presence and growth of pathogenic (capable of causing disease) biologic agents in an individual human or other animal host.

24. Correct option: (C) Are vectors for many diseases

Mosquitoes are known to transmit various diseases such as malaria, dengue fever, Zika virus, and more. Preventing mosquito breeding in our surroundings is crucial to reduce the spread of these diseases.

25. Correct option: (B) Tuberculosis

BCG, or bacille Calmette-Guerin, is a vaccine for tuberculosis (TB) disease. Many foreign-born persons have been BCG-vaccinated. BCG is used in many countries with a high prevalence of TB to prevent childhood tuberculous meningitis and miliary disease.

6. Improvement in Food Resources

Solutions

LEVEL-I

1. Correct option: (D) Phosphorous

Phosphorus is one of the three macronutrients essential for plant growth. It is required for the photosynthesis process, converting the sun's energy into food for the plant. It is also required for strong root development. A plant must be able to access phosphorus to ensure a healthy growing cycle.

2. Correct option: (D) Phosphorous

Phosphorous is used by plants to help form new roots, make seeds, fruit and flowers. It's also used by plants to help fight disease.

3. Correct option: (B) Black Gram

Black gram can be grown on variety of soils ranging from sandy soils to heavy cotton soils. The most ideal soil is a well-drained loam with pH of 6.5 to 7.8. Black gram cannot be grown on alkaline and saline soils. Land is prepared like any other kharif season pulse crop.

4. Correct option: (D) All of these

Intercropping is the practice of growing two or more crops in proximity. For instance, the spaces between the rows of the Jackfruit trees should be cultivated and used for the production of annual or perennial intercrops or both. Examples of such intercrops are Citrus, Banana, Pineapple, Corn, Peanut (Groundnut), and other Pulses, Spices, and vegetables.

5. Correct option: (D) Both A & B

Root nodules are mostly present in the leguminous plants. E.g. Pulses, beans, soybeans, etc. These nodules provide shelter to certain bacteria which aids in nitrogen fixation.

6. Correct option: (A) Rhizobia

Rhizobium is a soil bacteria that fixes nitrogen belonging to the genus of gram-negative bacteria. Rhizobium species from endosymbiotic nitrogen-fixing associations with legumes. The bacteria colonize plant cells in the roots, where they use the nitrogenase enzyme to convert atmospheric nitrogen to ammonia and then provide organic nitrogen compounds such as glutamine.

7. Correct option: (D) Both A & B

Manure is an organic substance obtained by the decomposition of animal and plant wastes. It helps to maintain the fertility of the soil without having deleterious effects on soil as well as the crops grown on it.

8. Correct option: (D) All of these

These concentrated manures are produced using basic materials that are either plant- or animal-derived. Oil cakes, bone meal, fish meal, blood meal, and meat meal are some of the concentrated organic manures that are frequently employed. These include Jatropha, Neem and Groundnuts.

9. Correct option: (D) Both A and C

Natural organic manures are generally divided into two types. One is concentrated organic manures and the other is macro-organic manure. Animal excreta, compost, deep litter are the examples of macro-organic manure.

10. Correct option: (C) Insecticides

Insecticides are pesticides that are formulated to kill, harm, repel or mitigate one or more species of insect. Insecticides work in different ways. Some insecticides disrupt the

nervous system, whereas others may damage their exoskeletons, repel them or control them by some other means.

11. Correct option: (D) All of these

The three options that have been used for increasing food production are (i) agro-chemical based agriculture (ii) organic agriculture; and (iii) genetically engineered crop-based agriculture.

12. Correct option: (B) Carbohydrate

Starch is a carbohydrate commonly found in nature and one of the primary sources of food energy for human beings. It is regularly eaten in the form of wheat, rice, potatoes, and other staple foods cultivated throughout the world. Alongside fiber and sugar, starch is one of the three main categories of carbohydrates.

13. Correct option: (C) Drip irrigation

Drip irrigation saves wastage of water because it supplies water directly to the roots.

14. Correct option: (B) Nitrogen

The primary macronutrients are Nitrogen (N), Phosphorus (P), and Potassium (K). Nitrogen is essential for plant development, since it plays a fundamental role in energy metabolism and protein synthesis. Nitrogen is absorbed by the plant in the form of a nitrate. This macronutrient is directly related to plant growth.

15. Correct option: (D) All of these

Fertilisers and manures contain plant nutrients such as nitrogen, phosphorous and potassium, etc. So, when fertilisers and manures are added to the soil in the fields, then the soil gets enriched with nutrients like nitrogen, phosphorous and potassium, etc.

Nutrients are replenished by certain different microorganisms that are found in the soil or also by the death of certain bacteria such as the rhizobium.

16. Correct option: (B) 50 to 150 kg

Values estimated for various legume crops and pasture species are often impressive, commonly falling in the range of 50 to 150 kg of N ha^{-1} $year^{-1}$.

17. Correct option: (D) All the above

Panchagavya or panchakavyam is a mixture used in traditional Hindu rituals that is prepared by mixing five ingredients. The three direct constituents are cow dung, urine, and milk; the two derived products are curd and ghee. These are mixed in proper ratio and then allowed to ferment.

18. Correct option: (B) 46%

The percentage of nitrogen present in urea is 46.6%. Urea contains four elements: carbon, hydrogen, nitrogen, and oxygen. The molecular formula for urea is CH4 N2 O. This compound is made up of a carbonyl (C=O) group attached to two nitrogen groups (NH2). A carbonyl carbon is one that is double bonded to oxygen.

19. Correct option: (A) Tungrovirus on paddy

Tungro virus disease is transmitted by leafhoppers, wherein the most efficient vector is the green leafhopper.

20. Correct option: (A) Rhizobium

Rhizobium is a soil bacteria that fixes nitrogen belonging to the genus of gram-negative bacteria. Rhizobium species from endosymbiotic nitrogen-fixing associations with legumes. The bacteria colonize plant cells in the roots, where they use the nitrogenase enzyme to convert atmospheric nitrogen to ammonia and then provide organic nitrogen compounds such as glutamine.

21. Correct Matching is: a-T, b-S, c-P, d-R, e-Q

(a) Kharif crop – (T) Agricultural crops that arise in the rainy season

In India, the season is popularly considered to start in June and end in October. Kharif crops are usually sown at the beginning of the first rains during the advent of the south-west monsoon season, and they are harvested at the end of monsoon season (October–November).

(b) Rabi crop – (S) Agricultural crops that raise in winter

Rabi Crops are harvested in the spring season while it is sown in winter. The rabi crops are sown around mid-November, preferably after the monsoon rains are over, and harvesting begins in April / May

(c) Mixed crop – (P) Cultivation of more than one crop in the same field

In agriculture, multiple cropping or multicropping is the practice of growing two or more crops in the same piece of land during one year, instead of just one crop. When multiple crops are grown simultaneously, this is also known as intercropping.

(d) Weedicides – (R) Chemicals used to kill weeds

Weedicides are the chemicals which are generally sprayed all around the field to prevent the growth of weeds. An example of common weedicide is 2,4- D (2,4 dichlorophenoxy acetic acid) which acts as a dicot weedicide.

(e) Herbicides – (Q) Chemicals used to kill herbs

Herbicide stops the growth of plants whereas weedicides skill weeds. They are designed to kill specific types of pests and are used mainly on plants.

22. Correct option: a-S, b-P, c-T, d-Q, e-R

(a) Earthworm – (S) Vermicompost

Vermicomposting is a natural process whereby earthworms convert waste material with rigid structures into compost. The compost produced in this green process is traditionally and popularly used as a natural fertilizer for enhancing plant growth.

(b) Urine of cow – (P) Panchagavya

It consists of nine products viz. cow dung, cow urine, milk, curd, jaggery, ghee, banana, Tender coconut and water. When suitably mixed and used, these have miraculous effects.

(c) Gingelly – (T) Concentrated organic manure

(d) Zinc – (Q) Micronutrient

Zinc (Zn) is an important micronutrient for plants since it is involved in many key cellular functions such as metabolic and physiological processes, enzyme activation, and ion homeostasis.

(e) Phosphorous – (R) Macronutrient

The primary macronutrients are Nitrogen (N), Phosphorus (P), and Potassium (K). Nitrogen is essential for plant development, since it plays a fundamental role in energy metabolism and protein synthesis. Nitrogen is absorbed by the plant in the form of a nitrate. This macronutrient is directly related to plant growth

LEVEL-II

1. Correct option: (C) Both A and B

Rice has approximately 2.69% of protein, and they have high carbohydrates. At the same time, millets are low in carbohydrates than rice and contain a higher amount of protein.

2. Correct option: (B) Green gram

The mung bean or Organic green gram is a legume family plant. Its super high protein content has made it a popular ingredient for making plant-based alternatives to animal-based food items like eggs.

3. Correct option: (A) Rainy

In India, the season is popularly considered to start in June and end in October. Kharif crops are usually sown at the beginning of the first rains during the advent of the south-west monsoon season, and they are harvested at the end of monsoon season (October–November).

4. Correct option: (C) Winter

Rabi Crops are harvested in the spring season while it is sown in winter. The rabi crops are sown around mid-November, preferably after the monsoon rains are over, and harvesting begins in April / May.

5. Correct option: (C) Peas

Examples of Kharif crops

Paddy, maize, bajra, jowar, soybean, castor, cotton, sugarcane, turmeric, chilly, bitter gourd, guar, okra, brinjal,turmeric, ragi.

6. Correct option: (B) Hybridization

Hybridization refers to crossing between genetically dissimilar plants. This crossing may be intervarietal (between different varieties), interspecific (between two different species of the same genus) or intergeneric (between different genera).

7. Correct option: D) All the above

Cultivation practices and crop yield are related to weather, soil quality and availability of water. Since weather conditions such as drought and flood situations are unpredictable, varieties that can be grown in diverse climatic conditions are useful.

8. Correct option: (C) Both A and B

Crop management is the set of agricultural practices performed to improve the growth, development and yield of crops. It begins with a seedbed preparation, sowing of seeds and crop maintenance; and ends with crop harvest, storage and marketing.

9. Correct option: (B) Manure

Manure is obtained when the animal and plant waste are dumped by farmers in an open pit and leave it for days. The waste starts to decompose by the action of certain microorganisms, this process is named decomposition. Finally, the decomposed organic matter becomes Manure.

10. Correct option: (D) Both A and B

Air supplies two nutrients to plants namely carbon and oxygen. Water supplies hydrogen and soil supplies thirteen nutrient elements to the plants. Among these thirteen nutrients, six are macronutrients and the other seven nutrients are micronutrients.

11. Correct option: (B) Soil

Plants derive the required nutrients from the air, water, and soil. There are sixteen nutrients essential for the growth of plants. Carbon and Oxygen are supplied by water. The remaining thirteen nutrients are supplied by soil.

12. Correct option: (A) Macronutrients

Nutrients which are needed in large quantities are called macro nutrients.

13. Correct option: (B) Calcium

These are the most important elements required for crops and examples include nitrogen (N), potassium (K), calcium (Ca), phosphorus (P), magnesium (Mg), sulphur (S), oxygen (O), carbon (C), and hydrogen (H). Calcium is an essential micronutrient in plants. It serves as a constituent of cell walls and membranes and thus contributes to the structure of cells and the upholding of physical barriers against pathogens.

14. Correct option: (B) Copper

Copper (Cu) is one of eight essential plant micronutrients. Copper is required for many enzymatic activities in plants and for chlorophyll and seed production. Deficiency of copper can lead to increased susceptibility to diseases like ergot, which can cause significant yield loss in small grains.

15. Correct option: (B) Inorganic

It is commonly prepared by decomposing plant and food waste, recycling organic materials, and manure. The resulting mixture is rich in plant nutrients and beneficial organisms, such as bacteria, protozoa, nematodes, and fungi.

16. Correct option: (A) Vermicompost

Vermicomposting is a natural process whereby earthworms convert waste material with rigid structures into compost. The compost produced in this green process is traditionally and popularly used as a natural fertilizer for enhancing plant growth.

17. Correct option: (C) Fertilizers

Fertilizers supply nitrogen, phosphorus and potassium. They are used to ensure good vegetative growth (leaves, branches and flowers), giving rise to healthy plants. Fertilizers are a factor in the higher yields of high-cost farming.

18. Correct option: (D) All the above

Excess use of fertilizers causes environmental pollution as their residual and unused amounts will become pollutants for air, water, and soil. It also initiates the process of eutrophication in the water bodies, decreasing the soil quality by killing the soil microflora, etc.

19. Correct option: (C) 3

There are three types of wells: dug, driven, and drilled. Drilled are the most common today.

20. Correct option: (A) Mixed cropping

Growing two or more crops simultaneously on the same piece of land is known as mixed cropping. This lowers the risk and provides some protection if one of the crops fails. For example - Wheat with gram.

21. Correct option: (B) Inter- Cropping

The practice of growing two or more crops simultaneously in definite rows in the same field is called inter cropping.

22. Correct option: (B) Crop rotation

Growing different crops on a piece of land in pre-planned succession is known as crop rotation.

23. Correct option: (D) All the above

Weeds are unwanted plants which grow along with the crop plants in the cultivated field. Some common weeds in India are Amaranthus (Chaulai), Parthenium (Gajar ghas), Cyperus rotundus (Motha), Chenopodium (Bathua), Xanthium (Gokhroo). Weeds compete with the crop plants for nutrients, space, light and water.

24. Correct option: (A) Animal

Animal husbandry is a scientific management of livestock, breeding and rearing of farm animals.

25. Correct option: (D) Both A and C

Most of the Indian beekeepers engaged in beekeeping activity are primarily doing the extraction of raw honey from the bee colonies maintained by them. Very few beekeepers are extracting wax & pollen.

7. Natural Resources

Solutions

LEVEL-I

1. Correct option: (B) Lithosphere

Soil is a part of the lithosphere. The lithosphere refers to the solid outer layer of the Earth, which includes the crust and uppermost part of the mantle. Soil is formed from the weathering and erosion of rocks and minerals in the Earth's crust. It is a mixture of organic matter, minerals, water, and air, and it supports the growth of plants. Soil plays a crucial role in various ecological processes, such as nutrient cycling, water filtration, and habitat for organisms. Therefore, it is considered a vital component of the lithosphere.

It covers the solid bedrock and is crucial for supporting plant life and various terrestrial ecosystems. While the atmosphere (A) is the layer of gases surrounding the Earth, the hydrosphere (C) consists of all the water bodies on Earth's surface, and the ionosphere (D) is a region of the Earth's upper atmosphere.

2. Correct option: (A) Troposphere

The maximum amount of air that we breathe is present in the (A) Troposphere. The troposphere is the lowest layer of Earth's atmosphere, extending from the Earth's surface up to an average altitude of about 8-15 kilometers (5-9 miles) depending on your location and the season. This layer contains the majority of the Earth's atmospheric gases, including the oxygen we breathe. It is also where weather events, such as clouds, storms, and precipitation, occur. As you ascend into higher atmospheric layers like the stratosphere (B), ionosphere (C), or mesosphere (D), the concentration of breathable air decreases significantly, making the troposphere the region with the highest concentration of the air we need for respiration.

3. Correct option: (C) Material cycles

The correct answer is (C) Material cycles. Biogeochemical cycles, also known as material cycles, are processes that describe the movement and transformation of essential elements and compounds in the Earth's ecosystems. These cycles include the circulation of nutrients like carbon, nitrogen, phosphorus, and sulfur through various reservoirs such as the atmosphere, lithosphere (Earth's crust), hydrosphere (water bodies), and living organisms. These cycles are called "biogeochemical" because they involve biological, geological, and chemical processes. For example, the carbon cycle involves the exchange of carbon dioxide between the atmosphere and plants during photosynthesis, the transfer of carbon through the food web, and the release of carbon back into the atmosphere through processes like respiration and decomposition. These cycles are crucial for sustaining life on Earth and maintaining ecological balance.

4. Correct option: (A) Azotobacter

Azotobacter is a free-living nitrogen-fixing bacteria that is commonly found in the soil. Nitrogen fixation is the process by which atmospheric nitrogen is converted into a usable form by bacteria. Azotobacter plays a crucial role in this process as it has the ability to convert atmospheric nitrogen into ammonia, which can then be used by plants for their growth and development. Unlike other nitrogen-fixing bacteria like Rhizobium, Azotobacter does not form a symbiotic relationship with plants and can survive and function independently in the soil. This makes Azotobacter an important contributor to the nitrogen cycle and soil fertility.

Nitrosomonas is a nitrifying bacterium that is also found in the soil. It oxidizes ammonia into nitrite. Rhizobium is a symbiotic bacterium that lives in the nodules of leguminous

plants. It fixes atmospheric nitrogen into ammonia, which is then used by the plant. Pseudomonas is a denitrifying bacterium that reduces nitrate into nitrogen gas.

5. Correct option: (A) Photosynthesis

The atmosphere's CO2 and O2 balance is primarily caused by photosynthesis. Plants, algae, and some bacteria use sunlight, water, and CO2 during the process of photosynthesis to create glucose and release oxygen as a byproduct. Plants absorb CO2 from the atmosphere during photosynthesis and transform it into glucose, which is then used as a source of energy for growth and other metabolic processes. By returning oxygen to the atmosphere during this process, the O2 balance is improved. As a result, photosynthesis is essential for preserving the equilibrium between CO2 and O2 in the atmosphere.

Respiration, or Option B, on the other hand, reverses this process by releasing CO2 as organisms break down glucose to produce energy. This process happens in plants, animals, and even some microorganisms. The overall balance of these gases in the Earth's atmosphere is achieved by photosynthesis, which ultimately absorbs more CO2 than is produced by respiration. The balance of CO2 and O2 in the atmosphere is not directly maintained by options C and D, leaf anatomy and photorespiration.

6. Correct option: (D) Both A and B

The right response is (D) A as well as B. The process by which atmospheric nitrogen gas (N2) is transformed into ammonia (NH3) and nitrates is known as nitrogen fixation (NO3-). Since atmospheric nitrogen is largely inert and cannot be utilized directly by the majority of plants and organisms, it is an essential step in the nitrogen cycle. Bacteria that fix nitrogen are in charge of this conversion. They begin by converting N2 into ammonia (A), a form of nitrogen that plants can quickly absorb. Other soil bacteria can then use a process known as nitrification to further transform ammonia into nitrates (B). Both ammonia and nitrates are crucial elements of the nitrogen cycling in ecosystems because they are both significant sources of nitrogen for plants.

7. Correct option: (B) Afforestation

The appropriate response is (B) Deforestation. Afforestation is the intentional planting of trees and other vegetation where none previously existed. It is a successful method for halting soil erosion. Large-scale root systems of trees and other plants bind soil particles together, stabilizing the soil and minimizing wind and water erosion. They also serve as a natural defense against the effects of rain by reducing runoff and allowing water to seep into the soil. Options (A) Deforestation, (C) Overgrazing, and (D) Removing vegetation, on the other hand, all cause soil erosion. When trees are cut down for deforestation, the soil can erode more quickly because the root systems that stabilize the soil are lost. excessive grazing

8. Correct option: (D) Trees

A source of energy that can naturally and sustainably replenish itself over time is referred to as being renewable. The correct response in this instance is (D) Trees. Because they can be planted and grown again after being harvested for various purposes, trees are regarded as a renewable source of energy. By transforming their organic matter into fuel, heat, or electricity, trees can produce biomass energy. The carbon dioxide released during combustion is balanced by the carbon dioxide absorbed by new trees as they grow, making this process carbon-neutral. As a result, trees are essential for lowering greenhouse gas emissions and fostering the development of a more sustainable energy future.

A fossil fuel that forms over millions of years is petroleum. Another fossil fuel that forms over millions of years is coal. The supply of nuclear fuel is limited and will eventually run out.

9. Correct option: (B) 78.09%

(B) 78.09 percent is the right response. Nitrogen (N2) gas, which makes up about 78.09 percent of the atmosphere's volume, is the main component of the atmosphere of the Earth. This indicates that nitrogen molecules make up almost 45% of the air we breathe. About 20.95 percent of the atmosphere is made up of oxygen (O2), which is the second most abundant gas after carbon dioxide. The remaining traces of gases include argon, carbon dioxide, and water vapor. In a variety of scientific and environmental contexts, such as climate studies, weather forecasting, and air quality monitoring, an understanding of the composition of the Earth's atmosphere is crucial.

10. Correct option: (B) Stratosphere

The stratosphere contains the ozone layer. The second layer of the Earth's atmosphere, the stratosphere, is situated below the mesosphere and above the troposphere. It rises between 10 and 50 kilometers above the surface of the Earth. A region of the stratosphere with a significant concentration of ozone molecules is known as the ozone layer. Most of the ultraviolet (UV) radiation from the Sun is absorbed by ozone, a gas with three oxygen atoms (O3). The negative effects of excessive UV radiation, such as skin cancer, cataracts, and harm to marine ecosystems, are lessened thanks to this absorption. Therefore, preserving a habitable environment on our planet depends on the ozone layer's presence in the stratosphere.

11. Correct option: (A) Nitrogen fixing bacteria

The bacteria that fix nitrogen is the proper response (A). Plants that are legumes, such as beans, peas, and clover, have nodules in their roots that are home to bacteria that fix nitrogen, particularly Rhizobia. These nodules coexist harmoniously with the plants. Using a process known as biological nitrogen fixation, the nitrogen-fixing bacteria inside the nodules change atmospheric nitrogen (N2) into ammonia (NH3). The legume plants can then use this ammonia as a source of nitrogen for growth. Sugars and other nutrients are given to the bacteria in exchange by the plants. With the help of this mutualistic relationship, which benefits both the bacteria and the legume plants, the soil is enriched with necessary nitrogen compounds. Sulphur fixing bacteria (Option B) and potassium fixing bacteria (Option C) are not typically found in nodules but may have other roles in soil fertility.

12. Correct option: (B) CFC

The correct answer is (B) CFC (Chlorofluorocarbons). CFCs are primarily to blame for the ozone layer's thinning. When CFCs are released into the atmosphere, they eventually ascend to the stratosphere where UV rays degrade them. The ozone layer is destroyed as a result of this breakdown because chlorine atoms are released, which then interact with ozone (O3) molecules. By absorbing and blocking harmful ultraviolet (UV) radiation from the sun, the ozone layer in the stratosphere is essential for safeguarding life on Earth. The ozone layer's reduction increases the amount of UV radiation that reaches the surface of the Earth, increasing the risk of skin cancer, cataracts, and ecosystem damage. The ozone layer is not primarily depleted by either nitrogen dioxide (Option C) or oxygen (Option A). Option (D) is incorrect because it implies that oxygen and nitrogen dioxide are important causes of the ozone layer's thinning, which is untrue.

13. Correct option: (D) (A) and (C)

The correct answer is (D) (A) and (C). Nitric acid (HNO3) and sulfuric acid are the main components of acid rain (H2SO4). When nitrogen oxides and sulphur dioxide are released into the atmosphere as a result of industrial processes and vehicle emissions, these acids are created. Acid rain forms when these pollutants react chemically with atmospheric moisture to produce nitric acid and sulfuric acid, which then fall to the Earth's surface.

The environment may suffer from acid rain, which may harm buildings, aquatic ecosystems, and vegetation. Acid rain is therefore characterized by the mixture of nitric acid and sulfuric acid, making option (D) the right decision. Acid rain does not contain a significant amount of hydrochloric acid (Option B).

14. Correct option: (A) Biotic resources

The correct answer is (A) Biotic resources. In ecology and environmental science, biotic resources include both plants and animals. All species of plant and animal life are considered biotic resources because they are living or have previously been living organisms in an ecosystem. As they provide food, shelter, and a wide range of other goods and services that promote both ecological and human well-being, these resources are crucial for the continuation of life on Earth. Abiotic resources, on the other hand, are non-living components of the environment like minerals, water, air, and sunlight that are still significant but do not include living things. In contrast to plants and animals, which are considered natural resources, machines (Option C) are man-made tools and equipment. Therefore, in the context of ecology and resource management, the term "biotic resources" appropriately describes plants and animals.

15. Correct option: (A) Exhaustible resource

The correct answer is (A) Exhaustible resource. Due to its limited supply and lack of potential for further growth, coal is regarded as an exhaustible resource. It is made up of organic material that was heated and compressed underground over millions of years to form the fossilized remains of ancient plants. In a time, frame that is relevant to humans, coal cannot be replaced once it has been mined and burned for energy production. So long as sustainable practices are not used, continued coal use could eventually exhaust this resource by depleting it. Exhaustible resources, on the other hand, are those that are practically limitless and can be replenished naturally over comparatively short periods of time, such as solar or wind energy (Option B). Potential resources (Option C) refer to resources that are not yet fully developed or exploited but have the potential to become useful in the future, but coal is already a well-established and extensively used resource. Therefore, the classification of coal as an exhaustible resource is appropriate.

16. Correct option: (D) 11-16 km above equator

The correct answer is (D) 11-16 km above the equator. The Earth's stratosphere contains an area known as the ozonosphere, also referred to as the ozone layer, where ozone (O3) concentrations are relatively high. It is not evenly distributed but is mainly found around the equatorial regions, between 11 and 16 kilometers (roughly 7 to 10 miles) above the Earth's surface. In this layer, where ozone concentrations are highest, a significant portion of the sun's harmful ultraviolet (UV) radiation is absorbed and blocked, helping to protect life on Earth. Because of the high concentration of ozone at this altitude, the harmful UV rays are efficiently absorbed and filtered out before they reach the Earth's surface. Option (A) and Option (C) represent incorrect altitudes for the ozonosphere, while Option (B) describes a specific altitude range above the equator where the ozone layer is most concentrated.

17. Correct option: (B) Living beings, lithosphere, hydrosphere, and atmosphere

The correct answer is (B) Living beings, lithosphere, hydrosphere, and atmosphere. The entire area of Earth where life is present is included in the biosphere. It includes all types of living things, including plants, animals, and microorganisms, as well as how they interact with the physical elements of the environment on Earth. The lithosphere (the solid Earth's crust), the hydrosphere (all of Earth's water, including the oceans, rivers, and lakes), and the atmosphere serve as representations of these physical elements (the layer of gases surrounding the Earth). These components work together to form a

complex and interconnected system where life flourishes and is impacted by geological, hydrological, and atmospheric processes. In order to study ecology and the interactions between living things and their environment, it is crucial to have a thorough understanding of the biosphere as the aggregate of all living things and the Earth's physical systems.

18. Correct option: (B) Afforestation

The correct answer is (B) Afforestation. Afforestation, which entails the purposeful planting of trees and vegetation in areas where they did not previously exist, is an efficient way to prevent and mitigate soil erosion. In order to prevent soil erosion brought on by wind and water, trees and other plants with deep root systems stabilize the soil by holding it together. They also serve as a natural blockade against rainfall, slowing water runoff and enabling it to seep into the soil. Options (A), (C), and (D), on the other hand, all contribute to soil erosion: deforestation, overgrazing, and vegetation removal. Deforestation involves the cutting down of trees, which can cause increased soil erosion because the roots are lost. Overgrazing and the removal of vegetation reduce the protective cover of plants, making the soil more susceptible to erosion. Therefore, afforestation is a key strategy for preventing soil erosion and maintaining soil health.

19. Correct option: (A) Water

The correct answer is (A) Water. Due to its role in the hydrological cycle, which sees water moving continuously between the atmosphere, land, and oceans through processes like evaporation, condensation, and precipitation, water is thought of as a renewable resource. Water becomes renewable as a result of this cycle, which ensures a steady supply over time. Metals (Option B), on the other hand, are typically regarded as non-renewable resources due to the fact that they are extracted from finite mineral deposits and are only occasionally available. Coal, oil, and natural gas are examples of fossil fuels (Option C), which are non-renewable resources because they were created over a long period of time—millions of years—and cannot be replenished. Water, a sustainable and replenishable resource used in Earth's natural processes, is the right response, and it is (A).

20. Correct option: (D) CO2

The correct answer is (D) CO2 (carbon dioxide). The main greenhouse gas causing climate change and global warming is carbon dioxide. CO2 builds up in the atmosphere and traps solar heat when human activities like the burning of fossil fuels (like coal, oil, and natural gas) and deforestation release it into the atmosphere. The greenhouse effect, a process that causes an increase in global temperatures, is the primary cause of global warming. Rising sea levels, more frequent and severe weather events, and disruptions to ecosystems and biodiversity are all effects of the atmosphere's increasing concentrations of CO2 and other greenhouse gases. In order to address the problem of global warming, efforts must be made to reduce CO2 emissions and lessen their effect on the climate.

LEVEL - II

1. Correct option: (A) Lithosphere

The correct answer is (A) Lithosphere. The rigid, outermost layer of the Earth is called the lithosphere. It includes the solid portion of the Earth, which is made up of the continents and the ocean floor. The Earth's crust and the topmost layer of the mantle make up the lithosphere, which is relatively thin in comparison to the other layers of the planet. It is divided into a number of sizable tectonic plates, each of which floats on the asthenosphere beneath it. Earth's land masses move and interact with these tectonic plates, causing natural events like earthquakes, volcanic eruptions, and the formation of

mountain ranges. Therefore, the lithosphere is an essential part of the Earth's structure and has a major impact on the geology and landscape of the planet.

2. Correct option: (B) Atmosphere

The correct answer is (B) Atmosphere. The layer of gases that envelops the Earth is known as the atmosphere. It is made up of a variety of gases, including carbon dioxide, nitrogen, and others. The air we breathe and the climate and temperature of the planet are both controlled by the atmosphere, which is essential to the survival of life on Earth. The troposphere, which is closest to the surface of the Earth, is one of its many distinct layers, along with the stratosphere, mesosphere, and thermosphere. Each layer has special properties and roles. The atmosphere also shields the planet from space debris and harmful solar radiation. It is a crucial part of the environment on Earth because it affects weather patterns, air circulation, and the overall ecosystem balance on our planet.

The lithosphere is the solid, outer layer of the Earth. It is made up of the crust and the upper mantle. The hydrosphere is the water on, above, and below the Earth's surface.

3. Correct option: (D) Interact between Lithosphere, Hydrosphere and Atmosphere.

The correct answer is (D) Interact between Lithosphere, Hydrosphere, and Atmosphere. The biosphere, which includes interactions between the lithosphere (the solid Earth's crust), the hydrosphere (all of the Earth's water bodies, including oceans, rivers, and lakes), and the atmosphere, is the region of the planet where life is present (the layer of gases that surrounds the Earth). Living things, such as plants, animals, and microorganisms, interact with these natural elements as well as with one another within the biosphere to build intricate ecosystems and support life. Processes like the water cycle, nutrient cycling, and the exchange of gases required for respiration and photosynthesis are examples of these interactions. The relationship between life and the environment of the Earth is intimate in the biosphere, which is a dynamic and interconnected system.

4. Correct option: (D) All the above

The correct answer is (D) All the above. In fact, air is a mixture of gases, with nitrogen and oxygen making up the majority (about 78 percent) of it (about 21 percent). However, it also includes small amounts of a number of other gases, such as carbon dioxide, argon, water vapor, and others, in trace amounts. While nitrogen and oxygen make up the majority of the atmosphere, other gases called trace gases are crucial for a number of environmental processes as well as maintaining the general makeup and characteristics of the atmosphere. For instance, carbon dioxide is essential to the greenhouse effect and the control of the climate. Therefore, option (D) is the appropriate decision because air is in fact a mixture of gases that includes nitrogen, oxygen, carbon dioxide, and more.

5. Correct option: (D) Both B and (C)

The correct answer is (D) Both B and (C). Green plants use sunlight and chlorophyll (Option B) to convert carbon dioxide (CO_2) into glucose and other organic compounds during photosynthesis (Option C). Chlorophyll is a green pigment that is present in plant cells, especially in chloroplasts, and it is essential for absorbing solar light energy. The chemical reactions that turn carbon dioxide and water into glucose are then propelled by this energy, which also produces oxygen as a byproduct. The energy needed for this process to take place is provided by sunlight, and chlorophyll serves as the light-absorbing molecule that starts the process of converting CO_2 into glucose through a series of intricate biochemical reactions known as photosynthesis. Therefore, the process of photosynthesis in green plants requires both sunlight and chlorophyll to function.

6. Correct option: (A) Air

The answer is (A) Air. Due to its poor suitability as a medium for the conduction of thermal energy, air is a poor heat conductor. This is mainly because air molecules do not have a strong atomic or molecular structure that would allow them to effectively transmit heat from one molecule to another, and they are also spread out fairly widely. Water, on the other hand, is a better conductor of heat through conduction because of its higher density and more ordered molecular structure. Due to the fact that air conducts heat poorly when compared to water or other materials, it is frequently used as an insulating material to trap heat or stop it from moving.

7. Correct option: (B) Heated

The answer is (B) Heated. Due to variations in their thermal characteristics, land warms up more quickly than the sea during the day. Land requires less energy to raise its temperature than water because it has a lower specific heat capacity. In addition, land is frequently darker and absorbs more sunlight, whereas water reflects a large amount of sunlight. As a result, during the day, the land's surface temperature rises more quickly, whereas the sea warms up more slowly due to its greater heat capacity and reflective qualities. The contrasting rates of heating and cooling between these two surfaces are what cause the temperature difference between land and sea, which helps to develop winds and regional weather patterns like sea breezes.

8. Correct option: (B) Evaporation

The correct answer is (B) Evaporation. A large amount of water evaporates when water bodies, such as lakes, rivers, and oceans, are heated by the sun's energy during the day. Water changes from a liquid state to a vapor or gas state during evaporation as a result of heat energy input. The water's molecules gain enough energy from the heat it absorbs to break their bonds and turn into water vapor, which rises into the atmosphere. The water cycle on Earth is fundamentally based on this process. When the conditions are right, the water vapor may eventually cool and condense to form clouds, which may then result in precipitation (Option C) in the form of rain, which replenishes the Earth's surface with water. So, evaporation is the key process responsible for the rising of water vapor from heated water bodies during the day.

9. Correct option: (A) Night

The correct answer is (A) Night. The sea cools down at a slower rate at night than the land does. This phenomenon results from the distinction between the heat-retention and -absorption capacities of water and land. Seawater has a higher heat capacity than other types of water, which allows it to absorb and hold onto heat for longer. As a result, the sea continues to release heat into the atmosphere long after the sun has set, maintaining the atmosphere's relatively constant temperature. Land, on the other hand, has a lower heat capacity and cools down more quickly at night as it rapidly loses heat. Because of the disparity in cooling rates between land and sea, temperature contrasts are created that may have an impact on local weather conditions, such as the development of land breezes at night when the land cools faster than the sea, causing air to move from the sea towards the land.

10. Correct option: (A) Condensation

The answer is (A) Condensation. As soon as water droplets are created, condensation causes them to enlarge. When water vapor in the air loses heat energy and changes into liquid water droplets, condensation takes place. This process, where water vapor condenses onto tiny particles known as cloud condensation nuclei to form visible water droplets, is frequently observed in clouds. These droplets enlarge and eventually get heavy enough to fall as precipitation like rain or snow as they continue to condense.

Condensation is the primary mechanism behind the expansion of water droplets in the atmosphere, which causes clouds to form and then precipitation to follow.

11. Correct option: (B) Rain

The correct answer is (B) Rain. Rain is created when water vapor in the atmosphere weighs enough to fall to the ground. When the teeny water droplets in a cloud combine and coalesce into larger drops that are heavy enough to overcome the upward air currents and gravity pulls them down, rain is a type of liquid precipitation. The replenishment of freshwater resources like rivers, lakes, and groundwater by rainfall is a crucial part of the Earth's water cycle because it supplies the moisture required for plant growth and ecosystem support. Weather and climate patterns are significantly influenced by rainfall patterns, which have an impact on both local and global climates. Rainfall therefore plays a crucial role in the distribution of water on Earth and has significant ecological and environmental effects.

12. Correct option: (B) Sulphur

The correct answer is (B) Sulphur. Small amounts of sulphur are present in fossil fuels like coal and petroleum. These fuels contain sulphur compounds that are released into the atmosphere as sulphur dioxide (SO2) and other sulphur oxides when they are burned to produce energy. When sulphur dioxide reacts with atmospheric moisture to form acid rain, these sulphur emissions can have negative effects on the environment, including air pollution. To lessen these negative environmental effects brought on by the combustion of fossil fuels, efforts to reduce sulphur emissions have been made, including the use of cleaner technologies and the development of low-sulfur fuels. In light of this, even though carbon and hydrogen make up the majority of fossil fuels, other elements are frequently present in trace amounts, including sulfur, which can have significant environmental consequences when released into the atmosphere.

13. Correct option: (A) Water

The correct answer is (A) Water. Oceans, seas, lakes, rivers, and other bodies of water cover a very large portion of the Earth's surface. In actuality, water covers about 71 percent of the Earth's surface. The majority of this water—roughly 97.5 percent of the water on Earth—is contained in the oceans alone. Our planet's basic element, water, is crucial to maintaining life as we know it. It is essential to many natural processes, such as the water cycle, regulating the climate, and providing habitat for a wide variety of species. The Earth's abundance of water is one of the planet's defining features and has a significant impact on its geology, climate, and ecosystems.

14. Correct option: (B) Water pollution

The answer is (B) Water pollution. Water pollution is the term used to describe the rise in the concentration of harmful substances in the water. When contaminants or pollutants, such as sewage, heavy metals, pesticides, or other dangerous substances, are introduced into bodies of water, such as rivers, lakes, oceans, or groundwater in quantities that are greater than natural levels or safe limits, water pollution results. These pollutants have the potential to harm human health, aquatic ecosystems, and the environment as a whole. In order to ensure the availability of clean, safe water for human consumption as well as the wellbeing of aquatic life, efforts are made to monitor, prevent, and mitigate water pollution. Water pollution is a major environmental concern around the world.

15. Correct option: (C) Both A and B

The answer is (C) Both A and B. Using pesticides and fertilizers too frequently can contaminate the water in farms. Pesticides can contaminate water and damage aquatic ecosystems when they are applied in excess or improperly, causing them to leach into the soil and eventually find their way into nearby water bodies through runoff. Similar to

this, using too much fertilizer can result in nutrient runoff, which is the process by which nutrients like nitrogen and phosphorus from fertilizers are washed into rivers, lakes, or streams. This may cause water pollution and contribute to issues like eutrophication, in which an abundance of nutrients causes harmful algal blooms and the depletion of oxygen in aquatic ecosystems. Therefore, when used improperly, pesticides and fertilizers can both cause water pollution on farms.

16. Correct option: (C) Both A and (B)

The answer is (C) Both A and (B). Both (A) the use of fertilizers and pesticides in farms and (B) the presence of sewage water can cause water pollution in villages. Through runoff, excessive or improper use of fertilizers and pesticides in agricultural practices can contaminate nearby water sources with chemicals that are harmful to aquatic life and human health. However, sewage water, which contains a variety of contaminants from domestic and human sources, can enter rivers, ponds, or other bodies of water if it is not properly treated or managed, which adds to water pollution. Therefore, as suggested by option, agricultural runoff and untreated sewage can combine to cause water pollution in rural areas (C).

17. Correct option: (C) Lichens

The answer is (C) Lichens. Unique living things known as lichens are created when fungi and algae or cyanobacteria form a cooperative relationship. They are renowned for their capacity to flourish on rocks and other surfaces regardless of the hostile surroundings. Acid-like substances are metabolic waste products produced by lichens. These acids, especially oxalic acid, can weather the rock surfaces on which they settle. Physical and chemical weathering are processes that occur over time as a result of the gradual buildup and action of these acids, which results in the surface of the rock disintegrating into smaller particles. Lichens are crucial pioneers in the formation of soil in areas with little vegetation because this deterioration of the rock surface can eventually result in the formation of soil.

18. Correct option: (B) Humus

The dark, organic material in soil known as humus is created when plant and animal matter break down. It is nutrient-rich and enhances the soil's capacity to hold onto nutrients and water. The Latin word "humus," which means "earth" or "ground," is the source of the English word "humus."

Humus is primarily made from decomposing organisms. When plants and animals pass away, bacteria, fungi, and other microorganisms decompose their remains. This procedure returns nutrients to the soil so that new plants can utilize them.

The ecosystem of the soil depends on humus. It aids in enhancing the soil's fertility, water retention, and structure. Additionally, it aids in reducing pests and plant diseases.

19. Correct option: (C) Soil pollution

The answer is (C) Soil pollution. The addition of harmful substances to the soil that have a negative impact on its fertility and general health is referred to as soil pollution. Industrial chemicals, heavy metals, pesticides, herbicides, and other pollutants are examples of these harmful substances. These pollutants can damage beneficial soil organisms, contaminate groundwater, and disrupt the natural balance of nutrients in the soil, all of which can reduce soil fertility and productivity. The introduction of harmful substances into the soil must be avoided and minimized because it can have a negative impact on agriculture, ecosystems, and human health. Soil pollution is a major environmental concern.

20. Correct option: (A) Soil erosion

The answer is (A) Soil erosion. The natural process of soil movement from one place to another brought on by the action of outside forces like water and wind is referred to as soil erosion. Wind erosion involves the lifting and transport of soil particles by strong

winds, whereas water erosion happens when rainfall or flowing water removes and carries away soil particles from the surface of the land. Human activities like deforestation, improper land use, and unsustainable agricultural practices can hasten soil erosion, which results in the loss of fertile topsoil, lowered soil quality, and environmental degradation. It poses a serious problem for agriculture and land management because it may have a negative impact on ecosystem sustainability and crop productivity.

21. Correct option: (B) Soil erosion

The answer is (B) Soil erosion. A good way to stop soil erosion is through afforestation, which involves planting trees and creating forests in places that had little to no previous tree cover. Because their roots bind soil particles together, trees and other vegetation are essential for stabilizing soil and preventing erosion. This reduces the effects of water and wind. They serve as organic barriers that slow down water flow and aid in capturing sediment to keep it from being washed away. Additionally, the tree canopy deflects raindrops, lessening their impact on the soil's surface. As a result, afforestation is a crucial strategy to prevent soil erosion and maintain the productivity and health of the soil in vulnerable areas.

22. Correct option: (B) Proteins

The answer is (B) Proteins. Through their roots, plants absorb nitrogen compounds from the soil, primarily in the form of nitrate ions (NO3-) and ammonium ions (NH4+). Nitrogen is a crucial nutrient for protein synthesis and is necessary for plant growth and development. Nitrogen is incorporated into amino acids, the building blocks of proteins, inside plant cells. These amino acids are then used to create a variety of proteins, such as transport proteins, structural proteins, and enzymes, which are essential for the structure, operation, and general growth of the plant. As a result, one of the primary uses of nitrogen in plant biology is the synthesis of proteins, which is accomplished primarily by the nitrogen compounds absorbed by plants.

23. Correct option: (A) Glucose

The answer is (A) Glucose. During the process of photosynthesis, plants use carbon from carbon dioxide (CO2) to create glucose and other carbohydrates. During photosynthesis, plants draw carbon dioxide from the atmosphere and use the energy from sunlight to catalyze a series of chemical reactions that turn it into glucose. A type of carbohydrate called glucose is the main source of energy for plants. Additionally necessary for plant development, it acts as the foundation for other organic compounds like starch and cellulose. Photosynthesis is crucial for maintaining life on Earth because it not only enables plants to store energy but also produces oxygen as a byproduct.

24. Correct option: (C) Respiration

The answer is (C) Respiration. The process of respiration is the primary use of oxygen in the Earth's atmosphere by all living things, including people, animals, and even plants. In order to release energy for different life processes, organisms use oxygen to break down organic molecules like glucose during respiration. This energy is necessary for cellular functions, growth, and upkeep. Carbon dioxide (CO2) is created during respiration as a waste product and is then released back into the atmosphere. All aerobic organisms must engage in the fundamental process of respiration in order to obtain the energy necessary for survival and daily activities.

25. Correct option: (B) Carbon

The answer is (B) Carbon. The majority of substances that deplete the ozone layer are referred to as ozone-depleting substances (ODS), and many of these substances, particularly chlorofluorocarbons (CFCs) and other halogenated hydrocarbons, contain carbon compounds. These ODS are released into the atmosphere, eventually ascending

to the stratosphere where ultraviolet (UV) radiation breaks them down. Chlorine and other halogen atoms are released as a result of this breakdown, and they go on to catalytically destroy ozone molecules in the ozone layer. The ozone layer thins due to ozone depletion, especially in places like the ozone hole over Antarctica. It's important to note that ODS are not limited to carbon compounds; nitrogen compounds, such as nitrous oxide (N2O), can also contribute to ozone layer damage, but carbon-containing ODS are some of the most potent contributors to ozone depletion. Efforts have been made to regulate and phase out the production and use of ODS to protect the ozone layer.

8. Heredity and Evolution

Solutions

LEVEL-I

1. Correct option: (B) Responsible for character

The alleles are responsible for character. Mendel studied peas and deduced that one is responsible for structure, the other for colour, and so on. All these are characters that are shown by alleles.

2. Correct option: (D) Both A & B

Natural selection is the process through which populations of living organisms adapt and change. Individuals in a population are naturally variable, meaning that they are all different in some ways. This variation means that some individuals have traits better suited to the environment than others.

3. Correct option: (B) Fossil evidence

A palaeontologist is a scientist who studies the history of life on earth. They study the history of life on earth with the help of fossil records as fossils are the evidence of past life on the planet and they can include animal bodies or their imprints.

4. Correct option: (C) Mendel

Gregor Johann Mendel is known as the father of genetics.

5. Correct option: (D) 22

In humans, each cell normally contains 23 pairs of chromosomes, for a total of 46. Twenty-two of these pairs, called autosomes, look the same in both males and females.

6. Correct option: (A) Lamarck

Inheritance of acquired characters is the once widely accepted idea that physiological modifications acquired by an organism can be inherited by the offspring. Lamarck presented the idea of the inheritance of acquired characters as self-evident.

7. Correct option: (D) All of the above

Vestigial organs can be considered as evidence of evolution because vestigial structures are often homologous to structures that are functioning normally in other species.

Homologous organs are organs which are similar in structure and developmental origin but perform different functions. For example, a human arm and a bat's wing. These homologous organs point towards a common ancestor from which these organisms evolved. Hence it is considered as evidence of evolution.

Organs which have different origin and dissimilar fundamental structure but have similar functions are called analogous organs. This means that analogous organs do not show common ancestor though they show convergent evolution. Hence analogous organs are considered as evidence of evolution.

Hence, homologous organs, analogous organs and vestigial organs is a set of evidence of evolution.

So, the correct answer is 'All of the above'.

8. Correct option: (B) Darwin

Charles Darwin was a British naturalist who proposed the theory of biological evolution by natural selection. Darwin defined evolution as "descent with modification," the idea that species change over time, give rise to new species, and share a common ancestor.

9. Correct option: (C) Pea plants

Mendel carried out his key experiments using the garden pea, Pisum sativum, as a model system. Pea plants make a convenient system for studies of inheritance, and they

are still studied by some geneticists today. Useful features of peas include their rapid life cycle and the production of lots and lots of seeds.

10. Correct option: (A) Male

The human being with XY pair of chromosomes is called a male. The sex of the human offspring is determined by the set of chromosomes inherited from father.

11. Correct option: (D) Part of a chromosome that transmits a trait

A gene is the basic physical and functional unit of heredity. Genes are made up of DNA. Some genes act as instructions to make molecules called proteins. Gene is a part of a chromosome that transmits a trait.

12. Correct option: (C) Phenotype

Visible characteristics of an organism are called phenotype.

13. Correct option: (A) One

The number of pair(s) of sex chromosomes in the zygote of humans is one.

14. Correct option: (A) Copies of the same chromosome

The two versions of a trait are brought in by the male and female gametes situated on copies of the same chromosome. During fertilization one of the chromosomes comes from the father through the male gamete and the other chromosome comes from the mother through the female gamete.

15. Correct option: (B) Mendel

Gregor Johann Mendel is known as the father of genetics.

16. Correct option: (A) Homologous organs

The organs that perform different functions but have the same basic structure are known as homologous organs. For example, forelimbs of humans and lizards are homologous.

17. Correct option: (B) Analogous organs

The organs which perform a similar function, but have different basic structures are called analogous organs. For example, wings of insects and birds are analogous.

18. Correct option: (A) Analogous organs

Wings of an insect and forelimbs of a bird are analogous organs, as they have different structures but perform the same function of flying.

19. Correct option: (C) S Miller

In 1953, Stanley L. Miller and Harold C. Urey, working at the University of Chicago, conducted an experiment which would change the approach of scientific investigation into the origin of life. Miller took molecules which were believed to represent the major components of the early Earth's atmosphere and put them into a closed system

20. Correct option: (D) Convergent evolution

The bodies of whales, seals, and sharks are streamlined and smooth. Whales, seals, and sharks have a single tail fin which is used for propulsion. These similar methods in whales, seals, and sharks for locomotion are examples of convergent evolution.

21. Correct option: (A) Possible origin of life by the combination of chemicals under suitable environmental conditions

During the course of evolution, the reducing environment of earth favoured the accumulation and reaction of elements to form bioorganic compounds which then formed the building blocks of life.

22. Correct option: (C) Australia

Every surviving marsupial is endemic to Australasia and the Americas. An unmistakable trademark normal to these species is that the majority of the youthful are conveyed in a pouch.

23. Correct option: (B) Homology

Homologous organs: The organs of different species of common descent which look different and perform different functions, but have the similar basic structure, similar topographic relationship, and similar embryonic origin.

24. Correct option: (A) i – b, ii – a, iii – d, iv – c

(i) Darwin - b) Origin of species

Charles Darwin which is considered to be the foundation of evolutionary biology is the author of 'The Origin of Species'. Darwin's book introduced the scientific theory that populations evolve over the course of generations through a process of natural selection.

ii) Lamarck - a) Use and disuse theory

The theory of use and disuse was proposed by Lamarck. He suggested that changes in traits or characters occur in organisms depending on whether an organ is used or disused continuously

iii) Hugo de Vries - d) Mutation theory

The mutation Theory of Evolution was proposed by Hugo de Vries.

iv) A. S. Oparin - c) Origin of life

Oparin believed that the life developed from microscopic spontaneously formed spherical lipid molecules held together by electrostatic forces, probably the earliest form of cells. These molecules most likely functioned as enzymes, essential for the biochemical metabolic reactions necessary for life's evolution.

25. Correct option: (B) Biogenetic law → Recapitulation theory

Biogenetic law, also called Recapitulation Theory, postulation, by Ernst Haeckel in 1866, that ontogeny recapitulates phylogeny—i.e., the development of the animal embryo and young traces the evolutionary development of the species.

26. Correct option: (A) Mesozoic

Amid the Mesozoic Era, life broadened quickly, and monster reptiles, dinosaurs, and different tremendous brutes meandered the Earth. The period, which ranges from around 252 million years back to around 66 million years prior, was otherwise called the time of reptiles or the time of dinosaurs.

27. Correct option: (B) Mutations are random and directionless

Mutations are random changes taking place at the level nucleotide sequence in DNA. Mutation is directionless means that nobody knows if it might result in something beneficial for the organism or harmful.

28. Correct option: (B) Analogous

The potato underground stem has evolved, allowing it to store food in the form of starch. Sweet potato roots, on the other hand, have been modified for the same purpose. The edible parts are the modified potato stem and the modified sweet potato root.

29. Correct option: (B)Oparin and Haldane

According to chemical theory of origin of life proposed by Oparin and Haldane, there formed the evolution of chemical substances that preceded the actual origin of life.

30. Correct option: (A) Charles Darwin

Darwin applied the phrase “survival of fittest” to refer to the unequal ability of individuals to survive and reproduce based on suitability of their inherited characters to prevailing environmental conditions. Lamarck viewed the inheritance of acquired characters as the basis of evolution.

31. Correct option: (D) All the above

The classical example of adaptive radiation during formation of new species is marsupials of Australia, Darwin's finches and giant Tortoise.

32. Correct option: (C) Darwin

English naturalist Charles Darwin developed the idea of natural selection after a five-year voyage to study plants, animals, and fossils in South America and on islands in the Pacific. In 1859, he brought the idea of natural selection to the attention of the world in his best-selling book, On the Origin of Species.

33. Correct option: (A) Both 'A' & 'R' are true, and 'R' is the correct explanation of 'A'

Assertion is correct, as natural selection is the outcome of difference in survival and reproduction among individuals that show variation in one or more traits. It is a key mechanism of evolution, the change in heritable traits of a population over time.

Reason is also correct and correct explanation of assertion as adaptive forms of a given trait tend to become more common, less adaptive ones become less common or disappear. The unequal ability of individuals to survive and reproduce will lead to a gradual change in population with favourable characteristics accumulating over the generation, which leads to the formation of new species.

34. Correct option: (C) Theory of mutations

The theory of mutation given by Hugo de Vries suggested that evolution is fuelled by the occurrence of the discontinuous process called mutation, that arises in the genetic material suddenly and are transmitted to the next generation.

35. Correct option: (A) Homologous organs

Tendrils of Cucurbita and thorns of Bougainvillea are homologous organs because they are similar in origin as they are modified branches and axial in position. Their different function is modified for protection and in Cucurbita, tendrils are for climbing.

36. Correct option: (D) All the above

Synthetic theory is considered as Modern theory and the main contributor of this theory was G.L.Stebbins. They considered 5 factors, responsible for organic evolution. They are as follows 1. Mutation 2. Genetic recombination 3. Chromosomal aberration 4. Reproductive isolation 5. Natural Selection

37. Correct option: (C) Adaptive radiation

Adaptive radiation is a developmental example described by a fast increment in the number of sorts of firmly related species.

38. Correct option: (C) Theory of natural selection

The theory of evolution by natural selection, first formulated in Darwin's book "On the Origin of Species" in 1859. It is the process by which organisms change over time as a result of changes in heritable physical or behavioural traits.

39. Correct option: (D) Darwin

On the Origin of Species', published on 24th November 1859, is a work of scientific literature by Charles Darwin which is the foundation of evolutionary biology. Its full title was 'On the Origin of Species by Means of Natural Selection or the Preservation of Favoured Races in the Struggle for Life'. Darwin's book introduced the scientific theory that natural selection and presented a body of evidence gathered from the Beagle expedition, stating that the diversity of life descended from a common ancestor through evolution.

40. Correct option: (D) Stanley Miller

In 1953 American chemists Harold C. Urey and Stanley Miller tested the Oparin-Haldane theory and successfully produced organic molecules from some of the inorganic components thought to have been present on prebiotic Earth. This became known as the Miller-Urey experiment.

LEVEL II

1. Correct option: (A) Cabbage

Wild cabbage is the common ancestor of broccoli, kohlrabi and kale.

2. Correct option: (B) Africa

According to the paleontological evidence, the fossils of the earliest man, Homo naledi was found in the regions of South Africa. The human species has genetic roots in Africa.

3. Correct option: (D) All

Excavating, time-dating and studying fossils, as well as determining DNA sequences – have been used for studying human evolution.

4. Correct option: (B) Big nose

Big nose is an inherited trait which can be passed from parents to offspring.

5. Correct option: (D) fossils

The remains (or impressions) of dead animals or plants that lived in the remote past are known as fossils.

6. Correct option: (B) dinosaur and fish (Knightian)

Fish (Knightian) and dinosaurs are now extinct and are studied from their fossils.

7. Correct option: (A) weight of a person

Height of a person depends on genes, that's why Our height depends on height of our parents. Genes also control many other things like colour of eye, body colour can also be inherited. Weight is acquired during life due to factors including lifestyle.

8. Correct option: (B) Size of body

Body size as it can change during the lifetime of a person (an individual may become fat or thin during his or her lifetime) but it cannot be passed to the next generation i.e., it is not inheritable.

9. Correct option: (B) self-pollination

Two pink coloured flowers on crossing resulted in 1 red, 2 pink, and 1 white flower progeny. The nature of the cross will be self-pollination.

10. Correct option: (A) 50%

Since the ratio of X chromosome and those containing Y chromosome is 50:50 in a male gamete, the statistical probability of male or a female offspring also becomes 50:50. In human beings, the statistical probability of getting either a male or female child is 50: 50.

11. Correct option: (B) girl

Father can donate either X or Y to the ovum with X chromosome, if zygote is of XX each from father and mother respectively, then the child must be a girl with XX chromosome.

12. Correct option: (D) Natural selection can alter frequency of an inherited trait.

The fourth statement, natural selection can alter the frequencies of an inherited character, is true with respect to theories of evolution and is not a direct conclusion of Mendelian experiments.

13. Correct option: (D) All

All of them can be called as characteristics.

14. Correct option: (A) Gall bladder

Because the gallbladder can be removed from humans without much effect and is not present in many animals, it is sometimes mistakenly classified as a vestigial organ—a body part that has lost all or most of its original function through evolution.

15. Correct option: (B) 9: 3: 3: 1

Mendel's cross on pea plants produced an observable ratio of 9: 3: 3: 1 as a result of the independent inheritance of two distinct traits, seed shape, and colour.

16. Correct option: (A) Chromosomes get halved during gamete formation

The number of chromosomes in parents and offsprings of a particular species remains constant due to the halving of chromosomes during gamete formation.

17. Correct option: (D) Equal chance of acquiring blood group A or blood group O.

The given information is not enough to determine the traits- blood group A or O is dominant.

In ABO blood grouping type, A is always dominant whereas, the blood type O is always recessive.

Here the father's blood group is A hence, it can either be homozygous (AA) or heterozygous (AO).

The mother's blood group is O and it is only possible if it is heterozygous (OO).

For the daughter to have blood group O the father must be heterozygous (AO) and mother must be (OO). but there are equal chances for acquiring blood group A and O.

18. Correct option: (B) Yeast fungus

Yeast fungus does not reproduce by spore formation method.

19. Correct option: (C) Genetic drift does not involve competition between members of a species.

Genetics drift is a sudden and random process, it may cause loss of some significant and vital genes in the process of evolution whereas natural selection is a non-random process due to which there is always the survival of the fittest and useful genes are added up in the process of evolution.

20. Correct option: (C) TtWW

A Mendelian experiment consisted of breeding tall pea plants bearing violet flowers with short pea plants bearing white flowers. The progeny all bore violet flowers, but almost half of them were short. This suggests that the genetic make-up of the tall parent can be depicted as TtWW.

21. Correct option: (D) Overproduction of offspring

22. Correct option: (A) A Chinese boy

We have more in common with a Chinese school-boy because both belongs to the same species of Homo sapiens, and thus, share a common ancestor.

23. Correct option: (D) birds have evolved from reptiles

Here, evolution means that birds have evolved from reptiles both mammals and birds evolved from reptiles, though not the same ones - mammals evolved from the mammal-like reptiles, whilst birds evolved from dinosaurs.

24. Correct option: (B) Round and yellow

If a round, green seeded pea plant (RRyy) is crossed with wrinkled, yellow seeded pea plant, (rrYY) the seeds produced in F1generation are round and yellow.

Parents: RRyy X rrYY

Gametes: Ry rY

F1: RrYy (Round yellow)

25. Correct option: (A) two individuals of a species

Species is the lowest level of the taxonomic classification hierarchy and the members of the species share the most common characteristics with themselves. Thus, any two individuals of a species share the maximum numbers of related or common characters.

26. Correct option: (C) All variations in a species have equal chance of survival

To survive, nature selects the most adaptable variant. This is Darwin's theory of Natural selection.

27. Correct option: (B) accumulation of variations over several generations

According to the evolutionary theory, the formation of a new species is generally due to accumulation of variations over several generations.

28. Correct option: (D) the extinction of organism has occurred thousands of years ago

If the fossil of an organism is found in the deeper layers of the Earth, then we can predict that the extinction of organisms occurred thousands of years ago.

29. Correct option: (A) Only 1

1. Weight of a person

The correct answer is weight of a person. Height of a person depends on genes, that's why Our height depends on height of our parents.

30. Correct option: (C) (iii) and (iv)

These unpaired chromosomes are: Y-chromosome and. X-chromosome.

9. Control and Coordination

Solutions

LEVEL-I

1. Correct option: (C) 24

Every mammal, including man, has 24 cranial nerves. Cranial nerves are a group of nerves that leave the brain immediately and are in charge of carrying sensory and motor information to and from the area of the head and neck. The cranial nerves in mammals, including humans, come in pairs of 12. Each pair performs a separate function, such as regulating taste and smell sensations, transferring visual information, directing facial expressions, and aiding eye and tongue movement. The total number of cranial nerves in a mammal, including man, is 24, and they all serve a significant role in preserving the proper operation of numerous sensory and motor functions in mammals.

2. Correct option: (D) Hypothalamus

The midbrain and the cerebral hemispheres are separated by a region of the brain called the diencephalon. The thalamus and hypothalamus are two of its many components. Particularly, the diencephalon and the hypothalamus share a strong relationship. It is a small but significant area that regulates numerous biological processes and preserves equilibrium. The hypothalamus regulates the pituitary glands hormone secretion, which in turn affects the actions of the body's other endocrine glands. In addition, it aids in the regulation of emotions, appetite, thirst, and sleep-wake cycles. Therefore, in terms of its role and significance in preserving general physiological balance, the diencephalon is intimately related to the hypothalamus.

Instead of the diencephalon, the corpora quadrigemina is a component of the midbrain. Instead of the diencephalon, the basal ganglia are a collection of nuclei found in the cerebrum. The pons is a part of the hindbrain, which is located below the diencephalon.

3. Correct option: (B) Nodes of Ranvier

The Nodes of Ranvier are the gaps in the myelin sheath. The myelin sheath, which surrounds and insulates the axons of nerve cells, is a protective coating. It facilitates a faster and more effective passage of nerve impulses. The myelin coating does not, however, extend the full length of the axon. The myelin sheath experiences irregular breaks known as Nodes of Ranvier. These nodes are essential for the transmission of nerve impulses. They enable electrical signals to move swiftly from one node to the next, hastening the movement of the nerve impulse along the axon. This saltatory conduction provides effective communication between nerve cells and aids in energy conservation.

The gap between one neuron's axon and another neuron's dendrite is known as the synaptic cleft. The enlarged end of the axon that releases neurotransmitters into the synaptic cleft is known as the synaptic knob. Schwann cells are the cells that produce the myelin sheath.

4. Correct option: (D) a – s, b – t, c – p, d – q

List – I	**List – II**
a) Cerebrum	s. Seat of intelligence
b) Cerebellum	t. Maintain body posture
c) Hypothalamus	p. Control pituitary
d) Midbrain	q. Control vision of hearing

a) Cerebrum - s. Seat of intelligence

The cerebrum, which makes up the majority of the brain, is in charge of higher-order mental processes including reasoning, thinking, and language. It is in charge of regulating free will movements as well.

b)Cerebellum - t. Maintain body posture

The cerebellum, which is situated near the back of the brain, controls balance and coordination. It contributes to motor learning as well.

c)Hypothalamus - p. Control pituitary

Body temperature, appetite, thirst, and sleep are just a few of the crucial processes that the hypothalamus, a tiny structure near the base of the brain, regulates. Additionally, it contributes to the functioning of the endocrine system by secreting hormones that regulate the release of hormones from other glands.

d)Midbrain - q. Control vision of hearing

The cerebrum, pons, and medulla oblongata are all connected by a little portion of the brain called the midbrain. It comprises a number of significant structures, including the inferior and superior colliculi, which regulate hearing and eye movements, respectively.

5. Correct option: (D) Cerebellum

The cerebellum, an essential component of the mammalian brain, regulates muscular coordination, equilibrium, and posture. It is situated immediately above the brainstem at the back of the brain. The cerebellum receives information from various sensory systems, such as the inner ear, to help maintain balance and coordination. It also receives input from the motor cortex and spinal cord, allowing it to fine-tune and coordinate muscle movements. Damage to the cerebellum can result in difficulties with balance, coordination, and posture, leading to problems with movement and motor skills. Overall, the cerebellum is essential for ensuring smooth and coordinated movements in mammals.

The cerebellum, an essential component of the mammalian brain, regulates posture, equilibrium, and muscular coordination. Just above the brainstem, it is situated at the rear of the brain. For balance and coordination, the cerebellum gets data from a variety of sensory systems, including the inner ear. It also gets information from the spinal cord and motor cortex, which enables it to coordinate and fine-tune muscle actions. Problems with movement and motor abilities can come from damage to the cerebellum, which can also cause balance, coordination, and posture issues. Overall, mammals' ability to move smoothly and in unison is dependent on the cerebellum.

6. Correct option: (A) Stimulus → Sensory →Motor →Response

Reflex action is a quick, unconscious reaction to a stimulus. The reflex arc, a particular process it follows, has four key steps: stimulus, sensory, motor, and reaction. First, specific sensory receptors in the body, such the skin or sense organs, detect a stimulus. Through sensory neurons, these receptors subsequently transmit messages to the brain or spinal cord. Motor neurons in the brain or spinal cord process the information and immediately transmit a motor command. Finally, effectors like muscles or glands carry out the response, resulting in a prompt and automated response to the stimuli. This reflex arc allows the body to respond swiftly to potential dangers or threats, without the need for conscious thought or decision-making.

7. Correct option: (D) All the above

A critical component of the brain that is essential for many physiological processes is the hypothalamus. It affects sleep regulation by modulating the sleep-wake cycle. The hypothalamus also has parts that control appetite and thirst, which aid in regulating food intake and preserving fluid balance. By regulating the body's thermoregulatory

systems, it also has a big impact on regulating body temperature. In order to keep the body's equilibrium, the hypothalamus is in charge of coordinating and integrating these vital processes.

8. Correct option: (D) Dura mater – arachnoid – pia mater

Dura mater, arachnoid, and pia mater are the proper order of the meninges of the brain from the outside to the interior. The Pia mater is the innermost layer, followed by the arachnoid, the Dura mater, and then the arachnoid. The brain is shielded by the dura mater, a robust and fibrous membrane. It creates many divisions within the cranial cavity and is joined to the inner surface of the skull. Under the Dura mater is the arachnoid, a thin and web-like membrane. It contains the cerebrospinal fluid and acts as a cushion for the brain. The Pia mater is a thin and delicate membrane that adheres closely to the surface of the brain, providing it with nourishment and protection.

9. Correct option: (C) Midbrain, pons, medulla

The midbrain, pons, and medulla are the three primary parts of the brain stem. The midbrain, which is situated near the top of the brain stem, is in charge of sending sensory data to various areas of the brain. Additionally, it is essential for regulating eye movement and coordinating motor activities. The pons, which is below the midbrain, serves as a link between various regions of the brain. It facilitates the control of breathing, sleep, and alertness. Last but not least, the medulla, which is found at the base of the brain stem, regulates critical processes including breathing, blood pressure, and heart rate. These three parts together make up the brain stem, which is crucial for sustaining fundamental physical functions and communicating information between the brain and the rest of the body.

10. Correct option: (A) Balancing

The main function of the cerebellum is balancing. At the back of the brainstem is a part of the brain called the cerebellum. It is essential for planning and controlling voluntary motions, such as preserving balance and posture. To ensure fluid and synchronized motions, it integrates information from multiple sensory systems, including the inner ear. Additionally, the cerebellum collaborates with other brain areas to polish motor abilities and modify motions in response to signals from the body. The cerebellum is crucial for preserving stability and balance throughout the body.

11. Correct option: (B) Olfactory lobes

The answer to the question "Centre for the sense of smell is" is (B) Olfactory lobes. The signals associated to the sense of smell are processed and interpreted by the olfactory lobes, also referred to as the olfactory bulbs. These lobes, which are found near the front of the brain, collect data from the nasal cavity's olfactory receptors and send it to other parts of the brain for additional processing and perception. The olfactory lobes play a crucial role in our ability to detect and distinguish different odors, contributing to our sense of taste, memory, and emotional responses.

The center for the sense of smell is not among the other choices. The cerebellum is responsible for coordination and balance. The midbrain is involved in vision and hearing. The cerebrum is the largest part of the brain and is responsible for many higher-order functions, such as thinking, reasoning, and language

12. Correct option: (A) Medulla oblongata

The brainstem's medulla oblongata is in charge of regulating vital processes like respiration, heartbeat, and peristalsis. It serves as the body's control center, regulating these operations by receiving signals from various body areas and sending out the proper signals in response. It is essential for preserving homeostasis and making sure

the body is operating properly. Although the cerebrum and cerebellum are significant brain structures as well, they are not directly responsible for these particular processes. The cerebrum, which makes up the majority of the brain and is in charge of numerous higher-order processes like reasoning, thinking, and language. The cerebellum, which is situated near the back of the brain, controls balance and coordination. They do not control respiration, heartbeat, or peristalsis

13. Correct option: (A) Corpora quadrigemina

Four little structures collectively known as the corpora quadrigemina are found in the midbrain. They are in charge of processing both auditory and visual information. The corpora quadrigemina's superior colliculi, which are involved in processing visual information and controlling eye movement, have an important function. The corpora quadrigemina's inferior colliculi, which are also involved in auditory processing, transfer auditory data to higher brain areas. Overall, the midbrain, in particular the corpora quadrigemina, is essential for processing and integrating sensory information.

Between the midbrain and the cerebrum is a part of the brain called the diencephalon. The greatest portion of the brain is called the cerebrum. The pons is a region of the brain that is located between the midbrain and the medulla oblongata

14. Correct option: (C) Hypothalamus

The hypothalamus is where hunger and thirst are controlled. A tiny area of the brain called the hypothalamus is essential for controlling several body processes, including appetite and thirst. It has specialized cells called neurons that can recognize alterations in the bodily environment, such low blood sugar or dehydration. When these neurons fire, they send messages to other regions of the brain and body to start the proper reaction, like boosting appetite or making someone want to drink more water. In order to make sure that the body's demands for food and water are met, the hypothalamus serves as the control center for hunger and thirst.

The largest portion of the brain, the cerebrum, is in charge of higher-order mental processes including thinking, reasoning, and language. Behind the cerebrum, in the cerebellum, which controls balance, coordination, and motor skills. The medulla, which is found near the base of the brain, controls automatic processes including breathing, heart rate, and blood pressure.

15. Correct option: (B) Medulla – temperature regulation

Breathing, heart rate, and blood pressure are controlled by the medulla, which is a structure in the brainstem. It doesn't directly affect how the body regulates temperature. The hypothalamus, which is located in the forebrain, plays a major role in temperature regulation. As the body's thermostat, the hypothalamus monitors temperature fluctuations and sets in motion reactions to keep homeostasis. As a result, (B) is the appropriate response due to the incompatibility between the medulla and temperature regulation.

All of the other claims are true. Memory, balance, and smell are all functions of the cerebrum, cerebellum, and olfactory lobes, respectively.

16. Correct option: (D) Vasopressin

Vasopressin, also referred to as antidiuretic hormone, is hypo secreted in diabetes insipidus (ADH). The posterior pituitary gland is where vasopressin is kept after it is created in the hypothalamus. By regulating the reabsorption of water in the kidneys, it controls the body's water balance. Vasopressin is a hormone that the kidneys need to adequately reabsorb water, therefore when there is a deficit it causes excessive urination and thirst, which are the main signs of diabetes insipidus. Numerous things, including harm to the hypothalamus or pituitary gland, interactions with particular

drugs, or inherited conditions, can contribute to this illness. To restore the body's normal vasopressin levels, treatment usually entails hormone replacement therapy.
The thymus gland produces a hormone called thymosin. It contributes to the immune system's growth. The posterior pituitary gland secretes the hormone oxytocin, which is made by the hypothalamus and released by it. A hormone called insulin is created by the pancreas. It has a role in controlling blood sugar levels.

17. Correct option: (A) Hormone of the adrenal gland

The adrenal gland produces the hormone epinephrine. It also goes by the name adrenaline. The body's reaction to stress and emergencies is greatly influenced by this hormone. Epinephrine speeds up heartbeat, tightens blood vessels, and widens airways when it enters the bloodstream, preparing the body for a "fight or flight" reaction. It also aids in controlling metabolism and blood sugar levels. Epinephrine is a vital hormone that aids the body in adjusting and reacting to a variety of stresses and difficulties.
The mesoderm's nephrostomal portion does not produce hormones. It is a section of the embryo where the kidneys are formed. Renal corpuscles are groups of glomeruli in mammalian kidneys. Nephridia's frontal lobe is also not a hormone. Nephridia are excretory organs found in some invertebrates.

18. Correct option: (A) Increases water absorption

The pituitary gland secretes ADH, also known as antidiuretic hormone, which is a hormone made by the hypothalamus. Its primary purpose is to improve renal water absorption. ADH levels increase in a dehydrated body, which prompts the kidneys to reabsorb more water from the urine into the bloodstream. This promotes water conservation and minimizes excessive fluid loss. Conversely, when the body is sufficiently hydrated, ADH levels fall, allowing for a greater amount of water to be passed through the urine. In general, ADH is essential for preserving the body's water balance and avoiding dehydration.

19. Correct option: (D) Kidney

Over the kidneys are the adrenal glands. These tiny glands, which have a triangle shape, are found above each kidney. They are a vital component of the endocrine system and produce and release hormones like cortisol, adrenaline, and aldosterone. The kidneys and adrenals can communicate and work together effectively because of their close closeness. The adrenal glands respond to brain impulses by releasing hormones that control a number of body processes, including as metabolism, blood pressure, and stress response. Therefore, D) Kidney is the right response.

20. Correct option: (B) Androgens are produced by the ovary

Males' testes are the primary source of a set of male sex hormones known as androgens, including testosterone. While the ovaries do make a minor quantity of androgens, the testes are where most of the hormone is made. Specifically, estrogen and progesterone, which are female sex hormones, are produced by the ovaries. Therefore, it is untrue to say that the ovary produces androgens.
All of the other claims are true. The pituitary secretes oxytocin, the thyroid secretes thyroxine, and the pancreas secretes glucagon.

21. Correct option: (A) Pineal body

The pineal gland, which is found in the brain, is the main gland responsible for producing the hormone melatonin. The sleep-wake cycle is one of many biological rhythms that the pineal gland controls. Melatonin is a hormone that the body produces in reaction to darkness to tell the body that it is time to rest. Additionally, it affects physiological systems like immunological performance and antioxidant activity. Melatonin is not predominantly secreted by the skin, despite the fact that the skin does

contribute to the production of several hormones. Similar to the hypothalamus, the pituitary gland secretes a number of hormones, although melatonin is not one of them. Therefore, (A) Pineal body is the appropriate response.

22. Correct option: (D) Melatonin

Melatonin is secreted by the pineal gland, a tiny endocrine gland that is situated in the brain. The circadian rhythm, generally known as the sleep-wake cycle, is regulated by the hormone melatonin. It is created in reaction to darkness and aids in the promotion of sleep by reducing body temperature and causing tiredness. The night is when melatonin levels are highest, and they drop during the day. This hormone is frequently used as a supplement to treat sleep disorders including insomnia because it is essential for maintaining a healthy sleep pattern.

The thyroid gland produces the hormone thyroxine. The testes in men and the ovaries in women both generate a group of hormones known as androgens. Prolactin is a hormone that is produced by the pituitary gland.

23. Correct option: (A) Glucagon

α-cells of the pancreas secrete the hormone called glucagon. Glucagon plays a crucial role in regulating blood sugar levels in the body. When blood sugar levels are low, glucagon is released into the bloodstream and signals the liver to break down stored glycogen into glucose, which is then released into the bloodstream to increase blood sugar levels. This hormone works in opposition to insulin, which is secreted by β-cells of the pancreas and helps lower blood sugar levels by promoting the uptake and storage of glucose by cells. Together, glucagon and insulin maintain a delicate balance of blood sugar levels in the body to ensure proper functioning and energy production.

The other options are incorrect. Insulin is secreted by beta cells in the pancreas. Androgens and testosterone are hormones produced by the testes in males.

24. Correct option: (D) Prolactin

The hormone known as prolactin stimulates females to make milk. In reaction to a newborn sucking, it is created by the pituitary gland. Prolactin is released when a baby latches onto the mother's breast and starts to suckle. The mammary glands in the breasts are then stimulated by this hormone to create and release milk. As long as the baby is nursing, prolactin levels stay high, maintaining a consistent supply of milk for the child. Prolactin is a crucial hormone for breastfeeding women because it is necessary for the production and secretion of milk.

The uterus contracts more forcefully during childbirth and breastfeeding thanks to oxytocin. The maintenance of pregnancy is accomplished by progesterone. LH, also known as luteinizing hormone, is what causes females to ovulate.

25. Correct option: (A) Childbirth

The hypothalamus produces the hormone oxytocin, which the pituitary gland then secretes. Its main purpose is to facilitate childbirth. Oxytocin induces uterine muscular contractions during birth, assisting with the progression of labor and baby delivery. Oxytocin also helps to release breast milk by stimulating the muscles around the milk ducts, which makes it possible for milk to be expelled. While oxytocin is most commonly linked to delivery, it also plays other roles in the body, including fostering social connections and bonding, controlling mood, and affecting sexual behavior. But without a question, its engagement in birthing plays the most significant function.

26. Correct option: (B) Adrenal

The adrenal gland is referred to as the body's emergency gland since it is essential to the body's reaction to stress and emergencies. The adrenal gland, which is situated above the kidneys, creates the stress-related chemicals cortisol and adrenaline that are

then released into the bloodstream. In order to get the body ready for a fight-or-flight response, these hormones help to raise energy levels, blood pressure, and heart rate. The adrenal gland also controls the body's immune system, electrolyte balance, and metabolism. Overall, the adrenal gland is crucial for preserving the body's general health and emergency reaction.

A male reproductive organ that makes testosterone and sperm is the testis. A gland in the chest called the thymus plays a role in the maturation of the immune system. Several hormones are produced by the pituitary gland, a gland at the base of the brain that regulates the activity of other glands.

27. Correct option: (C) β- cells

Insulin is a hormone that is secreted by the β-cells of the pancreas. These cells are responsible for producing and releasing insulin into the bloodstream. Insulin plays a crucial role in regulating blood sugar levels by allowing cells in the body to take in glucose from the bloodstream and use it as a source of energy. Without insulin, glucose cannot enter the cells, leading to high blood sugar levels and potentially causing diabetes. Therefore, the correct answer is (C) β-cells.

Alpha cells of the pancreas secrete glucagon, which raises blood sugar levels. S cells of the pancreas secrete somatostatin, which inhibits the secretion of glucagon and insulin.

28. Correct option: (C) a – iv, b – iii, c – ii, d – i

a. Testis - (iv) Testosterone

Sperm and testosterone are produced by the male gonads known as the testes. Men's secondary sex characteristics, such as facial hair and a deep voice, are brought on by the hormone testosterone.

b. Ovaries - (iii) Estrogen

Female gonads called ovaries produce estrogen and eggs. The hormone estrogen is in charge of the growth of female secondary sex organs including the breasts and hips.

c. Thymus - (ii) Atrophies in adult

A gland in the chest called the thymus plays a role in the maturation of the immune system. Adulthood causes the thymus to atrophy.

d. Melanin - (i) Pigmentation

The pigment melanin is what gives skin its color. Melanocytes are the type of cells that create it.

29. Correct option: (C) Differentiation of T lymphocytes

T lymphocyte differentiation is brought about by thymosin. White blood cells called T lymphocytes, sometimes referred to as T cells, are an essential component of the immune response. The thymus gland, which is found in the chest area, produces the hormone known as thymosin. It aids in T lymphocyte maturation and development, making them into functional immune cells with the ability to recognize and eradicate external invaders like bacteria or viruses. Thymosin is necessary for the differentiation of T cells; otherwise, the immune system would be compromised, making people more vulnerable to illnesses and infections.

30. Correct option: (A) Androgens

Androgens are a class of hormones that are produced by Leydig cells, which are only found in the testes of males. In the hormone class known as androgens, testosterone is one of the key players. It is essential for the growth and upkeep of male reproductive organs and secondary sexual traits. In addition to the development of sperm production and other aspects of the male reproductive system, such as voice deepening, facial and body hair growth, and muscular mass, these hormones also have a role in the growth and maturation of the male genitalia. So, androgens are the proper response (A).

The ovaries in females are the organs that create estrogens. In the adrenal cortex, aldosterone is made. The anterior pituitary gland makes gonad-stimulating hormones. So, the only option left is androgens, which are produced by Leydig cells.

31. Correct option: (C) Melatonin

Humans' sleep-wake cycles are regulated by the hormone melatonin. The pineal gland, a tiny gland in the brain, produces it. The body receives a signal to go to sleep when melatonin levels rise in the evening. This hormone aids in synchronizing the circadian rhythm, the body's internal clock, with the regular day-night cycle. Melatonin encourages alertness and wakefulness when levels are low during the day. However, melatonin levels rise in the evening, encouraging sluggishness and aiding in sleep. Melatonin is therefore essential for sustaining a regular cycle of sleep and wakefulness.

A hormone called epinephrine is released in the presence of stress or danger. It aids in raising heart rate, blood pressure, and blood sugar levels, putting us in a fight or flight position. A hormone called gastrin promotes the formation of stomach acid. A hormone called insulin aids in controlling blood sugar levels.

32. Correct option: (C) a - ii, b - iv, c - iii, d - i

a. Pineal - (ii) Melatonin

Melatonin is produced by the pineal gland in the brain, which also aids in controlling the body's sleep-wake cycle.

b. Thyroid - (iv) Tetraiodothyronine

In the body, tetraiodothyronine (T4) is converted to triiodothyronine (T3) by the thyroid gland, which is situated in the neck. The hormones T3 and T4 aid in controlling the body's metabolism.

c. Ovary - (iii) Estrogen

The ovaries, which are found in the female pelvis, create estrogen, a hormone that aids in controlling the menstrual cycle and the development of female sexuality.

d. Adrenal medulla - (i) Epinephrine

Above the adrenal glands, which are situated above the kidneys, lies the adrenal medulla. Epinephrine is a hormone that the adrenal medulla produces to aid the body's reaction to stress.

33. Correct option: (B) Thyroxine

The hormone thyroxine, sometimes referred to as T4, is important for maintaining basal metabolic rate (BMR). BMR is the term used to describe how much energy is used by the body while it is at rest to maintain essential physiological processes like breathing, circulation, and cell creation. The thyroid gland produces the hormone thyroxine, which boosts the body's ability to produce heat and consume oxygen, therefore regulating metabolism. It promotes the digestion of proteins, lipids, and carbs, enabling the body to turn these foods into energy. So, it is essential to maintain a healthy BMR and general metabolic function that thyroxine levels are adequate.

34. Correct option: (D) Thymus gland

Under the breastbone is where you'll find the thymus gland. It is a little organ with a significant impact on the immune system. The thymus gland is in charge of the development and production of T-cells, a subset of white blood cells that aid in the defense against infections and sickness. The thymus gland gradually shrinks and becomes less active as we age. However, it is extremely active during childhood and adolescence and necessary for the building of a robust immune system. Because of this, the thymus gland's location beneath the breastbone makes it convenient for its immune system tasks.

The adrenal glands are located on top of the kidneys. Calcitonin is a hormone produced by the thyroid gland. The thyroid gland is located in the neck.

35. Correct option: (B) Acromegaly

Acromegaly, a rare illness, is brought on by excessive growth hormone production by the pituitary gland (GH). A little gland called the pituitary is found near the brain's base. It creates a variety of hormones, such as GH, which aids in controlling growth and development. Enlarged hands and feet, facial abnormalities like a projecting jaw and an enlarged nose, joint pain, and thickening skin are all typical signs of acromegaly. Acromegaly can cause major health issues like cardiovascular disease, diabetes, and respiratory issues if it is not treated.

When growth hormone is oversecreted at the time of skeletal growth, it causes gigantism, which results in excessive bone growth and organ expansion. Due to a lack of growth hormone, a person with dwarfism has much shorter height than the average individual. Midget is a term that is no longer used to describe people with dwarfism.

36. Correct option: (B) Simple goitre

Simple goitre is a disorder marked by thyroid gland hypertrophy brought on by insufficient dietary iodine. Iodine is necessary for the thyroid gland to create thyroid hormones, which are crucial for controlling growth and metabolism. The thyroid gland enlarges when there is an iodine deficit in an effort to make up for the deficiency. Goitre is the medical term for this growth. Iodine shortage frequently leads to hypothyroidism, or an underactive thyroid, which is characterised by symptoms like fatigue, weight gain, and cold intolerance. It is crucial to remember that while iodine shortage is the root cause of simple goitre, other disorders including Graves' disease can also induce an enlarged thyroid.

The disorder known as gigantism is brought on by the overproduction of growth hormone throughout childhood, which causes abnormal and excessive bone and tissue growth. Exophthalmic goitre, or Graves' disease, is a condition that causes bulging eyes and hyperthyroidism (an overactive thyroid gland) (exophthalmos). Congenital hypothyroidism, another name for cretinism, is a condition in which the thyroid gland is not fully developed from birth.

37. Correct option: (A) Decreased level of glucose in the blood

A condition known as insulin shock occurs when the blood's glucose level falls. When the body produces too much insulin, the glucose is quickly absorbed by the cells and not enough remains in the bloodstream, which leads to this condition. Dizziness, confusion, sweating, and in severe situations, loss of consciousness, are just a few of the symptoms that can result from this. To quickly treat insulin shock and return blood sugar levels to normal, consume foods or beverages high in glucose.

Increased blood glucose levels are not the best course of action. Low blood sugar levels, not high blood sugar levels, are related to insulin shock. In the setting of insulin shock, glucose loss through urine excretion is incorrect. A sign of diabetes, more specifically diabetes mellitus, in which the body is unable to control blood sugar levels, is the loss of glucose through the urine.

38. Correct option: (C) Cortisol

Cortisol and adrenaline are the hormones that are released in reaction to stress and dangerous emergency conditions. The fight-or-flight response, often known as the body's natural stress response mechanism, includes several chemicals. Adrenaline, which raises heart rate, blood pressure, and energy levels under stressful or dangerous situations, is released by the adrenal glands and primes the body for quick action. Stress causes the body to release cortisol, which helps control blood sugar levels,

reduce inflammation, and improve the body's response to stress. When these hormones are together, the body is better able to handle and react to emergency conditions.
Androgens are hormones that are made by both the ovaries and the testes in both men and women. They are in charge of the growth of male secondary sex traits like muscular mass and facial hair. The thymus gland produces a hormone called thymosin. It aids in immune system regulation.

39. Correct option: (A) Isthmus

The thyroid gland is a neck-based, butterfly-shaped gland. Two lobes, one on either side of the trachea, are what make up the isthmus, a slender flap of connective tissue that connects the two lobes. The thyroid glands isthmus serves as a structural support and aids in holding the two lobes together. In order to ensure adequate hormone production and secretion, it also enables communication and coordination between the lobes. The thyroid lobes would be divided and unable to work together properly without the isthmus.
The thyroid gland's functional units are called follicles. Iodine and tyrosine are combined to create thyroid hormones in them. Diuresis is an increase in urine production. The thyroid gland does not contain it. Thyroid-stimulating hormone is abbreviated as TCT. The pituitary gland releases a hormone that prompts the thyroid to make thyroid hormones.

LEVEL-II

1. Correct option: (A) Early embryos

Only early embryos contain unipolar neurons. The neural system is still growing and diffusing in the early stages of embryonic development. One process protrudes from the cell body of unipolar neurons, a kind of sensory neuron. The axon and dendrites are eventually produced by the same mechanism. These unipolar neurons will undergo additional specialization and differentiation as the embryo develops to create the intricate neural networks observed in the mature nervous system. As a result, unipolar neurons are fragile and are only seen in early embryos during the beginning of the nervous system's development. As the embryo develops, the unipolar neurons differentiate into other types of neurons, such as bipolar neurons and multipolar neurons.
Bipolar neurons make form the retina of the eye and the olfactory epithelium. Multipolar neurons, which comprise a cell body with one axon and two or more dendrites, make up the spinal cord.

2. Correct option: (D)Malathion

Acetylcholinesterase is inhibited by malathion. Acetylcholine, a neurotransmitter used in signal transmission between nerve cells, is broken down by the enzyme acetylcholinesterase. Malathion blocks the breakdown of acetylcholine by blocking this enzyme, which causes the neurotransmitter to build up in the synaptic cleft. Because of the continual stimulation of the postsynaptic cell caused by this extended cholinergic presence, the nervous system is overstimulated. The consequences of this overstimulation might range from muscular spasms to paralysis to, eventually, death. As a result, malathion's capacity to block acetylcholinesterase makes it a potent pesticide since it prevents bugs' nerve systems from operating normally, which causes them to die.
Aldrin, BHC, and endosulfan are also pesticides, but they are not acetylcholinesterase inhibitors.

3. Correct option: (B)Voluntary motor impulses

Loss of voluntary motor impulses would come from the death of the spinal cord's anterior horn cells. The motor neurons, sometimes referred to as the anterior horn cells, are in charge of carrying messages from the brain to the muscles so that we may consciously regulate our actions. The connection between the brain and the muscles is hampered when these cells are injured or eliminated, which results in a lack of voluntary motor control. This may cause paralysis, muscular wasting, and the inability to carry out voluntary motions like walking, reaching, or grabbing.

Integrating impulses are utilized to synchronize muscular movements. Commissural impulses are impulses that go back and forth down the spinal cord between the left and right sides. The brain receives information from the body via sensory impulses.

4. Correct option: (C) Cerebellum

A component of the brain called the cerebellum is essential for maintaining proper posture and bodily balance. It gathers data from a variety of sensory systems, including the inner ear, the eyes, and the muscles, and uses this data to coordinate and control movement. To maintain stability and guarantee synchronized, smooth movements, the cerebellum continually modifies muscle tone and activity. Balance problems, unstable walking, and trouble maintaining good posture can all be brought on by damage to or malfunction of the cerebellum. In order to maintain a steady and balanced bodily posture, the cerebellum is essential.

Although the pons and medulla oblongata are significant brain structures, they are not the main generators of balance and posture. The cerebrum, which makes up the majority of the brain, is in charge of higher-order processes including reasoning, organizing, and language. Balance and posture are not its primary functions.

5. Correct option: (B) under secretion of thyroxin

Thyroxin, a hormone produced by the thyroid gland, is under secreted in cretinism, a disorder that results from this. Thyroxin is essential for controlling the body's development and metabolism. Particularly in children, a thyroxin shortage can result in different developmental defects and limited growth. Lack of this hormone impairs the body's ability to operate normally, particularly the brain, which causes mental retardation, postponed physical maturation, and impairments in speech and hearing. Furthermore, those who have cretinism may have physical traits such a small stature, enlarged facial features, and a protruding tongue. Early detection and thyroxin replacement medication can help treat the symptoms of cretinism and stop future issues from developing.

Excess secretion of growth hormone can lead to a condition called acromegaly. Acromegaly is characterized by enlargement of the bones and soft tissues. Excess secretion of adrenaline can lead to a condition called pheochromocytoma. Pheochromocytoma is characterized by high blood pressure and other symptoms.

6. Correct option: (C) Amacrine cell

One of the cell types present in the retina are amacrine cells. They are interneurons that are very important for the retina's visual processing and information integration. Bipolar and ganglion cells provide input to amacrine cells, which modulates the transfer of visual signals between these cells. They are in charge of lateral inhibition, which raises contrast and brings edges of visual objects into sharper focus. Amacrine cells also play a role in controlling the retina's sensitivity to various amounts of light and help us perceive motion and directionality. Generally speaking, amacrine cells are necessary for the retina's normal operation and the processing of visual information prior to transmission to the brain.

In the retina, neuroglial cells are present. The retina's neurons are supported and fed by these cells, which are the most prevalent cell type there. They do not, however, participate in the processing of visual data. Not the retina, but the peripheral nervous system contains Schwann cells. The neurons in the peripheral nervous system's axons are protected by them. Not the retina, but the cerebellum has Purkinje cells. They assist with motor control.

7. Correct option: (A)somatic N.S

The somatic nervous system regulates the skeletal muscles. This system is controlled consciously and is in charge of voluntary motions. Our ability to do numerous motions, like walking, running, and lifting items, is made possible by the somatic nervous system, which is made up of motor neurons that send signals from the brain and spinal cord to the skeletal muscles. Acetylcholine in particular, a neurotransmitter that promotes muscular contraction, is released to achieve this regulation. As a result, the somatic nervous system is essential for directing and coordinating the actions of our skeletal muscles.

The autonomic nervous system controls involuntary actions, such as breathing and digestion. Glial cells are supportive cells in the nervous system. They do not control movement.

8. Correct option: (A) Hypochondria

A psychosomatic condition, hypochondria. Psychosomatic disorders are diseases or physical symptoms that are brought on or made worse by psychological elements like stress, worry, or emotional anguish. The hallmark of hypochondria, sometimes referred to as sickness anxiety disorder, is extreme worry and dread about having a major medical condition, but having little to no medical evidence to back up these fears. Physical symptoms experienced by hypochondriacs are frequently exaggerated or misread, which results in ongoing concern and obsession with their health. These symptoms can seriously impair a person's daily functioning and general well-being. As a result of the effect of psychological variables on the physical symptoms, hypochondria is classified as a psychosomatic condition.

Other options are incorrect. Anxiety, sadness, and obsessive-compulsive behaviors and thoughts are frequently present in neuroses. A significant mental illness that affects a person's thoughts, feelings, and behaviors is schizophrenia. Hallucinations, delusions, and disordered thinking are common in people with schizophrenia. Numerous causes, including mental illnesses, substance misuse, and physical ailments, can lead to psychosis.

9. Correct option: (C) rapid cell divisions in tendriller cells that are away from support

Rapid cell divisions in the tendrilled cells that are furthest from the support cause tendrils to form on pea plants. Plants can climb and hold themselves thanks to unique structures called tendrils. The cells in the tendrils quickly divide, lengthening and growing as a result. The tendrils may stretch out and locate a structure to attach to thanks to these divisions that especially take place in the tendrilled cells that are furthest from the support. With the help of this growth mechanism, the pea plant may efficiently climb and increase its exposure to sunlight, which is necessary for photosynthesis and the general growth and development of the plant.

The other options are incorrect. Option (A) is incorrect because the effect of gravity is not involved in the growth of tendrils. Option (B) is incorrect because chemicals are not involved in the growth of tendrils. Option (D) is incorrect because rapid cell division in tendriller cells in contact with the support would cause the tendril to move away from the support.

10. Correct option: (A) Neurotransmitter

A synapse is a tiny space or gap between two adjacent neurons where nerve impulses are transmitted. When an electrical signal, known as an action potential, reaches the end of a neuron, it triggers the release of chemical messengers called neurotransmitters. These neurotransmitters then diffuse across the synapse and bind to specific receptors on the neighboring neuron, allowing the transmission of the signal to continue. The synapse acts as a junction or connection point between neurons, enabling the flow of information throughout the nervous system. It plays a crucial role in processes such as learning, memory, and coordination of bodily functions. Neurotransmitters are chemicals that are released by the axon of the first neuron and received by the dendrites of the second neuron. Dendrites are the branches of a neuron that receive signals from other neurons. Axons are the long fibers that transmit signals from one neuron to another.

11. Correct option: (C) Ear: Adequate of sylvius

The other options in the question are all related to specific structures within the body part mentioned. Nervous tissue: Node of Ranvier is associated with the nervous system; it is a gap in the myelin sheath that surrounds an axon. Bone: Volkman's refers to a type of blood vessel found in bones, it is a canal that runs through the bone and contains blood vessels and Eye: Canal of Schlemm is a structure in the eye, it is a canal that drains fluid from the eye. However, "Ear: Adequate of Sylvius" does not correspond to any known anatomical structure or term. The adequate of Sylvius is a structure found in the brain, not the ear. It is a small canal that connects the third and fourth ventricles.

12. Correct option: (B) PNS

The Peripheral Nervous System, or PNS, is the Peripheral Nervous System. All of the body's nerves that are not found in the brain or spinal cord make up the peripheral nervous system. It consists of the spinal nerves, which emerge from the spinal cord to innervate the rest of the body, as well as the cranial nerves, which link the brain to the head and neck. The PNS is in charge of delivering motor instructions from the brain to the muscles and organs as well as sensory data from the body to the brain. It is essential for the coordination and regulation of body processes like movement, sensation, and organ control. Overall, the PNS acts as a communication network that connects the central nervous system (brain and spinal cord) to the rest of the body.

13. Correct option: (B) 31

The answer is 31. A long, cylindric bundle of nerves, the spinal cord runs from the brain's base to the lower back. It is in charge of sending messages from the brain to the rest of the body. The 31 pairs of nerves that make up the spinal cord each originate from a distinct section of the cord. In order to transmit sensory data from the body to the brain and motor impulses from the brain to the muscles, these nerves branch out and innervate various parts of the body. There are 31 pairs of nerves that emerge from the spinal cord as a result.

14. Correct option: (D) Abscisic acid

Abscisic acid is the substance that triggers the fall of mature leaves and fruits from plants. This plant hormone plays a crucial role in regulating various physiological processes, including leaf and fruit abscission. When the concentration of abscisic acid increases in a plant, it signals the cells at the base of the leaf or fruit to break down the cell walls, leading to detachment. This process is important for plants to conserve energy and resources during unfavorable conditions, such as drought or winter.

Abscisic acid also helps in maintaining water balance, seed dormancy, and stress responses in plants."
Auxin, cytokinin, and gibberellin are other plant hormones, but they do not trigger the fall of mature leaves and fruits from plants.

15. Correct option: (A) Iron is essential for the synthesis of thyroxin

The statement that iron is essential for the synthesis of thyroxin is not true. Thyroxin, also known as thyroxine, is a hormone produced by the thyroid gland. It is synthesized from the amino acid tyrosine and iodine, not iron. Iron plays a crucial role in the body, particularly in the production of red blood cells and oxygen transportation, but it is not directly involved in the synthesis of thyroxin. The thyroid gland requires iodine to produce thyroxin, which is then released into the bloodstream to regulate metabolism, growth, and development. Therefore, the statement that iron is essential for the synthesis of thyroxin is incorrect.
The other statements are all true. Thyroxine regulates carbohydrates, protein, and fat metabolism in the body. It is also called thyroid hormone.

16. Correct option: (D) Photonasty

Photonasty is the term used to describe the response of dandelion flowers to light. This phenomenon refers to the opening and closing of the petals in response to the presence or absence of bright light. During the daytime when there is ample light, dandelion flowers open their petals to maximize exposure to sunlight for photosynthesis and pollination. However, at night or in darkness, the flowers close their petals to protect their reproductive structures and conserve energy. This adaptive behavior allows dandelion flowers to optimize their chances of successful reproduction and survival in varying light conditions.
Phototropism is a directional movement of plant parts in response to light. Chemotropism is a directional movement of plant parts in response to chemicals. Thigmonasty is a non-directional movement of plant parts in response to touch.

17. Correct option: (A) Scala vestibuli and Scala media

Scala vestibuli and scala media in the inner ear are divided by Reisner's membrane. The basilar membrane, which is found in the cochlea, is parallel to this membrane, which is a thin structure. Two of the three chambers of the cochlea that are filled with fluid are the scala media and scala vestibuli. The intermediate chamber is called the scala medium, while the top chamber is called the scala vestibuli. The fluids in each of these two compartments are kept apart by Reisner's membrane, which serves as a barrier between them. The capacity of the inner ear to operate properly and send sound impulses to the brain depends on this separation.
The other choices are false. Option (B) is erroneous because the eustachian tube divides the throat from the tympanic cavity. Option (C) is erroneous since sound waves can reach the inner ear through holes in the cochlea called the fenestra rotundas and fenestra ovalis. The basilar membrane divides the scala media and scala tympani, making option (D) erroneous.

18. Correct option: (C) Testosterone from testes and estrogen from ovary

The secretions of testosterone and estrogen from the ovary play a major role in the significant changes in physical characteristics that accompany puberty. As the body gets ready for reproductive maturity throughout puberty, considerable physical changes occur. The formation of secondary sexual traits including facial hair, a deeper voice, and muscular mass is caused by testosterone, which is generated by the testes in males. The development of secondary sexual features including the enlargement of the breasts, broadening of the hips, and the beginning of menstruation are all influenced by

the hormone estrogen, which is generated by the ovaries in females. In moulding people's physical development throughout puberty and separating them into their distinct genders, these hormones are essential.

19. Correct option: (D) Insulin

A lack of the hormone insulin causes diabetes in patients. Blood sugar levels are controlled by insulin, which is produced by the pancreas. In people with diabetes, either the pancreas does not generate enough insulin or the body does not utilize the insulin it does make properly. This results in elevated blood glucose levels, which, if addressed, can have a negative impact on a number of bodily functions. By replacing or enhancing the body's natural insulin production, insulin treatment is frequently used to control diabetes.

Thyroxine is a hormone produced by the thyroid gland that regulates metabolism. Testosterone is a hormone produced by the testes in men and the ovaries in women that is responsible for sexual development and reproduction. Estrogen is a hormone produced by the ovaries in women that is responsible for sexual development and reproduction.

20. Correct option: (B) Differentiate the smell of a perfume from that of agarbatti

When someone has a cold, they cannot tell the difference between the fragrance of perfume and agarbatti. This is due to the fact that a cold frequently results in nasal congestion, which blocks the nasal passages and impairs one's ability to smell. It is difficult for the nose's olfactory receptors to correctly detect and differentiate between many odours. As a result, the person has much less capacity to sense and distinguish between particular odours, such as the distinction between a perfume and agarbatti. A temporary loss or modification in one's sense of smell results from the obstruction and inflammation in the nasal passages making it difficult for the person to distinguish minute differences in aroma.

The other options are incorrect. Option (A) is incorrect because taste is detected by taste buds in the tongue. Option (C) is incorrect because vision is not affected by a cold. Option (D) is incorrect because touch is not affected by a cold.

21. Correct option: (A) CNS, PNS, the brain and spinal cord; nerves around the body

The central nervous system (CNS) and the peripheral nervous system (PNS) are two different parts of the nervous system. The brain and spinal cord, which are parts of the CNS, are in charge of processing and delivering information. The PNS, on the other hand, is made up of nerves that run the length of the body and connect the CNS to numerous muscles, organs, and sensory receptors. The control center, or CNS, is responsible for receiving information from the PNS and responding appropriately. In contrast, the PNS acts as a network of communication, transmitting sensory data to the CNS and transferring motor instructions from the CNS to the rest of the body. These two sections of the nervous system act in concert to control body processes, plan motions, and permit sensory experience.

Therefore, the nervous system is divided into the Central nervous system (CNS) and the peripheral nervous system (PNS); the former consists of the brain and spinal cord and the latter consists of nerves around the body.

22. Correct option: (C) Dendrite -> cell body -> axon -> nerve ending

Neural transmission is the movement of information inside a neuron. Dendrites, which are branch-like extensions of the neuron that receive impulses from neighboring neurons, are where it all begins. These signals are subsequently sent to the soma, the cell's body, where they are digested and integrated. The generation of an electrical

impulse known as an action potential depends on whether the signals are powerful enough to cross the threshold. After then, the action potential moves up the axon, a long, tubular structure that protrudes from the cell body. Myelin, a fatty material that coats the axon, aids in accelerating the transmission of the action potential.
The nerve ending, also referred to as the axon terminal, at the end of the axon releases chemical messengers known as neurotransmitters into the synapse, which is the minuscule space between the nerve ending and the dendrites of the following neuron. The signal is then sent to the following neuron in the circuit by these neurotransmitters binding to receptors on the dendrites of the neighboring neuron.

23. Correct option: (A) Bright light → receptors in eyes → sensory neuron –> spinal cord → motor neurons → eyelid closes

The option (A) correctly shows the order of events when a bright light is focused on our eyes. When a bright light enters our eyes, it first stimulates the receptors in our eyes, specifically the rods and cones. These receptors then send signals to the sensory neurons, which are responsible for transmitting information from the sensory organs to the central nervous system. The sensory neurons carry the signals to the spinal cord, where they are processed and interpreted. Once the spinal cord receives the signals, it activates the motor neurons, which are responsible for transmitting signals from the central nervous system to the muscles. In this case, the motor neurons signal the muscles responsible for controlling our eyelids, causing them to close as a protective reflex to shield the eyes from the bright light. Therefore, the correct order of events is bright light → receptors in eyes → sensory neuron spinal cord → motor neurons → eyelid closes.

24. Correct option: (D) A part of midbrain

The forebrain, not the midbrain, contains the thalamus. It is situated near the cerebrum at the apex of the brainstem. A crucial center for the coordination of sensory and motor signaling is the thalamus. It gathers sensory data from all across the body and transmits it to the cerebral regions that need it. The cerebrum also communicates motor commands to the rest of the body.
The midbrain is a small part of the brain that is located between the thalamus and the pons. It is involved in a variety of functions, including vision, hearing, and motor control.
So, the thalamus is not a part of the midbrain. It is a part of the forebrain and is located between the midbrain and the cerebrum.

25. Correct option: (C) Pons

A complex network of brain regions called the limbic system or limbic lobe is responsible for emotions, memory, and behavior. It contains a number of important regions, including the hippocampus, amygdala, and inner cerebral hemispheres. The pons, however, is not regarded as a component of the limbic system or lobe. The brainstem's pons is a structure that plays a role in signal transmission between various regions of the brain and spinal cord. The limbic system, which controls emotions and memory, is not directly affected by this structure, despite the fact that it is a crucial component of overall brain function. Therefore, the correct answer is C) PONS.
The other options are all parts of the limbic lobe. The amygdala is a small almond-shaped structure that is involved in emotion. The hippocampus is a seahorse-shaped structure that is involved in memory.

26. Correct option: (A) Spinal cord

A reflex arc is a neural pathway that controls a reflex. It is a simple pathway that does not involve the brain. The reflex arc consists of the following parts:

- Receptor: This is the part of the body that detects the stimulus.
- Sensory neuron: This neuron carries the signal from the receptor to the spinal cord.
- Interneuron: This neuron is optional and may be present in some reflex arcs. It helps to integrate the signal from the sensory neuron with other signals.
- Motor neuron: This neuron carries the signal from the spinal cord to the effector. The effector is the muscle or gland that produces the response.

The spinal cord is the part of the nervous system that carries signals between the brain and the rest of the body. It is the site of the reflex arc. So, the missing term in the reflex arc is the spinal cord

27. Correct option: (B) Ganglion cells - Bipolar cells - Photoreceptor cells

From inside to outside, the retina's cells should be arranged in the following order: B) Ganglion cells - Bipolar cells - Photoreceptor cells. Bipolar cells provide impulses to ganglion cells, which are the retina's deepest layer of cells. Photoreceptor cells send signals to bipolar cells, which in turn receive them. The retina's outermost layer of cells, the rods and cones, are photoreceptor cells that are in charge of detecting light and turning it into electrical impulses. Ganglion cells, bipolar cells, and ultimately photoreceptor cells are the proper sequence of cells in the retina.

28. Correct option: (D) Perikaryon and dendrites

The "perikaryon" (sometimes referred to as the "cell body" or "soma") and "dendrites" are the two components that set a nerve cell (or "neuron") apart from other types of cells. Dendrites are the branching extensions that receive messages from other neurons or sense receptors, whereas the perikaryon is where the nucleus and other cellular organelles are found. The distinct mix of features that distinguishes neurons from other cell types is a crucial characteristic of neurons.

Flagellum and medullary sheath are not found in all nerve cells. Flagellum is a whip-like structure that helps cells move, and medullary sheath is a fatty covering that insulates axons. Vacuoles are sacs that store nutrients and waste products, and they are found in all cells.

29. Correct option: (A) 1

A neuron has a 1:1 ratio of intracellular potassium to extracellular sodium. This indicates that the sodium ions outside the cell are balanced by the potassium ions present inside the neuron. Ion channels and pumps operating in the cell membrane help to maintain this ratio. To maintain a balanced concentration, these channels and pumps actively move potassium ions into the cell and sodium ions out of the cell. As it permits the creation and transmission of electrical impulses known as action potentials, this equilibrium is essential for the neuron to operate properly. Various neurological diseases and reduced neuronal function may result from any change in this concentration ratio.

30. Correct option: (C) 2,3

Electric synapses are faster: True. When compared to chemical synapses, electric synapses—also known as gap junctions—allow for direct electrical connection between cells, resulting in quicker information transmission.

It is more difficult to regulate an electric synapse w.r.t. a chemical synapse: True. In terms of regulation, chemical synapses are more complicated and adaptable. Greater control over signal transmission is made possible by the ability to fine-tune and modify the release and reception of neurotransmitters at chemical synapses. Electric synapses lack this degree of control and regulation since they are more direct.

Acetylcholine is released at the electric synapse: False. Acetylcholine is not released at the electric synapse. Electric synapses do not use neurotransmitters to transmit

signals. Instead, they use gap junctions, which are channels that directly connect the cytoplasm of two cells.

31. Correct option: (A)1 and 2

Humans sweat (1) and dilate their skin arteries (2) to lower their body temperature when it slightly rises. When sweat evaporates, it removes heat from the body and aids in cooling it down. Sweating helps to cool the body by evaporating. Moreover, widening the skin arteries boosts blood flow close to the skin's surface, promoting heat dissipation. Together, these processes control body temperature and keep it within a relatively small range.

Constricting the skin arteries and tightening the skin's muscles are two strategies utilized to retain heat rather than dissipate it. As a result, they are not engaged in reversing a little increase in body temperature.

32. Correct option: (C)2-1-4-3

This pattern captures the physiological process of stimulus-induced changes in the permeability of a neuron's cell membrane. Due to the opening of voltage-gated sodium channels in response to a stimulus, there is first an increase in permeability to sodium ions (2). The resting membrane potential (RMP) is depolarized by this inflow of sodium ions, which depolarizes the membrane potential (3). The restoration of RMP (4) then occurs as a result of the diffusion of potassium ions (1) outside the cell, which is made possible by the opening of voltage-gated potassium channels. For neurons to operate properly in conveying electrical impulses, this sequence of events—in which sodium ions enter the cell first, followed by potassium ions exiting and ultimately restoring the RMP—must occur.